KCOR: A depletion-neutralized framework for retrospective cohort comparison under latent frailty

Manuscript metadata

- **Article type**: Methods / Statistical method
- **Running title**: KCOR under selection-induced cohort bias

- **Author**: Steven T. Kirsch
- **Affiliations**: Independent Researcher, United States
- **Corresponding author**: stk@alum.mit.edu

- **Word count**: 11,200 (excluding Abstract and References; approximate after identifiability revisions)
- **Keywords**: selection bias; frailty model; gamma mixture model; frailty inversion; frailty heterogeneity; selection-induced depletion; non-proportional hazards; cumulative hazard; hazard normalization; cumulative hazards; estimands; gamma frailty; negative controls; observational studies; observational cohort studies

Abstract

We propose KCOR, a framework for retrospective cohort comparison with irreversible outcomes under latent frailty heterogeneity. Selection-induced depletion can generate non-proportional hazards and curvature in observed cumulative hazards, biasing standard survival estimands in retrospective cohort studies using registry and administrative data. KCOR is a depletion-neutralized cohort comparison framework that uses a gamma-frailty working model with a Gompertz baseline to estimate a separate enrollment-time frailty variance $\theta_{0,d}$ for each cohort after a prespecified stabilization skip. The core estimator proceeds in four steps. It fits a seed $(k_d, \theta_{0,d}^{(0)})$ in the nearest quiet window, reconstructs the effective cumulative hazard over the full trajectory, aligns multiple quiet windows through persistent wave-offset terms, and refits $\theta_{0,d}$ jointly across all prespecified quiet windows within that cohort before applying gamma-frailty inversion. In epidemic-wave applications, KCOR also includes an optional non-proportional-hazard (NPH) exponent module that models cross-cohort hazard amplification as a function of depletion geometry and localizes a single shared exponent α using cross-cohort structure together with **externally supplied VE or other prespecified intervention-effect scale** under minimal aggregated data. Identification of depletion geometry is curvature-based and invariant to constant multiplicative hazard effects when scaling is common or approximately common across cohorts, while α is not separately identifiable from cohort-specific multiplicative effects if that external scale is not fixed. Wave-period NPH adjustment is applied after inversion only to **frailty-neutral cumulative hazard**, not to raw observed hazard (§2.7.3). Because identification comes from relative cohort behavior rather than absolute hazard levels, α is invariant to unknown common wave amplitudes under the working model once the external scale is specified. This extension remains optional, diagnostic, and assumption-dependent rather than part of the universal KCOR core. Across θ_0 recovery simulations, negative and positive controls, and Cox comparison exercises, KCOR remains a diagnostic and descriptive framework: it targets cumulative contrasts after depletion normalization rather than counterfactual effects. This revised architecture replaces the earlier quiet-window fit under a flatter baseline specification with a more explicit identification strategy for retrospective cohort comparisons under selection-induced hazard curvature. We illustrate the method using national COVID-era registry data as a worked example.

1. Introduction

1.1 Retrospective cohort comparisons under selection

Randomized controlled trials (RCTs) are the gold standard for causal inference, but are often infeasible, underpowered for rare outcomes, or unavailable for questions that arise after rollout. As a result, observational cohort comparisons are widely used to estimate intervention effects on outcomes such as all-cause mortality.

Although mortality is used throughout this paper as a motivating and concrete example, the method applies more generally to any irreversible event process observed in a fixed cohort, including hospitalization, disease onset, or other terminal or absorbing states. Mortality is emphasized here because it is objectively defined, reliably recorded in many national datasets, and free from outcome-dependent ascertainment biases that complicate other endpoints.

However, when intervention uptake is voluntary, prioritized, or otherwise selective, treated and untreated cohorts are frequently **non-exchangeable** at baseline and evolve differently over follow-up. This problem is not limited to any single intervention class; it arises whenever the same factors that influence treatment uptake also influence outcome risk.

While motivated by epidemic-era registry data, the method applies to general retrospective cohort comparisons where selection-induced depletion of susceptibles generates curvature in cumulative hazard trajectories.

This manuscript is a methods paper. Real-world registry data are used solely to illustrate estimator behavior, diagnostics, and failure modes under realistic selection-induced non-proportional hazards; no policy conclusions are drawn. KCOR does not attempt to recover counterfactual survival curves or causal vaccine-effect estimates, and readers seeking causal VE should not expect them here.

1.2 Curvature (shape) is the hard part: non-proportional hazards from frailty depletion

Selection does not merely shift mortality **levels**; it can alter mortality **curvature**—the time-evolution of cohort hazards. Frailty heterogeneity and selection-induced depletion (depletion of susceptibles) naturally induce curvature of the cumulative hazard even when individual-level hazards are simple functions of time. When selection concentrates high-frailty individuals into one cohort (or preferentially removes them from another), the resulting cohort-level hazard trajectories can be strongly non-proportional.

This violates core assumptions of many standard tools:

- **Cox PH**: assumes hazards differ by a time-invariant multiplicative factor (proportional hazards).
- **IPTW / matching**: can balance measured covariates yet fail to balance unmeasured frailty and the resulting depletion dynamics.
- **Age-standardization**: adjusts levels across age strata but does not remove cohort-specific time-evolving hazard shape.

KCOR is designed for this failure mode: **cohorts whose hazards are not proportional because selection induces different depletion dynamics (curvature).** In the revised estimator, curvature is interpreted through an explicit working model: a Gompertz baseline hazard combined with gamma-frailty depletion and an enrollment-time frailty variance $\theta_{0,d}$ defined at rebased time $t = 0$ after the stabilization skip. Approximate linearity after normalization remains a diagnostic, but it no longer serves as the sole identification anchor.

The methodological problem addressed here is general. The COVID-19 period provides a natural empirical regime characterized by strong selection heterogeneity and non-proportional hazards, and it serves as a useful illustration for the proposed framework. The **core** KCOR estimator identifies $\theta_{0,d}$ from prespecified quiet windows but applies depletion normalization to the full post-enrollment trajectory. For epidemic-wave applications, KCOR also supports an **optional** NPH exponent model in which a shared amplification parameter α captures how external hazards interact with depletion geometry across cohorts. That module is context-specific and is not part of the universal definition of KCOR. KCOR is therefore not specific to COVID, vaccination, or infectious disease, even though COVID-era registries motivate one important extension module.

Two mechanisms often lumped as the 'healthy vaccinee effect' (HVE) are distinguished here:

- **Static HVE:** baseline differences in latent frailty distributions at cohort entry (e.g., vaccinated cohorts are healthier on average). In the KCOR framework, this manifests as differing enrollment-time frailty variance $\theta_{0,d}$ and is the primary target of frailty normalization.

- **Dynamic HVE:** short-horizon, time-local selection processes around enrollment that create transient hazard suppression immediately after enrollment (e.g., deferral of vaccination during acute illness, administrative timing, or short-term behavioral/health-seeking changes). Dynamic HVE is operationally addressed by prespecifying a skip/stabilization window (§2.7), which also defines the rebased time origin at which $\theta_{0,d}$ is interpreted.

In epidemic-wave applications, an additional complication arises from external non-proportional hazards that interact with baseline frailty. In this manuscript, that issue is handled through an optional NPH exponent module described in Methods §2.7.1–§2.7.4 and revisited in Limitations §5.4.

Box 1. Two fundamentally different strategies for cohort comparability

- **Traditional matching and regression approaches:** attempt to construct comparable cohorts by matching or adjusting *characteristics of living individuals* at baseline or over follow-up, and then estimating effects via a fitted hazard model (e.g., Cox proportional hazards). This implicitly assumes that sufficiently rich covariate information can render cohorts exchangeable with respect to unobserved mortality risk.
- **Problem under latent frailty:** even meticulous 1:1 matching on observed covariates can fail to equalize mortality risk trajectories. In such settings, cohort differences arise not from mismeasured covariates, but from **selection-induced depletion of susceptibles**, which alters hazard curvature over time.
- **KCOR strategy:** rather than equating cohorts based on characteristics of the living, KCOR equates cohorts based on how they die in aggregate. It estimates cohort-specific enrollment-time depletion geometry through a Gompertz-plus-frailty working model anchored at rebased $t = 0$, aligns prespecified quiet windows through a structured delta-iteration estimator, adjusts for that geometry via analytic inversion under the working model, and compares cohorts on the resulting depletion-neutralized cumulative hazard scale.
- **Inferential distinction:** Cox-type methods are **model-based and individual-level**, conditioning on survival and fitting covariate effects, whereas KCOR is **measurement-based and cohort-level**, operating directly on aggregated mortality trajectories without fitting covariate models. The inferential target is cumulative outcome accumulation rather than an instantaneous hazard ratio conditional on survival.

1.3 Related work (brief positioning)

KCOR builds on the frailty and selection-induced depletion literature in which unobserved heterogeneity induces deceleration of cohort-level hazards over follow-up (a standard working model is gamma frailty)[1]. KCOR's distinct contribution is not additional hazard flexibility, but a **diagnostics-driven normalization** of selection-induced depletion geometry in cumulative-hazard space prior to defining a cumulative cohort contrast. Related approaches that address non-proportional hazards (time-varying effects, flexible parametric hazards, additive hazards) or time-varying confounding (MSM/IPW/g-methods) target different estimands and typically require richer longitudinal covariates than are available in minimal registry data[2–9]. Additional discussion is provided in the Supplementary Information (SI).

Time-varying coefficient Cox models allow hazards to change over time but do not adjust for frailty-induced depletion in the marginal cumulative sense targeted by KCOR because estimation remains conditional on survival; they therefore address non-proportionality without adjusting for selection-induced curvature in marginal cumulative comparisons.

Marginal structural models target causal effects under exchangeability using longitudinal covariates and weighting; KCOR instead targets descriptive cumulative contrasts under minimal data, so the estimands, assumptions, and failure modes differ.

Flexible parametric survival models improve baseline fit but do not resolve depletion-induced selection bias when frailty heterogeneity is present.

1.4 Evidence from the literature: residual confounding despite meticulous matching

Motivating applied studies suggest that even careful matching and adjustment can leave substantial residual differences in non-COVID mortality and time-varying "healthy vaccinee effect" signatures, consistent with selection and depletion dynamics not captured by measured covariates[10–13].

1.5 Contribution of this work

This work makes four primary contributions: (i) it formalizes selection-induced depletion under latent frailty heterogeneity as a source of non-proportional hazards and curvature that can bias common survival estimands; (ii)

it replaces a single-window constant-baseline fit with a Gompertz-based delta-iteration estimator that targets a cohort-specific enrollment-time frailty variance $\theta_{0,d}$ across aligned quiet windows; (iii) it separates a universal KCOR core from an optional epidemic-wave NPH exponent module with a shared amplification parameter α localized, if at all, from cross-cohort structure together with **externally supplied VE or other prespecified intervention-effect scale**, and a post-inversion cumulative correction on **frailty-neutral cumulative hazard** (§2.7); and (iv) it validates the revised estimator using a broader stack that includes θ_0 recovery, control analyses, Cox mismatch demonstrations, and explicit failure signaling.

A central implication is identifiability: in minimal-data retrospective cohorts, interpretability depends not on any one quiet window alone but on a coherent combination of Gompertz curvature, rebased enrollment-time anchoring, consistency across quiet windows, and diagnostics that indicate structured depletion geometry has been identified rather than absorbed into residual time-varying effects. Under minimal aggregated data, $\theta_{0,d}$ is identified from curvature when multiplicative hazard scaling is common or approximately common across cohorts; heterogeneous multiplicative effects can induce second-order curvature differences through differential depletion and are assessed via diagnostics, while additive or time-varying hazards remain primary threats to curvature interpretation. The optional NPH exponent α is not separately identifiable from cohort-specific multiplicative structure in minimal aggregated data (Limitations §5.4).

Together, these contributions position KCOR not as a replacement for existing survival estimands, but as a prerequisite normalization step that addresses a source of bias arising prior to model fitting in many retrospective cohort studies.

1.6 Target estimand and scope (non-causal)

Box 2. Target estimand and scope (non-causal)

- **Primary estimand (KCOR)**: For two fixed enrollment cohorts A and B, it is a **depletion-adjusted marginal estimand**, defined as
$$\mathrm{KCOR}(t) = \tilde{H}_{0,A}(t)/\tilde{H}_{0,B}(t),$$
where $\tilde{H}_{0,d}(t)$ is cohort d's **depletion-neutralized baseline cumulative hazard** obtained by estimating enrollment-time frailty variance $\theta_{0,d}$ from prespecified quiet windows and applying the gamma-frailty inversion (Methods §2). When the optional NPH cumulative correction is used, the same ratio employs $H_{\mathrm{corr},d}(t)$ in place of $\tilde{H}_{0,d}(t)$ (§2.7.3).
- **Operational summary**: KCOR proceeds by (i) freezing cohorts and rebasing time after a prespecified stabilization skip, (ii) estimating $\theta_{0,d}$ with a Gompertz-based delta-iteration procedure across prespecified quiet windows, (iii) applying gamma-frailty inversion to the full post-enrollment cumulative hazard trajectory, (iv) optionally applying the NPH correction to **frailty-neutral cumulative hazard** after inversion when prespecified (§2.7.3), not to raw hazard, and (v) comparing cohorts on that scale. Steps (i)–(iii) and (v) define the **core** pipeline; step (iv) is **optional**. When α is not identified, step (iv) is **inactive** and does not affect reported results. In epidemic-wave settings, localization of α requires **externally supplied VE or other prespecified intervention-effect scale** (§2.7.1–§2.7.2).
- **Interpretation**: KCOR is a time-indexed **cumulative** contrast on the depletion-neutralized scale. Values above/below 1 indicate greater/less cumulative event accumulation in cohort A than B by time t after depletion normalization. KCOR is not an instantaneous hazard ratio. It should be read as a **diagnostic** signal about persistent contrasts after working-model depletion adjustment, not—by itself—as evidence of causal harm or benefit.
- **What it is not**: KCOR is **not** a causal effect estimator (no ATE/ATT) and does not recover counterfactual outcomes under hypothetical interventions.
- **When interpretable**: Interpretation is conditional on explicit assumptions (fixed cohorts; shared external hazard environment; adequacy of the working frailty model; identifiability of $\theta_{0,d}$ from prespecified quiet windows; structured additivity of persistent wave offsets when the delta estimator is used) **and** on internal diagnostics (quiet-window fit quality; post-normalization linearity within quiet windows; multi-window consistency; parameter stability to perturbation; and, when relevant, identifiability and stability of the optional NPH exponent model).
- **If diagnostics indicate non-identifiability**: the analysis is treated as not identified and KCOR is not

reported as a "normalized contrast".

1.7 Paper organization and supporting information (SI)

The main text introduces the KCOR estimator, provides a canonical demonstration of Cox bias under frailty-driven depletion, and presents two primary validation examples (a negative control and a stress test). Additional validations—including positive controls—along with extended diagnostics, empirical registry-based nulls, and detailed simulation specifications are provided in the Supplementary Information (SI; Sections S2 and S4.2–S4.3; Tables S1–S3).

2. Methods

Mortality is used as the primary example throughout this section because it is objectively defined and reliably recorded in many administrative datasets.

Table 4 defines the notation used throughout the Methods section.

For COVID-19 vaccination analyses, intervention count corresponds to the number of vaccine doses received; more generally, this can index any discrete exposure level.

2.1 Conceptual framework and estimand

Retrospective cohort differences can arise from two qualitatively different components:

- **Level differences**: cohort hazards differ by an approximately time-stable multiplicative factor (or, equivalently, cumulative hazards have different slopes but similar shape).
- **Depletion (curvature) differences**: cohort hazards evolve differently over time because cohorts differ in latent heterogeneity and are **selectively depleted** at different rates.

Contrast with RMST and model-free cumulative summaries. Restricted mean survival time (RMST), cumulative incidence, and similar integrated-survival summaries characterize outcome accumulation on the **observed** scale without adjusting for selection-induced depletion curvature in the hazard trajectory. KCOR instead defines a **geometry-consistent descriptive contrast** on the depletion-neutralized cumulative-hazard scale after frailty inversion when diagnostics support identification of $\theta_{0,d}$. The estimands differ when non-proportionality is driven by frailty depletion: RMST-style summaries retain that geometry in the marginal comparison, whereas KCOR attempts to partial it out under the stated working model.

Non-causal estimand. $\mathrm{KCOR}(t)$ summarizes a depletion-adjusted contrast in cumulative outcomes on the working-model scale. It does not, by itself, identify causal effects. Interpretation as causal requires additional assumptions beyond the KCOR framework. Accordingly, KCOR should be interpreted as a **diagnostic signal** indicating whether outcome differences persist after adjusting for depletion geometry under the assumed frailty structure, not as evidence of causality.

This framework targets the second failure mode. Under latent frailty heterogeneity, high-risk individuals die earlier, so the surviving risk set becomes progressively "healthier." This induces **downward curvature** (deceleration) in cohort hazards and corresponding concavity in cumulative-hazard space, even when individual-level hazards are simple and even under a true null treatment effect. When selection concentrates frailty heterogeneity differently across cohorts, the resulting curvature differences produce strong non-proportional hazards and can drive misleading contrasts for estimands that condition on the evolving risk set.

KCOR does not assert a causal interpretation of the resulting contrast; rather, it highlights residual differences that cannot be attributed to frailty-driven selection under the working model, flagging regimes that demand substantive explanation beyond selection artifacts. KCOR does not identify constant multiplicative hazard effects as separate parameters; they are not separately identifiable from baseline scale. Under minimal aggregated data, $\theta_{0,d}$ is nonetheless identifiable from curvature structure when multiplicative scaling is common or approximately common across cohorts, and such scaling does not alter that curvature-based identification under the working model (§2.4.4 for heterogeneous multiplicative scaling and diagnostics). Non-multiplicative effects (additive or time-varying) may

distort curvature and reduce identifiability. Interpretation is therefore conditional on the adequacy of the gamma-frailty working model, the Gompertz structure, and the associated diagnostics.

The strategy is therefore:

1. **Estimate the cohort-specific depletion geometry** (via curvature) during prespecified epidemiologically quiet periods.
2. **Map observed cumulative hazards into a depletion-neutralized space** by inverting that geometry.
3. **Compare cohorts only after normalization** using a prespecified post-adjustment estimand; ratios of depletion-neutralized cumulative hazards (KCOR) are used here.

KCOR operates by first estimating depletion geometry ($\theta_{0,d}$) from prespecified quiet windows, then optionally adjusting wave-period amplification (α) on the **frailty-neutral cumulative-hazard scale after gamma-frailty inversion** (§2.7.3), and finally comparing cumulative outcomes on the normalized scale.

The normalization logic is illustrated empirically in Figure 1 using a representative cohort comparison drawn deterministically from the Czech analysis pipeline.

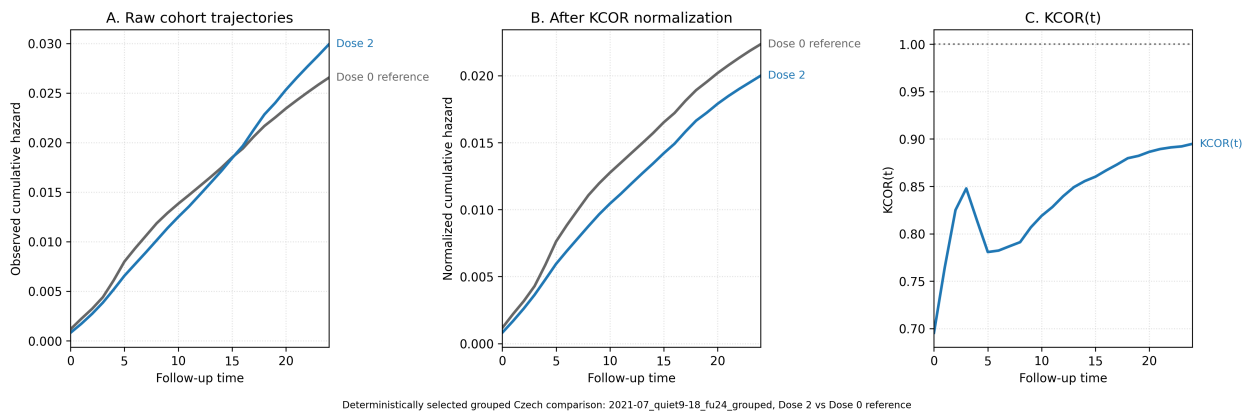

Figure 1: Empirical illustration of the KCOR workflow on a deterministically selected grouped Czech cohort comparison (2021-07 enrollment, grouped dose 2 versus dose 0 reference). (A) The raw observed cumulative hazards differ in both level and curvature. (B) After estimating cohort-specific depletion geometry and applying depletion normalization, the cumulative hazards become more nearly linear and directly comparable, illustrating how depletion normalization adjusts for curvature attributed to selection-induced depletion under the working model. (C) KCOR(t) is then computed as the cumulative contrast on that normalized scale. This figure is intended to illustrate the geometric effect of the workflow on a representative analysis example rather than to serve as a standalone substantive result.

KCOR deliberately targets cumulative mortality contrasts rather than instantaneous hazard ratios. This choice is intentional: cumulative outcome accumulation is invariant to transient hazard-ratio oscillations induced purely by depletion geometry. It is therefore not intended as a replacement for Cox proportional hazards models or flexible parametric survival models that estimate time-varying hazard effects. Instead, KCOR provides a geometry-based normalization of cumulative risk in settings where selection-induced depletion distorts marginal comparisons.

All analyses are performed using discrete weekly time bins; continuous-time notation is used solely for expositional convenience. See Table 4 for the full symbol list.

Notation preview.
Throughout this paper, t denotes event time since cohort enrollment and d indexes cohorts. For frailty identification, time is rebased after the stabilization skip so that $t_{\text{rebased}} = 0$ at the first accumulating interval. Unless otherwise specified, modeling and parameter estimation are carried out in rebased time. Let $H_{\text{obs},d}(t)$ denote the observed cohort-level cumulative hazard, computed from fixed risk sets after the required preprocessing steps, and let $\tilde{H}_{0,d}(t)$ denote the corresponding *depletion-neutralized baseline cumulative hazard* obtained after frailty normalization. Let $H_{\text{corr},d}(t)$ denote the optional NPH-corrected cumulative hazard after inversion (§2.7.3). Hats denote estimated

quantities; tildes denote depletion-neutralized quantities. Individual hazards are modeled using a latent multiplicative frailty term z, with cohort-specific enrollment-time variance $\theta_{0,d}$, which governs the strength of selection-induced depletion and resulting curvature in observed cumulative hazards. Each cohort's $\theta_{0,d}$ is estimated separately; there is no pooled/shared frailty variance across comparison groups in the core estimator. Full notation is summarized in Table 4.

2.1.1 Target estimand

Scope and interpretation are summarized in Box 2 (§1.6); the formal definition used throughout is provided here.

Let $\tilde{H}_{0,d}(t)$ denote the **depletion-neutralized baseline cumulative hazard** for cohort d at event time t since enrollment (Table 4). For two cohorts A and B, KCOR is defined as

$$\mathrm{KCOR}(t) = \frac{\tilde{H}_{0,A}(t)}{\tilde{H}_{0,B}(t)}. \tag{1}$$

When the optional NPH cumulative correction is active, $H_{\mathrm{corr},d}(t)$ replaces $\tilde{H}_{0,d}(t)$ in this ratio (§2.7.3).

KCOR is therefore a constructed cumulative contrast defined on a depletion-normalized scale rather than a directly observable quantity in the raw data.

For visualization, an **anchored KCOR** is sometimes reported to show post-reference divergence:

$$\mathrm{KCOR}(t; t_0) = \mathrm{KCOR}(t)/\mathrm{KCOR}(t_0),$$

with prespecified t_0 (e.g., 4 weeks). Anchoring is used only when explicitly stated to remove pre-existing level differences and emphasize post-reference divergence.

2.1.2 Identification versus diagnostics

Scope and interpretation are summarized in Box 2 (§1.6). A working-model identification proposition for $\theta_{0,d}$ is stated immediately after §2.4.4, together with a diagnostic interpretation disclaimer.

Interpretability of a KCOR trajectory is assessed via prespecified diagnostics (Supplementary Information §S2; Tables S1–S3). When diagnostics indicate non-identifiability, the analysis is treated as not identified and results are not reported. Checks include:

- stability of $(\hat{k}_d, \hat{\theta}_{0,d})$ and the aligned quiet-window geometry under small perturbations,
- approximate linearity of $\tilde{H}_{0,d}(t)$ within the quiet window,
- absence of systematic residual structure in hazard space.

Diagnostics corresponding to each assumption are summarized in Supplementary Information §S2 (Tables S1–S3).

2.1.3 KCOR assumptions and diagnostics

These assumptions define when KCOR normalization is interpretable.

The KCOR framework relies on the following assumptions, which are framed diagnostically:

1. **Fixed cohort enrollment.** Cohorts are defined at a common enrollment time and followed forward without dynamic entry or rebalancing.
2. **Multiplicative latent frailty.** Individual hazards are assumed to be multiplicatively composed of a baseline hazard and an unobserved frailty term, with cohort-specific frailty distributions.
3. **Quiet-window stability.** A prespecified epidemiologically quiet period exists during which external shocks to the baseline hazard are minimal, allowing depletion geometry to be estimated from observed cumulative hazards. Quiet windows need not be perfectly free of shocks; they should represent intervals where no large transient perturbations dominate baseline mortality. Residual baseline drift can be absorbed into the frailty

parameter if quiet-window conditions are violated; empirical robustness of fitted frailty parameters to quiet-window placement (12-month windows shifted monthly) is demonstrated in Supplementary Figure S5 using Czech registry data, supporting stability when this condition holds approximately. Epidemic-wave periods with sharp hazard shocks fall outside this assumption and are excluded from normalization by design.

4. **Independence across strata.** Cohorts or strata are analyzed independently, without interference, spillover, or cross-cohort coupling.
5. **Sufficient event-time resolution.** Event timing is observed at a temporal resolution adequate to estimate cumulative hazards over the quiet window.

If no candidate quiet window satisfies diagnostic criteria, KCOR is not interpretable and analysis should terminate without reporting contrasts. Multiple disjoint quiet windows may be pooled provided fitted depletion parameters are stable and diagnostics are consistent across windows. Quiet-window validity diagnostics are summarized in Supplementary Information §S2 (Tables S1–S3).

These assumptions are evaluated empirically using post-normalization diagnostics. Violations are expected to manifest as residual curvature, drift, or instability in adjusted cumulative hazard trajectories.

2.2 Cohort construction

KCOR is defined for **fixed cohorts at enrollment**. Required inputs are minimal: enrollment date(s), event date, and optionally birth date (or year-of-birth) for age stratification. Analyses proceed in discrete event time t (e.g., weeks) measured since cohort enrollment.

Cohorts are assigned by intervention state at the start of the enrollment interval. In the primary estimand:

- **No post-enrollment switching** is allowed (individuals remain in their enrollment cohort),
- **No censoring** is applied (other than administrative end of follow-up),
- analyses are performed on the resulting fixed risk sets.

Censoring or reclassification due to cohort transitions (e.g., moving between exposure groups over time) is not permitted, because such transitions alter the frailty composition of the cohort in a time-dependent manner. Allowing transitions would introduce additional, endogenous selection that changes cohort mortality trajectories in unpredictable ways, confounding depletion effects that KCOR is designed to normalize.

This fixed-cohort design is intentional. It avoids immortal-time artifacts and prevents outcome-driven switching rules from creating time-dependent selection that is difficult to diagnose under minimal covariate availability. Extensions that allow switching or censoring are treated as sensitivity analyses (§5.2) because they change the estimand and introduce additional identification requirements.

Conceptual requirements of the KCOR framework are distinguished from operational defaults, which are reported separately for reproducibility (Supplementary Section S4).

Throughout this manuscript the failure event is **all-cause mortality**. KCOR therefore targets cumulative mortality hazards and is not framed as a cause-specific competing-risks analysis.

2.3 Hazard estimation and cumulative hazards in discrete time

For each cohort d, let $N_d(0)$ denote the number of individuals at enrollment. Let $d_d(t)$ denote deaths occurring during interval t, and let

$$D_d(t) = \sum_{s \le t} d_d(s)$$

denote cumulative deaths up to the end of interval t.

Define the risk set size at the start of interval t as

$$N_d(t) = N_d(0) - \sum_{s<t} d_d(s) = N_d(0) - D_d(t-1).$$

In the primary estimand, individuals do not switch cohorts after enrollment and there is no loss to follow-up; therefore $N_d(t)$ is the risk set used to define all discrete-time hazards and cumulative hazards in this manuscript.

Define the interval mortality ratio

$$\mathrm{MR}_{d,t} = \frac{d_d(t)}{N_d(t)}.$$

The discrete-time cohort hazard is computed as

$$h_{\mathrm{obs},d}(t) = -\ln\big(1 - \mathrm{MR}_{d,t}\big) = -\ln\left(1 - \frac{d_d(t)}{N_d(t)}\right). \tag{2}$$

This transform is standard: it maps an interval event probability into a continuous-time equivalent hazard under a piecewise-constant hazard assumption. For rare events, $h_{\mathrm{obs},d}(t) \approx \mathrm{MR}_{d,t} = d_d(t)/N_d(t)$, but the log form remains accurate and stable when weekly risks are not negligible. The exact transform $h_{\mathrm{obs},d}(t) = -\log(1 - \mathrm{MR}_{d,t})$ is used throughout; the rare-event approximation is provided for intuition only. This discrete-time hazard transform under a piecewise-constant hazard assumption is standard; see, e.g., Kalbfleisch and Prentice (*The Statistical Analysis of Failure Time Data*)[14].

All hazard and cumulative-hazard quantities used in KCOR are discrete-time integrated hazard estimators derived from fixed-cohort risk sets; likelihood-based or partial-likelihood formulations are not used for estimation or for the subsequent frailty-based normalization.

Observed cumulative hazards are accumulated over event time after preprocessing (§2.7), where preprocessing is the **stabilization skip** only: $h_d^{\mathrm{adj}}(t) = h_d^{\mathrm{eff}}(t)$. The optional NPH exponent module does **not** alter this accumulation; it adjusts frailty-neutral cumulative hazards **after** gamma-frailty inversion (§2.7.3).

$$H_{\mathrm{obs},d}(t) = \sum_{s \le t} h_d^{\mathrm{adj}}(s), \qquad \Delta t = 1. \tag{3}$$

Discrete binning accommodates tied events and aggregated registry releases. Bin width is chosen based on diagnostic stability (e.g., smoothness and sufficient counts per bin) rather than temporal resolution alone.

In addition to the primary implementation above, $\hat{H}_{\mathrm{obs},d}(t)$ was computed using the Nelson–Aalen estimator $\sum_{s \le t} d_d(s)/N_d(s)$ as a sensitivity check; results were unchanged.

2.4 Selection model: gamma frailty and depletion normalization

2.4.1 Individual hazards with multiplicative frailty

Let $z_{i,d}$ denote an individual-specific latent frailty term with mean 1 and variance $\theta_{0,d}$, defined at rebased time $t_{\mathrm{rebased}} = 0$ after the stabilization skip. Let $\tilde{h}_{0,d}(t)$ denote the depletion-neutralized baseline hazard for cohort d. Individual hazards are modeled as

$$h_{i,d}(t) = z_{i,d}\,\tilde{h}_{0,d}(t), \qquad z_{i,d} \sim \mathrm{Gamma}(\mathrm{mean} = 1,\ \mathrm{var} = \theta_{0,d}). \tag{4}$$

Here $\theta_{0,d}$ is an **enrollment-time** parameter: it characterizes the cohort's frailty heterogeneity at the start of the accumulating follow-up, not the time-varying variance of the surviving population after depletion has acted. Larger $\theta_{0,d}$ produces stronger early depletion of high-frailty individuals and therefore stronger cohort-level hazard deceleration.

Gamma frailty is used because it yields a closed-form link between observed and baseline cumulative hazards via the Laplace transform[1]. In KCOR, gamma frailty is a **working geometric model** for depletion normalization, not a claim of biological truth. KCOR does **not** require the gamma frailty model to be structurally correct; it uses this specification to extract depletion-consistent curvature. If the true heterogeneity deviates from gamma, $\theta_{0,d}$ should be interpreted as an **effective parameter** summarizing depletion geometry rather than as the literal variance of a gamma distribution. Deviations from the gamma assumption are expected to manifest as diagnostic failures (e.g., poor fit,

unstable $\hat{\theta}_{0,d}$, or sub-nominal coverage), providing an internal check on model adequacy. Adequacy is evaluated empirically through fit quality, multi-window consistency, and post-normalization diagnostics.

2.4.2 Gamma-frailty identity and inversion

Let

$$\tilde{H}_{0,d}(t) = \int_0^t \tilde{h}_{0,d}(s)\, ds \tag{5}$$

denote the depletion-neutralized baseline cumulative hazard. Under the gamma-frailty working model, the observed cohort-level cumulative hazard satisfies

$$H_{\text{obs},d}(t) = \frac{1}{\theta_{0,d}} \log\big(1 + \theta_{0,d}\tilde{H}_{0,d}(t)\big)\,, \tag{6}$$

and the corresponding inversion is

$$\tilde{H}_{0,d}(t) = \frac{\exp\big(\theta_{0,d}H_{\text{obs},d}(t)\big) - 1}{\theta_{0,d}}. \tag{7}$$

Differentiating Eq. (6) gives the hazard-level relation

$$h_{\text{obs},d}(t) = \frac{\tilde{h}_{0,d}(t)}{1 + \theta_{0,d}\tilde{H}_{0,d}(t)}, \qquad \tilde{h}_{0,d}(t) = h_{\text{obs},d}(t)\,\big(1 + \theta_{0,d}\tilde{H}_{0,d}(t)\big)\,,$$

so the Step 2 recursion is the discrete-time reconstruction implied by this inversion geometry. This relation is exact only under the assumed gamma-frailty working model; it is not a model-free algebraic identity for arbitrary cohort hazards. The inversion is the **normalization operator**: given an estimate $\hat{\theta}_{0,d}$, it maps the observed cumulative hazard $H_{\text{obs},d}(t)$ into a depletion-neutralized cumulative hazard scale. This transformation is monotonic in $H_{\text{obs},d}(t)$ for fixed $\theta_{0,d}$, preserving ordering of cumulative risk while altering curvature. KCOR normalization is not model-free: if the gamma-frailty working model or baseline specification is materially misspecified, the inversion may introduce bias rather than reduce bias under the assumed frailty structure, in which case diagnostic failure rather than estimator output is the primary signal. We use a tilde (e.g., $\tilde{H}_{0,d}(t)$) to denote depletion-neutralized baseline quantities obtained after frailty normalization; observed cohort-aggregated quantities are written without a tilde (e.g., $H_{\text{obs},d}(t)$).

2.4.3 Gompertz baseline and enrollment-time interpretation

To identify $\theta_{0,d}$, the revised KCOR estimator uses a Gompertz baseline hazard over rebased event time:

$$\tilde{h}_{0,d}(t_{\text{rebased}}) = k_d e^{\gamma t_{\text{rebased}}}, \qquad H_{\text{gom},d}(t_{\text{rebased}}) = \frac{k_d}{\gamma}\left(e^{\gamma t_{\text{rebased}}} - 1\right). \tag{8}$$

This Gompertz form supplies the structured age-like trend against which depletion-induced curvature is identified. Here γ is a fixed Gompertz slope and k_d is a cohort-specific scale parameter. This choice imposes more structure than the earlier flatter-baseline specification, but it aligns the estimator with the biological age-slope assumption used in the v7.5 implementation and makes the parameter of interest explicit: $\theta_{0,d}$ is the frailty variance at rebased $t = 0$, not a late-window summary of an already depleted cohort. The Gompertz specification is used as **minimal structure sufficient to separate baseline growth from depletion curvature**—a baseline trend anchor—while preserving identifiability of $\theta_{0,d}$; more flexible baselines would absorb curvature and weaken that identification. Sensitivity to the Gompertz slope parameter γ is evaluated in the Supplementary Information; results are stable over a prespecified range, indicating that identification is not driven by a single slope choice. Identification is driven by curvature structure rather than the exact functional form of the baseline hazard.

Misspecification of the Gompertz slope γ primarily rescales the inferred frailty parameter $\hat{\theta}_{0,d}$. Because KCOR(t) is constructed from ratios of depletion-normalized cumulative hazards, **to first order** the effects of γ misspecification tend to **cancel in ratios** unless misspecification differs systematically across cohorts. Sensitivity analyses varying γ confirm that KCOR(t) is driven primarily by curvature structure rather than by the exact baseline parameterization.

2.4.4 Quiet-window validity and identifiability

Frailty parameters are identified using only bins whose corresponding calendar weeks lie inside prespecified quiet windows (defined in ISO-week space). These windows are chosen to avoid sharp external shocks that would confound depletion-geometry identification. In the revised estimator, however, no one quiet window is assumed sufficient on its own. Identification depends on:

1. adequate Gompertz-plus-frailty fit within quiet windows,
2. consistency of $\hat{\theta}_{0,d}$ across aligned quiet windows after structured offset correction,
3. stability to small boundary perturbations, and
4. plausible reconstruction of persistent wave offsets when the delta-iteration path is used.

Refinement: effect-scale invariance of curvature-based identification.
Identification of $\theta_{0,d}$ relies on curvature in cumulative-hazard space rather than absolute hazard levels. A constant multiplicative hazard effect (i.e., $h_{\text{obs}}(t) = c \cdot h_{\text{true}}(t)$) rescales cumulative hazards without altering their curvature and therefore does not alter curvature-based identification of $\theta_{0,d}$ under the working model when multiplicative scaling is **common or approximately common across cohorts** within the identification regime. Constant multiplicative effects are not separately identifiable from baseline scale.

In contrast, additive or time-varying effects can alter curvature and may be absorbed into $\theta_{0,d}$, leading to confounding. Accordingly, $\theta_{0,d}$ should be interpreted as capturing curvature attributable to depletion geometry plus any non-multiplicative residual effects present within the identification window.

Heterogeneous multiplicative effects across cohorts may introduce second-order curvature differences through differential depletion, but these are typically small relative to primary depletion geometry and are assessed via diagnostics.

When any of the conditions in items 1–4 fails, when non-multiplicative contamination or Gompertz misspecification dominates, or when diagnostics indicate material breakdown of the common-scale approximation, the analysis is treated as not identified rather than reported as a stable normalized contrast. Full operational diagnostics are summarized in Supplementary Information §S2.

Proposition (Identification under curvature).
Within the **parametric Gompertz–gamma frailty** working model with enrollment-time variance $\theta_{0,d}$, the mapping from the observed cumulative hazard $H_{\text{obs},d}(t)$ to $\theta_{0,d}$ is **injective** on a **non-degenerate** interval provided $H_{\text{obs},d}(t)$ exhibits **non-zero curvature** over that interval (equivalently, depletion-linked curvature with $\theta_{0,d} > 0$). Consequently, $\theta_{0,d}$ is **parametrically** identifiable from quiet-window data up to sampling variability under the stated model.

This follows from the **one-to-one** link between depletion-linked curvature in cumulative-hazard space and the frailty variance parameter under the Gompertz–gamma specification (derivation omitted). This is a **parametric** identification statement under the working model. KCOR does **not** claim nonparametric identification; identifiability is evaluated **empirically** via curvature diagnostics and multi-window coherence (§2.1.2; SI §S2).

While other latent heterogeneity structures could in principle generate similar curvature in cumulative hazard trajectories, KCOR does **not** rely on uniqueness of this mapping. Instead, identifiability is assessed empirically through diagnostic fit, stability across quiet windows, and consistency under stress tests. The parameter $\theta_{0,d}$ should therefore be interpreted as a **depletion-consistent effective frailty parameter** under the working model, rather than a uniquely identified structural quantity.

Accordingly, KCOR recovers a **geometry-consistent representation** of depletion dynamics rather than a uniquely identified biological or mechanistic frailty parameter.

2.5 Estimating $\theta_{0,d}$ by delta iteration

KCOR estimates $(\hat{k}_d, \hat{\theta}_{0,d})$ independently for each cohort d using a structured four-step procedure. Throughout this section, let $h_d^{\text{adj}}(t)$ denote the preprocessed hazard used for accumulation and fitting: $h_d^{\text{adj}}(t) = h_d^{\text{eff}}(t)$ after the stabilization skip (§2.7) only. The optional NPH module applies **after** gamma-frailty inversion in cumulative-hazard space relative to the fitted Gompertz baseline path (§2.7.3) and does not enter $h_d^{\text{adj}}(t)$. Pooling in this procedure occurs only across prespecified quiet windows within the same cohort; there is no shared θ_0 parameter across cohorts.

Let $W_{0,d}$ denote the nearest quiet window after enrollment and let $\bigcup_j W_{j,d}$ denote the union of all prespecified quiet windows for cohort d, both evaluated on rebased time with $t_{\text{rebased}} \geq 0$.

Intuition. The estimator reconstructs the latent baseline trajectory across the full follow-up, then aligns quiet windows by accounting for persistent cumulative deviations introduced by wave periods before the final frailty refit. Additivity is specified in cumulative-hazard space, where independent hazard contributions accumulate over time under the working model, making persistent offsets interpretable as additive deviations in $H(t)$.

Step 1: joint seed fit in the nearest quiet window.
Estimate $(k_d, \theta_{0,d}^{(0)})$ from the nearest quiet window by nonlinear least squares in hazard space:

$$(\hat{k}_d, \hat{\theta}_{0,d}^{(0)}) = \arg\min_{k_d>0,\ \theta_{0,d}\geq 0} \sum_{t\in W_{0,d}} \left[h_d^{\text{adj}}(t) - \frac{k_d e^{\gamma t_{\text{rebased}}}}{1 + \theta_{0,d} H_{\text{gom},d}(t_{\text{rebased}})} \right]^2. \tag{9}$$

This objective anchors the Gompertz scale parameter $\hat{k}_d$ close to enrollment and avoids identifying $\theta_{0,d}$ from a late, already depleted population.

Step 2: reconstruct the effective cumulative hazard over the full trajectory.
Given the current iterate $\hat{\theta}_{0,d}^{(m)}$, reconstruct the effective individual-level hazard and cumulative hazard over **all** observed weeks by applying the discrete-time version of the gamma-frailty hazard inversion from §2.4.2:

$$h_{0,d}^{\text{eff}}(t) = h_d^{\text{adj}}(t) \left(1 + \hat{\theta}_{0,d}^{(m)} H_{0,d}^{\text{eff}}(t)\right), \qquad H_{0,d}^{\text{eff}}(t+1) = H_{0,d}^{\text{eff}}(t) + h_{0,d}^{\text{eff}}(t),$$

starting from $H_{0,d}^{\text{eff}}(0) = 0$.

This recursion reconstructs the hidden baseline trajectory over all observed weeks rather than only inside the fit window. It should be read as the working-model inversion step implied by the gamma-frailty geometry, not as a model-free update rule.

Step 3: compute persistent wave offsets.
For each prespecified wave end time t_i, compute the incremental offset

$$\delta_{i,d} = \left[H_{0,d}^{\text{eff}}(t_i) - H_{\text{gom},d}(t_i) \right] - \sum_{j<i} \delta_{j,d}, \qquad \Delta_d(t) = \sum_{i:\, t_i \leq t} \delta_{i,d}.$$

This offset calculation summarizes how wave periods shift later quiet windows through persistent cumulative deviations. It encodes the structural assumption that wave effects are additive in cumulative-hazard space and persist after the wave ends. Implausible offset behavior is treated diagnostically; detailed fallback rules are provided in the Supplementary Information.

Step 4: pooled refit across all quiet windows.
Holding $\hat{k}_d$ fixed, refit $\theta_{0,d}$ across the union of all quiet windows:

$$\hat{\theta}_{0,d} = \arg\min_{\theta_{0,d}\geq 0} \sum_{t\in\bigcup_j W_{j,d}} \left[h_d^{\text{adj}}(t) - \frac{\hat{k}_d e^{\gamma t_{\text{rebased}}}}{1 + \theta_{0,d}\left(H_{\text{gom},d}(t_{\text{rebased}}) + \Delta_d(t)\right)} \right]^2.$$

This pooled refit updates $\theta_{0,d}$ using all aligned quiet-window information on a common reconstructed geometry. The reconstruction and pooled refit are iterated to convergence, which in the reference implementation typically occurs within a small number of iterations. When curvature is weak, the delta structure is inapplicable, or events are too sparse, the estimator may collapse toward the $\theta_{0,d} \to 0$ limit; this is interpreted as weak identifiability rather than as evidence for or against any particular cohort.

All analyses use a prespecified reference implementation with fixed operational defaults; full derivations, convergence notes, and edge-case rules are provided in Supplementary Section S4.

2.6 Normalization (depletion-neutralized cumulative hazards)

After fitting, KCOR computes the depletion-neutralized baseline cumulative hazard for each cohort d by applying the inversion to the full post-enrollment trajectory:

$$\tilde{H}_{0,d}(t) = \frac{\exp\left(\hat{\theta}_{0,d}\, H_{\text{obs},d}(t)\right) - 1}{\hat{\theta}_{0,d}}. \tag{10}$$

This inversion adjusts for the curvature attributed to the estimated depletion geometry under the working model and places cohorts on a common cumulative-hazard scale. The normalization maps each cohort into a depletion-neutralized baseline-hazard space in which the contribution of gamma frailty parameters $(\hat{\theta}_{0,d}, \hat{k}_d)$ to hazard curvature has been factored out. The inversion uses the full observed cumulative hazard trajectory built from $h_d^{\text{eff}}(t)$ after the stabilization skip. Optional wave-period NPH correction applies only **after** this inversion to frailty-neutral $\tilde{H}_{0,d}(t)$ (§2.5; §2.7.3), never as preprocessing on raw hazard. This normalization defines a common comparison scale in cumulative-hazard space; it is not equivalent to Cox partial-likelihood baseline anchoring, but serves an analogous geometric role for cumulative contrasts. The core identities used in KCOR are given in Equations (2), (9), (10), and (1). Normalization defines a common comparison scale; the scientific estimand is then computed on that scale (Box 2).

2.6.1 Computational considerations

KCOR operates on aggregated event counts in discrete time and cumulative-hazard space. Computational complexity scales linearly with the number of time bins and strata rather than the number of individuals, making the method feasible for very large population registries. In practice, KCOR analyses on national-scale datasets (millions of individuals) are memory-bound rather than CPU-bound and can be implemented efficiently using standard vectorized numerical libraries. No iterative optimization over individual-level records is required. **Numerical stability at vanishing frailty variance.**
When the fitted frailty variance satisfies $\hat{\theta}_{0,d} < \varepsilon$ (with ε set to a small numerical tolerance, e.g., 10^{-8}), the gamma-frailty inversion in Eq. ((10)) is evaluated in the $\theta_0 \to 0$ limit, yielding

$$\tilde{H}_{0,d}(t) = H_{\text{obs},d}(t),$$

which corresponds to the standard Nelson–Aalen cumulative hazard. This avoids numerical instability from evaluating $(\exp(\theta_0 H) - 1)/\theta_0$ near machine precision and reflects the fact that near-zero fitted θ_0 indicates negligible detectable depletion curvature.

2.6.2 Internal diagnostics and 'self-check' behavior

KCOR includes internal diagnostics intended to make model stress visible rather than hidden.

1. **Post-normalization linearity in quiet periods.** Within prespecified quiet windows (see §2.4.4), the depletion-neutralized cumulative hazard should be approximately linear in rebased event time after inversion. Systematic residual curvature indicates contaminated windows, misspecified depletion geometry, or failure of the offset structure.
2. **Fit residual structure in hazard space.** Define residuals over the pooled fit set:

$$r_d(t) = h_d^{\mathrm{adj}}(t) - \frac{\hat{k}_d e^{\gamma t_{\mathrm{rebased}}}}{1 + \hat{\theta}_{0,d}\left(H_{\mathrm{gom},d}(t_{\mathrm{rebased}}) + \Delta_d(t)\right)}. \tag{11}$$

KCOR expects residuals to be small and not systematically time-structured within or across quiet windows. Strongly patterned residuals indicate that the curvature attributed to depletion is instead being driven by unmodeled time-varying hazards.

3. **Multi-window stability.** Under valid quiet-window selection, the reconstructed offsets and fitted parameters should be stable to small perturbations of quiet-window boundaries and to omission of individual quiet windows. Large changes under minor perturbations signal that the fitted geometry is being driven by transient dynamics rather than by stable depletion structure.
4. **Weak identifiability manifests through boundary behavior.** Near-zero $\hat{\theta}_{0,d}$, unstable $\Delta_d(t)$, or failure of the offset reconstruction all indicate that θ_0 is weakly identified. In such cases, KCOR should be interpreted primarily as a diagnostic rather than as a strong normalization.

These diagnostics are reported alongside KCOR(t) curves. The goal is not to assert that a single parametric form is always correct, but to ensure that when the form is incorrect or the window is contaminated, the method signals this explicitly rather than silently producing a misleading normalized estimate. When diagnostics indicate non-identifiability, the depletion-based normalization is inappropriate and KCOR should not be interpreted.

2.7 Stabilization (early weeks)

The optional NPH module (§2.7.1–§2.7.4) extends the framework to settings with wave-induced hazard amplification but is **not** required for core KCOR estimation. Core KCOR(t) and depletion-normalized cumulative hazards $\tilde{H}_{0,d}(t)$ are defined and computed without α. When α is not identified, that module is **inactive** and does not affect reported results.

In many applications, the first few post-enrollment intervals can be unstable due to immediate post-enrollment artifacts (e.g., rapid deferral, short-term sorting, administrative effects). KCOR supports a prespecified stabilization rule by excluding early weeks from accumulation and from quiet-window fitting. The skip-weeks parameter is prespecified and evaluated via sensitivity analysis to exclude early enrollment instability rather than to tune estimates.

In discrete time, define an effective hazard for accumulation:

$$h_d^{\mathrm{eff}}(t) = \begin{cases} 0, & t < \mathrm{SKIP_WEEKS} \\ h_{\mathrm{obs},d}(t), & t \geq \mathrm{SKIP_WEEKS}. \end{cases} \tag{12}$$

For frailty identification, event time is rebased so that

$$t_{\mathrm{rebased}} = t_{\mathrm{raw}} - \mathrm{SKIP_WEEKS},$$

placing $t_{\mathrm{rebased}} = 0$ at the first accumulating interval. This rebased origin is the point at which $\theta_{0,d}$ is defined. All modeling is performed in rebased time unless otherwise specified.

2.7.1 Optional non-proportional hazard (NPH) exponent model

Throughout §2.7, the **operational NPH correction** (§2.7.3) is applied only to **frailty-neutral cumulative hazard** $\tilde{H}_{0,d}(t)$ after gamma-frailty inversion—**not** to raw observed instantaneous hazard. The hazard-level relationships below motivate the frailty-dependent excess structure used to **localize** α during wave periods; they do not describe the domain of the cumulative correction operator.

In some settings, especially COVID-era mortality analyses, external hazards interact with cohort-specific frailty in a non-proportional manner. In such cases, scalar rescaling is not rich enough to align trajectories across cohorts because amplification may depend on depletion geometry itself. KCOR therefore treats NPH as an **optional exponent module**, not as part of the universal KCOR core.

Because α is not separately identifiable from cohort-specific multiplicative intervention effects in minimal aggregated data, application of this module requires **externally supplied VE or other prespecified intervention-effect scale**, not learned from the same KCOR wave-period contrasts used to identify α. KCOR does not estimate that intervention scale from the same aggregated wave-period contrasts used to localize α. This separation matters because frailty-dependent amplification and intervention-associated hazard reduction can otherwise generate similar cross-cohort excess-hazard ratios, producing flat objectives and boundary-seeking estimates.

Let the stabilized hazard during an NPH period be represented schematically as

$$h_d(t) = h_{0,d}(t) + A(t)\, E_d[z^\alpha \mid t],$$

where $A(t)$ is an unknown common external intensity, z denotes latent frailty under the gamma-frailty working model, and the exponent α is treated as global across cohorts within the analyzed period. Equivalently, the excess-hazard component satisfies

$$h_{\text{excess},d}(t) \propto E_d[z^\alpha \mid t].$$

Under the gamma-frailty working model, this moment depends on the cohort's current depletion state, so the NPH module links external-hazard amplification to cumulative-hazard geometry rather than to a cohort-specific scalar multiplier. When $\alpha = 1$, amplification is proportional to baseline risk and the NPH module is effectively inactive. The parameters $\theta_{0,d}$ and α are conceptually distinct: $\theta_{0,d}$ governs how each cohort depletes over time, while α governs how external hazards interact with that depletion.

Because $A(t)$ is common across cohorts, absolute excess hazards are not directly comparable across cohorts. However, ratios eliminate the unknown common amplitude:

$$\frac{h_{\text{excess},i}(t)}{h_{\text{excess},j}(t)} = \frac{E_i[z^\alpha \mid t]}{E_j[z^\alpha \mid t]}.$$

This cancellation is the geometric core for relating observed cross-cohort excess structure to $F_d(t;\alpha)$ once an **externally supplied VE or other prespecified intervention-effect scale** is fixed (§2.7.2). Identification of α still requires sufficient cross-cohort variation in depletion state and excess-hazard measurement; when cohorts are too similar, objectives flatten. Failure of identification manifests as flat objective functions, boundary-seeking estimates, or disagreement between estimators; such patterns are treated as weak signal or model misspecification rather than as valid α estimates.

2.7.2 Identification of α given an externally supplied intervention-effect scale

Under minimal aggregated data, the NPH exponent α is not separately identifiable from cohort-specific multiplicative intervention effects if both are left free. Accordingly, when the optional NPH module is used to localize α, the intervention-effect scale must be supplied from outside the KCOR core—for example from external clinical evidence, randomized trials, or other prespecified analyses—**as externally supplied VE or other prespecified intervention-effect scale**. KCOR does not identify that scale from the same aggregated wave-period contrasts used to localize α.

Let x denote an externally specified efficacy-style summary for the relevant wave-period hazard contrast between two cohorts A and B, coded so that the **true** frailty-neutral excess hazards satisfy

$$\Delta h_A^{\text{true}}(t) = (1 - x)\, \Delta h_B^{\text{true}}(t),$$

where $\Delta h_d^{\text{true}}(t)$ denotes frailty-neutral excess hazard during the NPH period for cohort d. Under the gamma-frailty working model, the **observed** excess hazard satisfies

$$\Delta h_d^{\text{obs}}(t) = \Delta h_d^{\text{true}}(t)\, F_d(t;\alpha), \qquad F_d(t;\alpha) = E_d[z^\alpha \mid t].$$

Therefore,

$$\frac{\Delta h_A^{\text{obs}}(t)}{\Delta h_B^{\text{obs}}(t)} = (1 - x)\,\frac{F_A(t;\alpha)}{F_B(t;\alpha)}.$$

This relation localizes α, if at all, from the mismatch between the observed cross-cohort excess-hazard ratio and the ratio predicted from the prespecified scale $(1 - x)$ together with the cohort-specific depletion geometry encoded in $F_d(t;\alpha)$. Equivalently, α is chosen so that the model-implied amplification ratio matches the observed excess ratio after adjustment for the external scale.

Operationally, pairwise or collapse objective functions analogous to those in Supplementary Information §S2.1 are evaluated with x (hence $(1 - x)$) treated as **fixed external input** rather than estimated jointly from the same contrasts. If diagnostics indicate weak localization, estimator disagreement, or boundary-seeking behavior, α is treated as not identified.

2.7.3 Application of the NPH correction once α is specified

Applying the NPH correction **after** gamma-frailty inversion ensures that wave-period excess is defined relative to a **frailty-neutral** baseline path in cumulative-hazard space. Applying the correction to raw observed hazards would confound depletion geometry with wave-period amplification and yield mis-scaled excess.

The NPH module **does not** modify quiet-window estimation of $\theta_{0,d}$ or $\hat{k}_d$; depletion geometry is identified from prespecified quiet windows **before** any wave-period cumulative correction.

Once α has been specified or localized, the correction is applied **after** Eq. (10) to $\tilde{H}_{0,d}(t)$. In discrete weekly time (§2.3), let $t_{0,d}$ index the first rebased week inside the prespecified NPH window for cohort d. Define a **no-wave** frailty-neutral reference cumulative hazard $H_{\text{ref},d}(t)$ by anchoring

$$H_{\text{ref},d}(t_{0,d}) = \tilde{H}_{0,d}(t_{0,d})$$

and, for subsequent weeks t inside the same window, accumulating fitted Gompertz weekly increments with $\Delta t = 1$:

$$H_{\text{ref},d}(t) = H_{\text{ref},d}(t - 1) + \hat{k}_d \,\exp\!\big(\gamma\, t_{\text{rebased}}(t)\big)\,,$$

where $t_{\text{rebased}}(t)$ is the rebased time index at week t and $\hat{k}_d, \gamma$ are the fitted quiet-window Gompertz parameters. Outside the prespecified NPH window, no NPH correction is applied.

The wave-induced excess in cumulative-hazard space is $\Delta H_{\text{wave},d}(t) = \tilde{H}_{0,d}(t) - H_{\text{ref},d}(t)$. Only **wave-attributable** excess strictly above the frailty-neutral Gompertz baseline path is rescaled: when $\Delta H_{\text{wave},d}(t) > 0$,

$$H_{\text{corr},d}(t) = H_{\text{ref},d}(t) + \frac{\Delta H_{\text{wave},d}(t)}{F_d(t;\alpha)},$$

and when $\Delta H_{\text{wave},d}(t) \leq 0$ the corrected path equals $\tilde{H}_{0,d}(t)$ at that week. The implementation enforces that $H_{\text{corr},d}(t)$ is **non-decreasing** in event time (cumulative maximum repair) so corrected trajectories remain valid cumulative-hazard geometry.

Weekly corrected hazard increments are obtained by differencing $H_{\text{corr},d}(t)$ if a hazard stream is required; the downstream KCOR contrast uses $H_{\text{corr},d}(t)$ in place of $\tilde{H}_{0,d}(t)$ when the module is active (§2.8). Thus the optional module adjusts only positive wave excess above the fitted frailty-neutral baseline path; it does not replace the core KCOR estimator.

2.7.4 Practical workflow for the optional NPH module

1. Estimate cohort-specific depletion geometry from prespecified quiet windows to obtain $\hat{\theta}_{0,d}$ and $\hat{k}_d$.
2. Obtain or prespecify **externally supplied VE or other prespecified intervention-effect scale** for the relevant wave-period hazard contrast (not from the same contrasts used to localize α).
3. Using observed excess hazards during NPH periods together with the fixed external scale, localize α via §2.7.2, if diagnostics support identification.
4. Apply gamma-frailty inversion (Eq. (10)) to obtain $\tilde{H}_{0,d}(t)$.
5. During the NPH window, construct the discrete Gompertz reference path and correct only strictly positive wave excess using $F_d(t;\alpha)$ as in §2.7.3.
6. Continue with the standard KCOR pipeline on the corrected cumulative-hazard scale (§2.8).
7. If identification diagnostics fail, report α as not identified and treat the NPH module as inactive.

2.8 KCOR estimator

With depletion-neutralized cumulative hazards in hand, the primary KCOR trajectory is defined as

$$\mathrm{KCOR}(t) = \frac{\tilde{H}_{0,A}(t)}{\tilde{H}_{0,B}(t)}. \tag{13}$$

When the optional NPH cumulative correction is active (§2.7.3), the same ratio is computed with $H_{\mathrm{corr},d}(t)$ in place of $\tilde{H}_{0,d}(t)$ for the affected cohorts and time range.

When an anchored presentation is used, KCOR is normalized to its mean over a prespecified short reference window after the anchor time. The primary estimand, however, remains the ratio of (optionally NPH-corrected) depletion-neutralized cumulative hazards rather than an instantaneous hazard ratio.

This ratio is computed after depletion normalization and is interpreted conditional on the stated assumptions and diagnostics (Box 2; §2.1.2).

Normalized cumulative contrasts at clinically relevant horizons (e.g., 90-day or 1-year mortality) are obtained directly from the cumulative-hazard scale used in Eq. (13) without requiring proportional hazards assumptions.

2.9 Uncertainty quantification

Uncertainty is quantified using stratified bootstrap resampling as **variance propagation** for the aggregated counting-process pipeline: event stochasticity and refitting uncertainty in $(\hat{k}_d, \hat{\theta}_{0,d})$ are carried through inversion (and, when prespecified, the optional NPH correction on **frailty-neutral cumulative hazard**, §2.7.3) to $\mathrm{KCOR}(t)$. The bootstrap is **not** asserted to yield universally calibrated inference under arbitrary frailty or baseline misspecification; interpretation remains tied to diagnostic pass status (§2.9.1; Table 11; §5). Formal bootstrap theory for this stacked construction is **left to future work**.

2.9.1 Stratified bootstrap procedure

Throughout this manuscript, cohorts define the primary comparison groups and are the units at which frailty parameters $(k_d, \theta_{0,d})$ are estimated. Strata (e.g., age bands or sex) are treated as independent realizations of the same cohort-level process and are used solely for aggregation, age standardization, and uncertainty propagation; frailty parameters are not estimated separately by stratum. This use of stratification follows standard survival-analysis practice (e.g., Kalbfleisch and Prentice, The Statistical Analysis of Failure Time Data)[14].

The stratified bootstrap procedure for KCOR proceeds as follows:

1. **Resample cohorts using a block-preserving aggregated bootstrap.**
   For each cohort and stratum, bootstrap replicates are generated by resampling **contiguous blocks of event-time bins** with replacement, preserving the internal temporal dependence of the counting process. Within each resampled block, deaths $d_d(t)$ and corresponding risk sets $N_d(t)$ are carried forward sequentially so that

$$N_d(t) = N_d(0) - \sum_{s<t} d_d(s)$$

holds identically within each replicate. This preserves the martingale covariance structure of the cumulative hazard estimator while operating on aggregated cohort–time data.

2. **Re-estimate frailty parameters.** For each bootstrap replicate, re-estimate $(\hat{k}_d, \hat{\theta}_{0,d})$ independently for each cohort d using the resampled data, applying the same delta-iteration pipeline and quiet-window definitions as in the primary analysis.
3. **Recompute normalized cumulative hazards.** Using the bootstrap-estimated frailty parameters, recompute $\tilde{H}_{0,d}(t)$ for each cohort using Eq. (10) applied to the resampled observed cumulative hazards.
4. **Optional NPH post-inversion correction.** When the primary analysis applies the optional NPH module, repeat the §2.7.3 correction on each replicate's frailty-neutral $\tilde{H}_{0,d}(t)$ using the same prespecified α, external intervention-effect scale, and wave window as in the primary analysis (treating those inputs as fixed by design in the bootstrap).
5. **Recompute KCOR.** Compute $\mathrm{KCOR}(t)$ for each bootstrap replicate as the ratio of the per-replicate cumulative hazards used in Eq. (13) (i.e., $\tilde{H}_{0,d}$ or $H_{\mathrm{corr},d}$ when applicable).
6. **Form percentile intervals.** From the bootstrap distribution of $\mathrm{KCOR}(t)$ values at each time point, form percentile-based confidence intervals (e.g., 2.5th and 97.5th percentiles for 95% intervals).

Importantly, event-time bins are **not resampled independently**. Any bootstrap procedure that breaks the sequential dependence between $d_d(t)$ and $N_d(t)$ would underestimate uncertainty in cumulative hazards. The block-preserving aggregated bootstrap used here maintains the temporal structure required for valid uncertainty propagation in cumulative-hazard space.

Bootstrap resampling is performed at the cohort-count level in the aggregated representation by resampling contiguous blocks of $(d_d(t), N_d(t))$ event-time pairs within cohort-time strata with replacement, preserving within-cohort temporal structure, rather than resampling individual-level records.

Uncertainty intervals reflect event stochasticity and model-fit uncertainty in $(\hat{k}_d, \hat{\theta}_{0,d})$ and are interpreted as sampling variability of the aggregated counting process, not uncertainty in individual-level causal effects.

Bootstrap intervals are calibrated under the working gamma frailty model. Simulation studies (Table 11) indicate that coverage remains near nominal when diagnostic criteria are satisfied, but becomes mildly anti-conservative under frailty misspecification or sparse-event regimes. Estimates are therefore intended to be interpreted conditionally on diagnostic pass status.

The bootstrap is used as a **variance-propagation** device for the aggregated process rather than a universally calibrated inferential procedure. Sub-nominal coverage arises in regimes where model assumptions are violated or curvature is insufficient for identification. In those regimes, KCOR diagnostics explicitly flag non-identifiability and results are not interpreted. Reported percentile intervals are **conditional on passing identifiability diagnostics**. Sub-nominal coverage occurs **primarily** in weak-identification regimes (e.g., low curvature or model misspecification), which are flagged by KCOR's diagnostic criteria; **in well-identified settings, empirical coverage is near nominal** (Table 11; §5). **Inference is therefore interpreted conditional on diagnostic acceptance.**

2.10 Algorithm summary and reproducibility checklist

Table 5 summarizes the complete KCOR pipeline.

2.11 Relationship to Cox proportional hazards

Cox proportional hazards models estimate an instantaneous hazard ratio under the assumption that hazards differ by a time-invariant multiplicative factor. Under selection on frailty with latent heterogeneity, this assumption is typically violated, yielding time-varying hazard ratios induced purely by depletion dynamics. This reflects an estimand mismatch: Cox targets an instantaneous hazard ratio conditional on survival, whereas KCOR targets a cumulative hazard contrast after depletion normalization. The resulting Cox bias is structural: it arises from conditioning on survival in the presence of latent heterogeneity and cannot be removed by increasing sample size

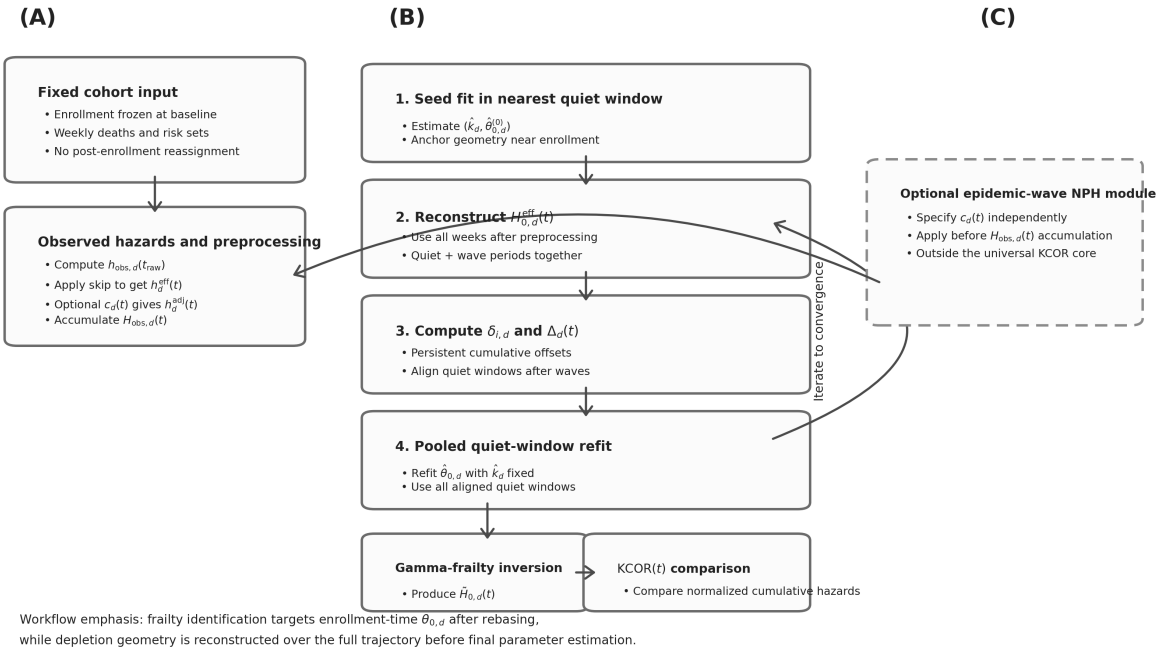

Figure 2: **KCOR workflow: four-step $\theta_{0,d}$ estimation, then normalization and comparison.** **(A)** Fixed-cohort cumulative hazards are stabilized after enrollment, and event time is rebased so that frailty identification targets enrollment-time $\theta_{0,d}$. **(B)** Steps 1-4 estimate $\theta_{0,d}$: a seed Gompertz fit is obtained in the nearest quiet window, the effective cumulative hazard is reconstructed over the full trajectory using all weeks, persistent offsets are aligned across quiet windows, and $\theta_{0,d}$ is refit across the pooled quiet windows with $\hat{k}_d$ held fixed. Reconstruction, offset alignment, and pooled refitting iterate to convergence before gamma-frailty inversion and KCOR computation. **(C)** In epidemic-wave applications, an optional NPH exponent module may localize α using **externally supplied VE or other prespecified intervention-effect scale** and apply a cumulative-hazard correction **after** gamma-frailty inversion (§2.7); it is not part of the universal KCOR core and is interpreted only when identification diagnostics are credible. *This schematic is illustrative rather than empirical. In the schematic, $\tilde{H}_{0,d}(t)$ denotes the depletion-neutralized baseline cumulative hazard and $\theta_{0,d}$ denotes enrollment-time frailty variance after the stabilization skip.*

or covariate adjustment alone. KCOR operates on Nelson–Aalen–type cumulative hazards without individual-level frailty observables.

Accordingly, Cox results are presented here as a diagnostic demonstration of estimand mismatch, not as a competing intervention-effect estimator. This limitation is consistent with earlier work by Deeks showing that increasing covariate adjustment in non-randomized analyses can exacerbate bias and imprecision when selection effects and measurement error dominate. Deeks further noted that, despite the widespread reliance on covariate adjustment in non-randomized studies, there is no empirical evidence that such adjustment reduces bias on average[15].

Even when Cox models are extended with shared frailty to accommodate heterogeneity, they continue to estimate instantaneous hazard ratios conditional on survival. KCOR instead uses a cohort-level working model with Gompertz curvature, cohort-specific enrollment-time $\theta_{0,d}$, and quiet-window alignment to normalize selection-induced depletion geometry before computing a cumulative contrast on the depletion-neutralized scale.

Distinction from shared-frailty Cox models.
Cox models with shared frailty incorporate unobserved heterogeneity as a random effect during partial-likelihood maximization, but continue to estimate an instantaneous hazard ratio conditional on survival. Frailty variance is treated as a nuisance parameter and depletion geometry is absorbed into the fitted hazard ratio trajectory. KCOR differs fundamentally: enrollment-time frailty variance $\theta_{0,d}$ is estimated explicitly from aligned quiet-window geometry under a Gompertz working model and is then analytically inverted *prior* to comparison. The resulting estimand is a cumulative hazard contrast on a depletion-neutralized scale, rather than a conditional instantaneous hazard ratio. Shared-frailty Cox **remains conditional on survival** and **does not adjust for depletion-induced selection effects in marginal cumulative comparisons** in the sense targeted by KCOR. Consequently, even when shared-frailty Cox models reduce bias relative to standard Cox regression, they do not target the same estimand as KCOR and may still exhibit residual non-null behavior under selection-only regimes (Table 9).

2.11.1 Demonstration: Cox bias under frailty heterogeneity with no treatment effect

A controlled synthetic experiment was conducted in which the **true effect is known to be zero by construction**, isolating latent frailty heterogeneity as the sole driver of depletion-induced non-proportional hazards. Cox and KCOR were applied to the same simulated datasets under identical information constraints.

Data-generating process.

Two cohorts of equal size were simulated under the same baseline hazard $h_0(t)$ over time (constant or Gompertz). Individual hazards were generated as $z\,h_0(t)$, with frailty

$$z \sim \text{Gamma}(\theta_0^{-1}, \theta_0^{-1}),$$

with mean 1 and variance θ_0.

Cohort A was generated with $\theta_0 = 0$ (no frailty heterogeneity), while Cohort B was generated with $\theta_0 > 0$. **No treatment or intervention effect was applied**: conditional on frailty, the two cohorts have identical hazards at all times. Thus, the hazard ratio between cohorts is 1 by construction for all t.

Simulations were repeated over a grid of frailty variances $\theta_0 \in \{0, 0.5, 1, 2, 5, 10, 20\}$.

Cox analysis.

For each simulated dataset, a standard Cox proportional hazards model was fitted using partial likelihood (statsmodels `PHReg`), with cohort membership as the sole covariate (no time-varying covariates or interactions). The resulting hazard ratio estimates and confidence intervals therefore reflect **only differences induced by frailty-driven depletion**, not any treatment effect.

KCOR analysis.

The same simulated datasets were analyzed using KCOR. Observed cumulative hazards were estimated from cohort-level event counts, cohort-specific frailty parameters were estimated using the same structured pipeline described in §2.5, and normalized cumulative hazards were then obtained using Eq. (10) prior to computing $\text{KCOR}(t)$. Although the data-generating process specifies individual hazards, the KCOR analysis mirrors the aggregated

information available in registry studies rather than exploiting simulator-only knowledge. Post-normalization slope and asymptotic KCOR(t) values were examined as one component of the broader validation stack rather than as the sole validation criterion.

Expected behavior under the null.

Because the data-generating process includes **no treatment effect**, the appropriate benchmark is null cumulative behavior. In this setting:

- **Cox regression** is expected to produce apparent non-null hazard ratios as θ_0 increases, reflecting differential selection-induced depletion and violation of proportional hazards induced by frailty heterogeneity.
- **KCOR** is expected to remain centered near unity with negligible post-normalization slope across all θ_0, which would be consistent with the expected null behavior under the working model and therefore serves as a diagnostic check rather than as proof of correctness.

Summary of findings.

Across increasing values of θ_0, Cox regression produced progressively larger apparent deviations from a hazard ratio of 1. The direction and magnitude of the apparent effect depended on the follow-up horizon and degree of frailty heterogeneity. In contrast, KCOR(t) trajectories remained stable and centered near unity, with post-normalization slopes approximately zero across the simulated conditions examined here.

These results show that frailty heterogeneity alone can induce spurious hazard ratios in Cox regression, while KCOR behaves close to the null benchmark expected under the same working-model conditions.

Bias in Cox-based estimators increases monotonically with frailty variance on this grid, while KCOR remains near-unbiased (cumulative-scale bias near zero) across the same parameter range.

Table 6 reports numerical summaries of the Cox-vs-KCOR behavior across the frailty grid. Small residual deviations in the asymptotic level are expected under extreme frailty heterogeneity and are negligible relative to the distortions observed under Cox; the diagnostic criterion is slope stability rather than exact unity.

Figure 5 summarizes **relative bias** (estimate minus true target) for standard Cox, an **approximate reference curve** for shared-frailty Cox constructed from null-case bias in Table 9 (see caption—not a refit at each θ_0), and KCOR asymptotes on the same θ_0 grid. This approximation is included to illustrate **expected bias attenuation** relative to standard Cox; a full simulation of shared-frailty Cox across the θ_0 grid is left for future work. Additional Cox HR results from the same synthetic-null grid are shown in Figure 3.

A compact summary of KCOR bias as a function of frailty variance θ_0 is provided in the Supplementary Information (Figure S2).

Interpretation.

This controlled synthetic null shows that Cox proportional hazards regression can report highly statistically significant non-null hazard ratios even when the true effect is identically zero, purely due to frailty-driven depletion and induced non-proportional hazards. In the same simulations, KCOR remains near unity after normalization, which is the diagnostic behavior expected when the working model matches the data-generating process but does not by itself prove that the model is correct.

2.12 Worked example (descriptive)

A brief worked example is included to illustrate the KCOR workflow end-to-end. This example is descriptive and intended solely to illustrate the mechanics of cohort construction, hazard estimation, θ_0 estimation, depletion normalization, and KCOR computation.

The example proceeds from aggregated cohort counts through cumulative-hazard estimation, stabilization and rebasing, delta-iteration fitting across prespecified quiet windows, gamma inversion, and KCOR(t) construction, accompanied by diagnostic plots assessing post-normalization linearity and parameter stability.

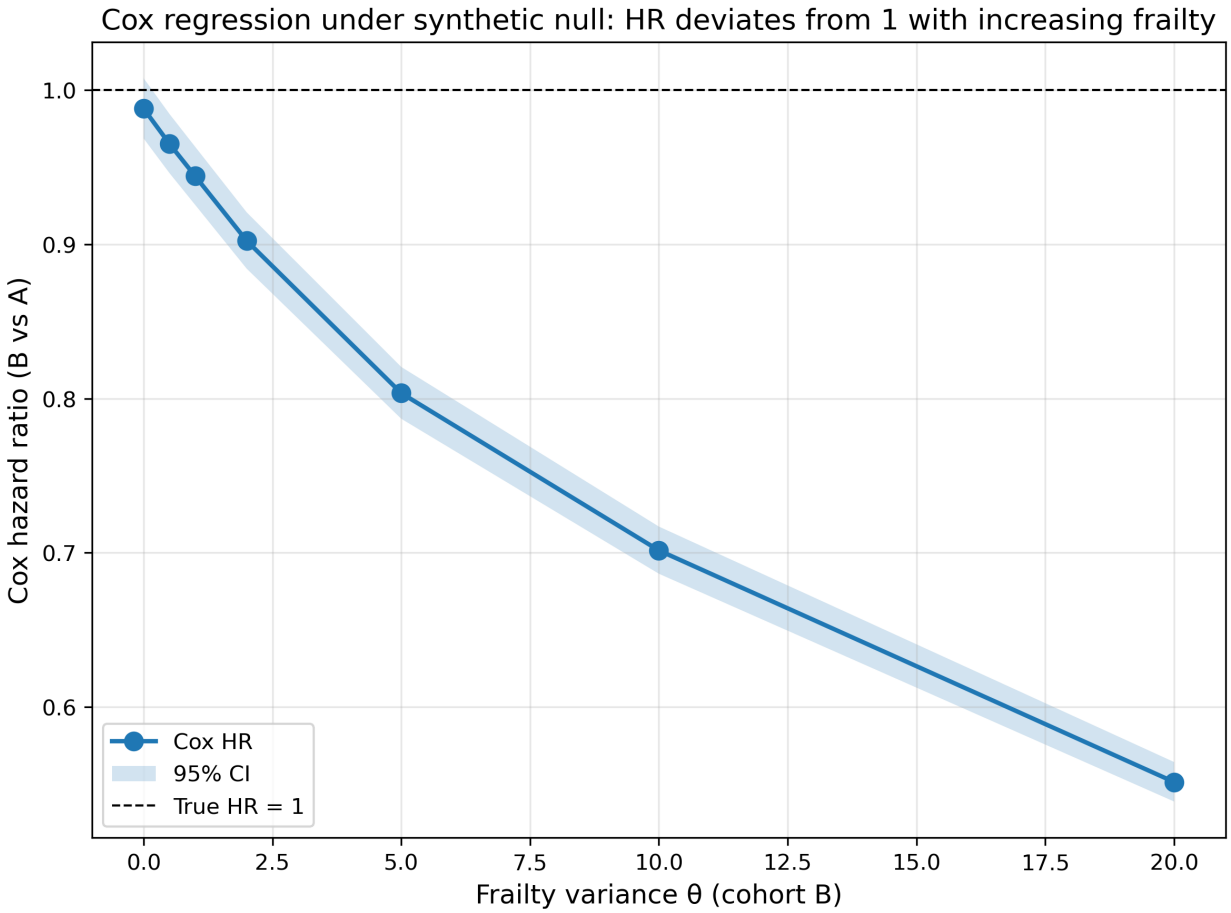

Figure 3: Cox regression produces spurious non-null hazard ratios under a *synthetic null* as frailty heterogeneity increases. Hazard ratios (with 95% confidence intervals) from Cox proportional hazards regression comparing cohort B to cohort A in simulations where the true treatment effect is identically zero and cohorts differ only in enrollment-time frailty variance (θ_0). Deviations from HR=1 arise solely from frailty-driven depletion and associated non-proportional hazards.

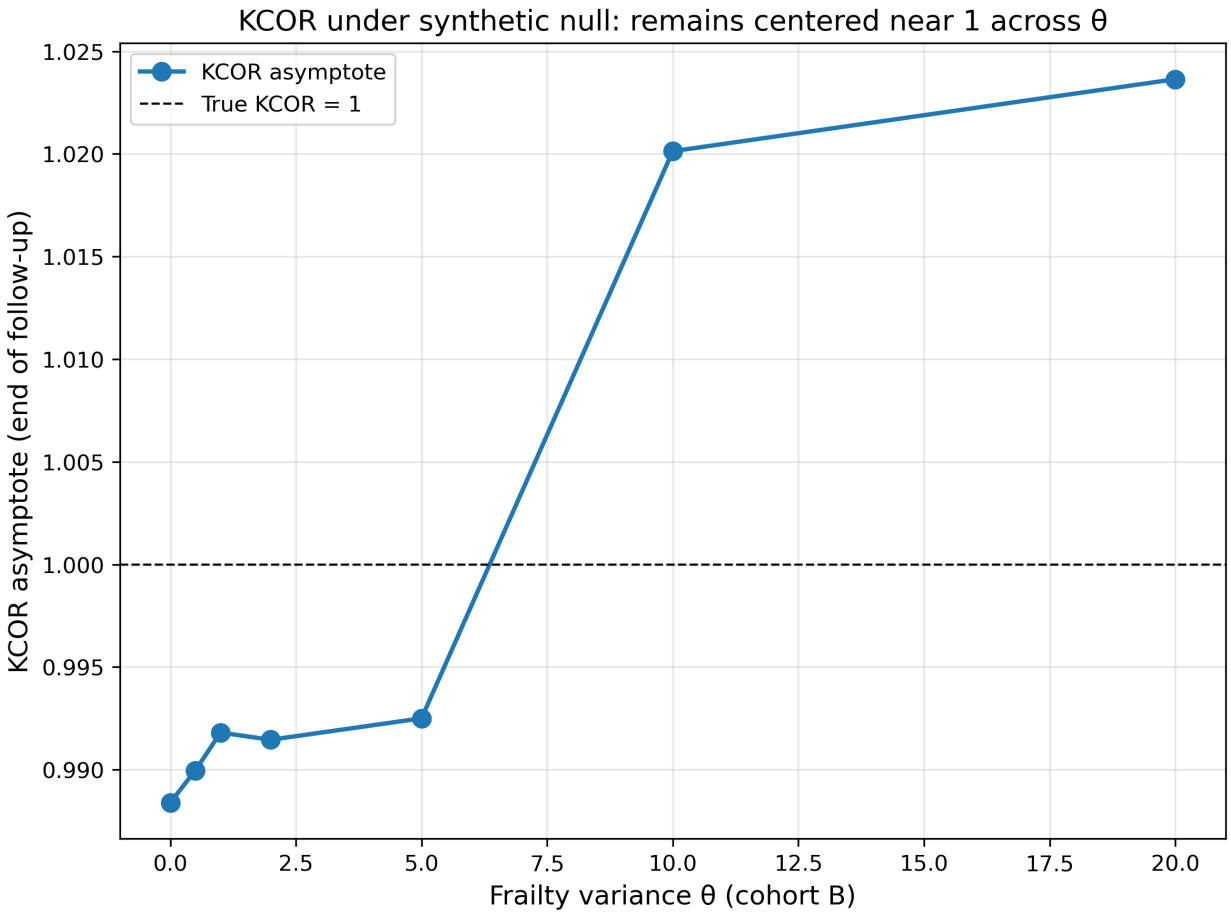

Figure 4: KCOR(t) stays near the null benchmark under a synthetic null across increasing frailty heterogeneity. In these simulations, KCOR(t) asymptotes remain near 1 across θ_0, which is the diagnostic behavior expected under the working model rather than proof that the model is true or that all confounding has been removed. Uncertainty bands (95% bootstrap intervals) are shown but are narrow due to large sample sizes.

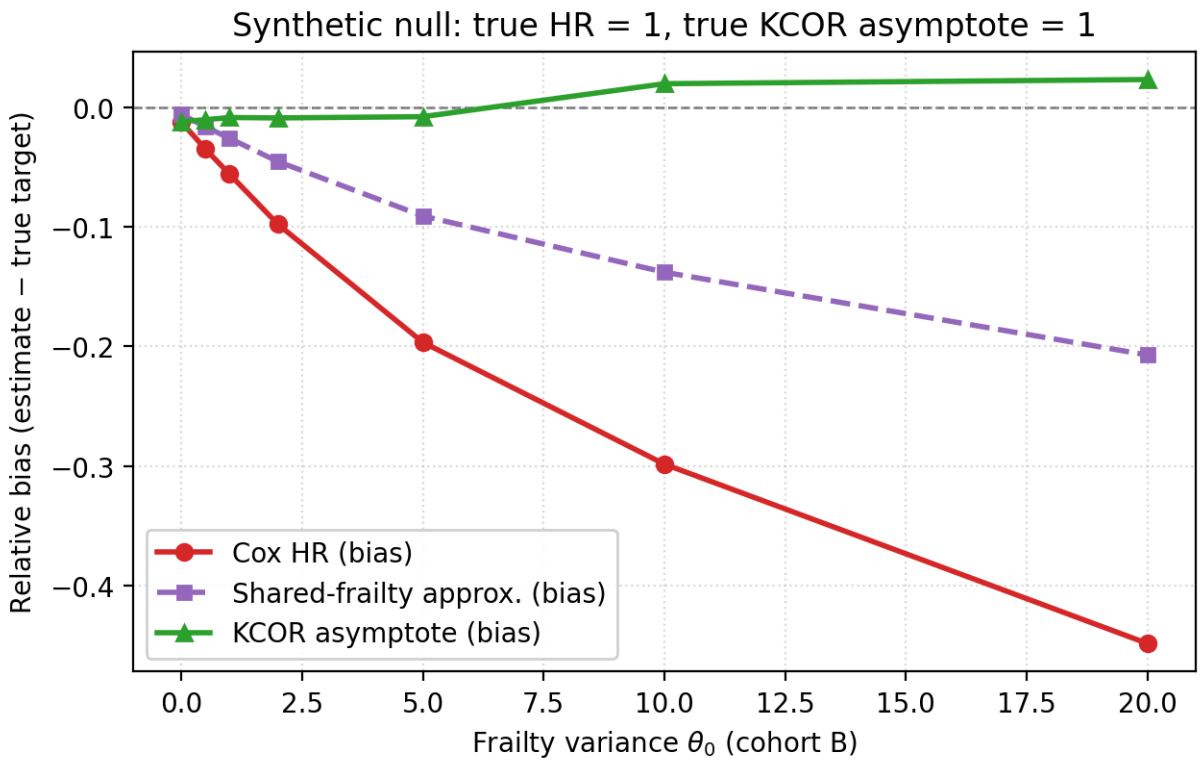

Figure 5: Estimator bias as a function of enrollment-time frailty variance θ_0 on the synthetic-null grid (true hazard ratio 1; true KCOR asymptote 1). **Cox:** standard Cox hazard ratio bias, $\widehat{\mathrm{HR}} - 1$. **KCOR:** asymptotic KCOR bias, $\widehat{\mathrm{KCOR}}_\infty - 1$. **Shared-frailty (approx.):** the shared-frailty Cox curve is an **approximate reference** constructed from the null-case bias observed in Table 9 (proportional attenuation of each standard Cox $\widehat{\mathrm{HR}}$ toward 1 using the ratio of deviations from 1 in that table's gamma-frailty null row), **not** a full re-estimation of shared-frailty Cox at each frailty level. Generated by `test/sim_grid/code/plot_estimator_bias_vs_theta.py`.

2.13 Reproducibility and computational implementation

All figures, tables, and simulations can be reproduced from the accompanying code repository. The manuscript is built from `documentation/preprint/paper.md` using the root `Makefile` build target `make paper-pdf`.

Additional environment and runtime details are provided in the Supplementary Information (SI); code and archival links are provided in Code/Data Availability.

2.14 Data requirements and feasible study designs

KCOR is designed for settings where outcomes are ascertained repeatedly over time but individual-level covariates, visit schedules, or counterfactual exposure histories are unavailable or unreliable. This section clarifies the data structures required for valid application, as well as designs for which the method is not appropriate.

Minimum data requirements. KCOR requires (i) a well-defined cohort entry or enrollment rule, (ii) repeated outcome ascertainment over calendar or follow-up time (which may be passive), (iii) sufficient event counts to estimate cumulative hazards with reasonable stability, and (iv) the presence of at least one diagnostically identifiable "quiet" window during which baseline risk is approximately stable. Individual-level covariates, treatment assignment models, or visit-based follow-up schedules are not required.

What KCOR does not require. Unlike causal estimators, interrupted time-series methods, synthetic controls, or digital twin approaches, KCOR does not require exchangeability assumptions, individual-level confounder measurement, continuous exposure tracking, or explicit modeling of counterfactual untreated trajectories. The method operates entirely on aggregated cumulative hazard information and is therefore compatible with registries and administrative systems where outcomes are captured far more frequently than exposures or visits.

Well-suited data sources and designs. KCOR is particularly well matched to vital statistics–linked cohorts, administrative or registry-based studies with passive outcome capture, program enrollments with irregular follow-up but reliable endpoints, and population-based datasets where cohorts are defined by eligibility, vulnerability, or sociodemographic characteristics rather than repeated clinical encounters.

Designs where KCOR is not appropriate. KCOR is not suitable for settings with extremely sparse events, rapidly shifting baseline hazards with no diagnostically quiet interval, highly fluid cohorts with frequent switching between exposure groups, or outcome ascertainment that is strictly conditional on clinic visits. In such cases, the required identifiability diagnostics will fail and results should not be reported.

3. Results

This section evaluates the revised KCOR estimator in five layers: (i) recovery of enrollment-time frailty variance θ_0 and corresponding null behavior in synthetic settings, (ii) empirical negative controls, (iii) positive controls with injected effects, (iv) failure signaling under model stress, and (v) estimation of the optional NPH exponent α. Synthetic-null flatness remains informative as a diagnostic implication under the working model, but it is no longer the sole validation anchor.

The central validation claim is therefore broader than in earlier versions of the paper:

- **Identification performance:** the estimator should recover the cohort-specific enrollment-time frailty parameters used to generate the geometry and align quiet windows coherently under the stated working model.
- **Negative-control behavior:** under a true null effect, KCOR should remain approximately flat at 1 after normalization, subject to sampling variability and model adequacy; this is interpreted as a diagnostic consistency check rather than as proof that all confounding has been removed.
- **Positive-control behavior:** when known harm/benefit is injected, KCOR should deviate in the expected direction after the same normalization pipeline.
- **Failure signaling:** when key assumptions are violated or the working model is stressed, diagnostics should degrade and the analysis should be treated as not identified rather than reported as a stable contrast.
- **Optional NPH behavior:** when the NPH module is invoked, α should be recoverable in synthetic settings, stable only when cross-cohort signal is adequate, and explicitly non-interpretable when diagnostics fail.

Throughout, curvature in cumulative-hazard plots reflects selection-induced depletion or external hazard structure, while linearity after normalization is interpreted as consistent with removal of that curvature under the working model.

In vaccinated–unvaccinated comparisons, large early differences in $\mathrm{KCOR}(t)$ may reflect baseline risk selection rather than intervention-attributable effects; in such cases, $\mathrm{KCOR}(t; t_0)$ is emphasized to report deviations relative to an early post-enrollment reference while preserving time-varying divergence.

3.1 Negative controls (selection-only null)

3.1.1 Structured synthetic validation: θ_0 recovery and null behavior

We first evaluate KCOR under synthetic settings in which the data-generating process matches the revised working model. The primary synthetic target is no longer only whether $\mathrm{KCOR}(t)$ is flat under a null; it is also whether the estimator recovers the **enrollment-time** frailty variance θ_0 that generated the cohort geometry.

Under correct specification, the revised estimator should recover θ_0 with small error, align the quiet windows coherently after offset reconstruction, and then produce $\mathrm{KCOR}(t)$ trajectories that remain approximately constant at 1 under a true null effect. Near-1 behavior in this setting is treated as a diagnostic implication of correct working-model alignment, not as a stand-alone proof of estimator validity. Full design details, recovery grids, and figures are provided in the Supplementary Information, including the θ_0 recovery simulations and the synthetic negative-control figures (Section S4.2.1 and related figures).

3.1.2 Empirical negative control using national registry data (Czech Republic)

This application is presented solely to illustrate KCOR's diagnostic behavior on real registry data; it uses an age-shift construction (pseudo-cohorts) that is a negative control by design rather than an observational treatment-effect analysis. No causal interpretation of vaccine effects is implied.

The repository includes a pragmatic negative control construction that repurposes a real dataset by comparing "like with like" while inducing large composition differences (e.g., age band shifts). In this construction, age strata are remapped into pseudo-doses so that comparisons are, by construction, within the same underlying category; the expected differential contrast is near zero, but the baseline hazards differ strongly.

These age-shift negative controls deliberately induce extreme baseline mortality differences (10–20 year age gaps) while preserving a pseudo-null by construction, since all vaccination states are compared symmetrically. The $\mathrm{KCOR}(t)$ trajectories are expected to be approximately horizontal under the null, subject to sampling stochasticity, if the normalization is handling the induced selection geometry adequately. This is an empirical negative control of **end-to-end behavior**, not a proof that every source of confounding has been removed.

The purpose is to verify **pipeline and normalization behavior under strong composition shift** when the working model and windowing assumptions hold (**model-consistent conditions**), not to establish robustness to every form of misspecification. Deliberate model stress and frailty misspecification are addressed in §3.3.

This negative control evaluates end-to-end pipeline behavior under model-consistent conditions with **minimal true treatment contrast** (pseudo-null by construction). It is **not** intended as a stress test of depletion correction under strong latent heterogeneity. Robustness to strong depletion geometry and frailty misspecification is assessed separately in simulation stress tests (§3.3).

For the empirical age-shift negative control (Figure 6), aggregated weekly cohort summaries derived from the Czech Republic administrative mortality and vaccination dataset are used and exported in KCOR_CMR format.

Notably, KCOR estimates frailty parameters independently for each cohort without knowledge of exposure status; the observed asymmetry in depletion normalization arises entirely from differences in hazard curvature rather than from any vaccination-specific assumptions.

Figure 6 provides a representative illustration; additional age-shift variants are provided in the Supplementary Information (SI).

Table 7 provides numeric summaries.

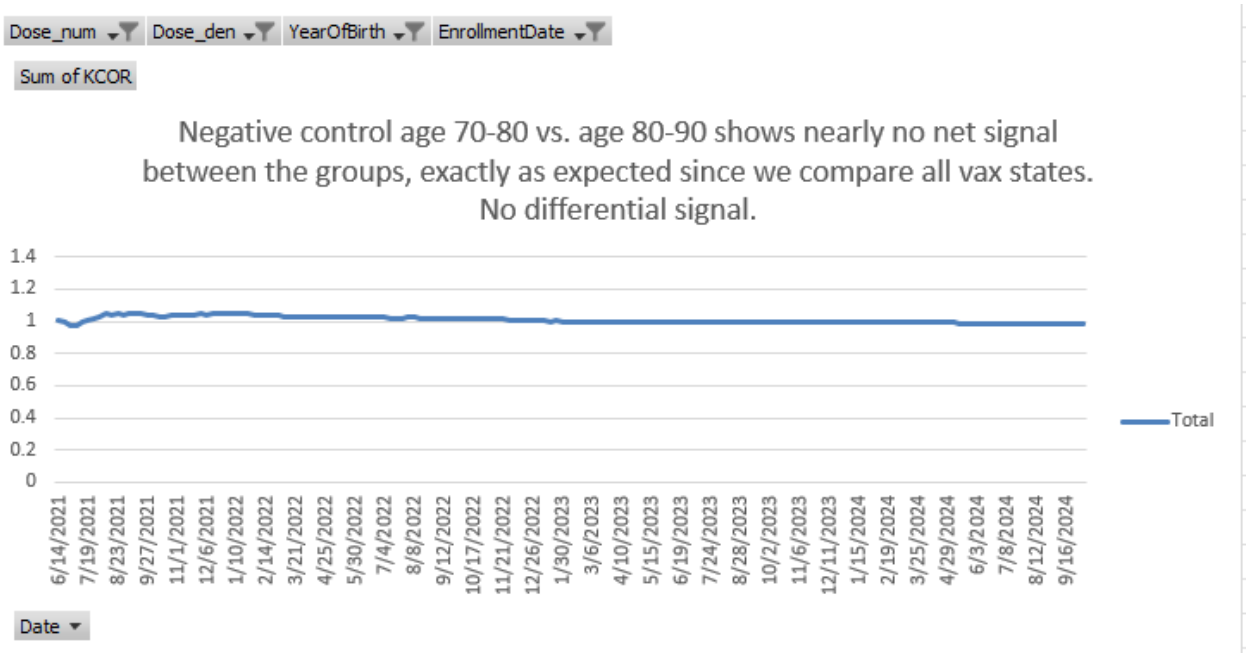

Figure 6: Empirical negative control with approximately 10-year age difference between cohorts. Despite large baseline mortality differences, $\mathrm{KCOR}(t)$ is expected to be approximately horizontal under the pseudo-null, subject to sampling stochasticity. Curves are shown as anchored $\mathrm{KCOR}(t;t_0)$, i.e., $\mathrm{KCOR}(t)/\mathrm{KCOR}(t_0)$, which removes pre-existing cumulative differences and displays post-anchor divergence only. KCOR curves are anchored at $t_0 = 4$ weeks (i.e., plotted as $\mathrm{KCOR}(t;t_0)$). Cumulative hazard is indexed from week 4 following enrollment. Uncertainty bands (95% bootstrap intervals) are shown. Data source: Czech Republic mortality and vaccination dataset processed into KCOR_CMR aggregated format (negative-control construction; see Supplementary Information (SI)).

3.2 Positive controls (injected effects)

Positive controls (injected harm/benefit) are provided in Supplementary Section S3. They test whether, after the same preprocessing and θ_0 estimation pipeline, KCOR deviates in the expected direction and with magnitude broadly consistent with the injected effect (up to discretization and sampling noise).

In positive-control simulations with injected multiplicative hazard shifts, KCOR detects both harm and benefit in the expected direction, with estimated $\mathrm{KCOR}(t)$ trajectories tracking the imposed effects. This complements the null validations by showing that the revised estimator is not only flattening trajectories indiscriminately; it can also preserve post-normalization divergence when an effect is present.

3.3 Stress tests (frailty misspecification)

3.3.1 Frailty misspecification robustness

To assess robustness to departures from the gamma frailty assumption, simulations were conducted under alternative frailty distributions while maintaining the same selection-induced depletion geometry. These stress tests also probe the new estimator's vulnerability to failure of the offset structure, weak curvature, and model overfit. Simulations were performed for:

- **Gamma** (baseline reference)
- **Lognormal** frailty
- **Two-point mixture** (discrete frailty)
- **Bimodal** frailty distributions
- **Correlated frailty** (within-subgroup correlation)

For each frailty specification, bias (deviation from true cumulative hazard ratio), variance (trajectory stability), coverage (proportion of simulations where uncertainty intervals contain the true value), and non-identifiability rate (proportion of simulations where diagnostics indicated non-identifiability) are reported.

Under frailty misspecification, KCOR can degrade by attenuating toward unity or by not meeting diagnostic criteria, rather than by producing large unsupported contrasts. When the alternative frailty distribution produces similar depletion geometry to gamma frailty, KCOR normalization remains approximately valid, with bias remaining small and diagnostics indicating acceptable identification. When the alternative frailty structure produces substantially different depletion geometry, or when the structured offset model fails to align windows coherently, diagnostics such as poor fit, residual structure, offset instability, or weak identification signal that the working approximation is inadequate. The pathological/non-gamma frailty mixture simulation in the Supplement provides a concrete stress test of this regime. Additional validation results—including full simulation grids, quiet-window robustness catalogs, dynamic-selection checks, and extended comparator analyses—are provided in the Supplementary Information (SI).

Additional derivations, simulation studies, robustness analyses, and implementation details are provided in the Supplementary Information.

3.4 Estimation of the NPH exponent α

The optional NPH module is a **prespecified generalization** of the core pipeline, validated synthetically when cross-cohort depletion geometry carries identifying signal. We evaluated estimation of the NPH exponent α using a staged validation framework consisting of synthetic recovery, pooled empirical estimation, and diagnostic sensitivity analyses. Throughout, α is treated as a parameter of the working model describing how excess hazard scales with latent frailty, rather than as a causal or biological quantity. **When α is not identified, the cumulative correction on frailty-neutral cumulative hazard (§2.7.3) is not applied for inference—that outcome is a successful diagnostic, not a method failure.** The empirical pooled exercise in §3.4.2 is **illustrative**: under Methods §2.7.2, localization of α in minimal aggregated data requires **externally supplied VE or other prespecified intervention-effect scale** not learned from the same wave-period contrasts.

3.4.1 Synthetic validation of α

We first assessed whether the estimation procedure can recover known values of α under controlled conditions. Synthetic cohorts were generated under the gamma-frailty working model with specified α, varying frailty variance and noise structure.

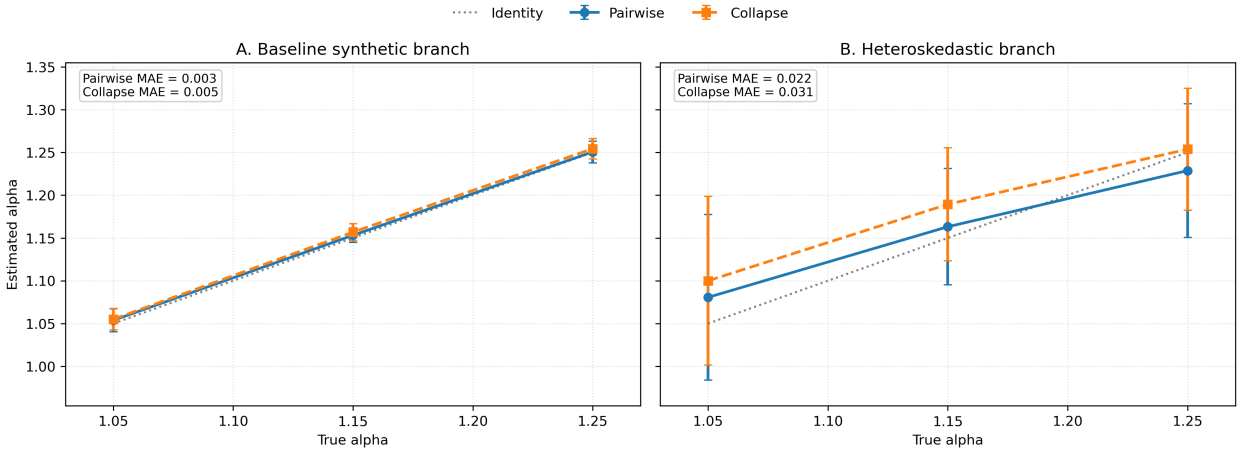

Figure 7: Synthetic validation of NPH exponent recovery. Estimated α versus true α under the working model, shown separately for the baseline synthetic branch (A) and a harder heteroskedastic branch (B). Both pairwise and collapse estimators recover α with low bias when cross-cohort depletion geometry differs. The heteroskedastic branch shows reduced precision rather than structural failure. Error bars show replicate standard deviations.

Figure 7 shows estimated versus true α under two regimes. In the baseline synthetic setting, both the pairwise and collapse estimators recover α with low bias across the tested range. Under heteroskedastic noise, recovery remains directionally correct but exhibits increased dispersion, reflecting reduced information rather than structural failure.

These results confirm that the estimation procedure is capable of recovering α when the working model holds and sufficient cross-cohort variation is present.

3.4.2 Czech registry application

We next applied the same estimation procedure to the Czech primary analysis using pooled cohorts over the main wave period.

Figure 8 shows the corresponding objective functions for the pairwise and collapse estimators. Both exhibit interior minima in the vicinity of $\alpha \approx 1.18$–1.19, and the two estimators are numerically close (pairwise 1.190, collapse 1.185), indicating a consistent preferred region under the working model.

However, the objective functions are shallow, with normalized curvature metrics well below the prespecified identifiability threshold. In particular, curvature values of approximately 7.8×10^{-4} and 8.4×10^{-4} indicate that the data provide limited information to localize α precisely.

Accordingly, while the pooled analysis suggests a region of plausible α values, it does not satisfy the criteria required to report α as identified.

3.4.3 Diagnostics and identifiability of α

We evaluated identifiability using bootstrap, leave-one-out, and sensitivity analyses.

Bootstrap resampling shows substantial dispersion and strong boundary-seeking behavior, with a large fraction of resampled estimates accumulating at the limits of the search grid. Leave-one-out analysis reveals instability, with several cohort omissions producing large shifts in the estimated optimum.

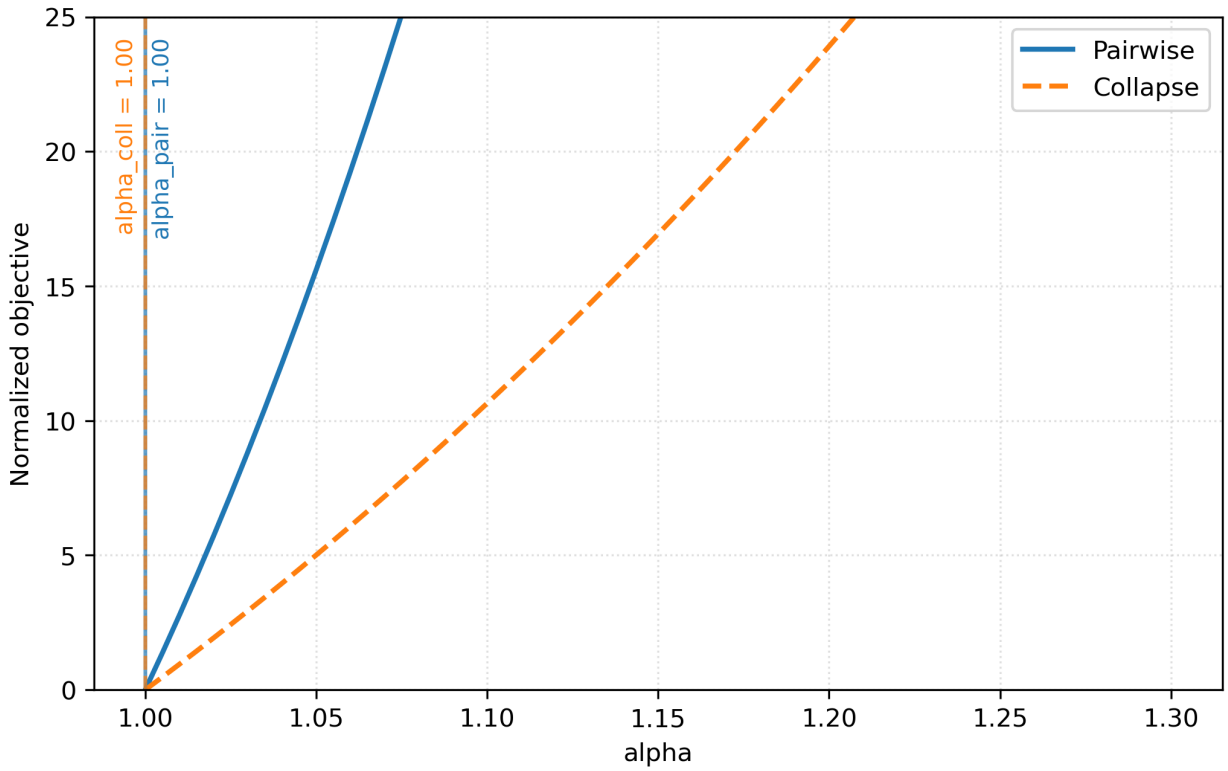

Figure 8: Objective functions for α under the pooled Czech primary specification. The absolute scales of the two objectives differ by construction, so each curve is shifted so that its minimum is zero for visual comparability. Both estimators exhibit interior minima near $\alpha \approx 1.18$–1.19, but the curves remain shallow by the prespecified curvature criterion, indicating a preferred region rather than a strongly identified point estimate.

Sensitivity analyses across anchor choice, excess-hazard definition, and time segmentation further demonstrate inconsistency. In particular, certain configurations produce boundary-seeking optima or materially different estimates, and segmented analyses show reduced agreement relative to the pooled specification. These diagnostic patterns are summarized in Figure 9.

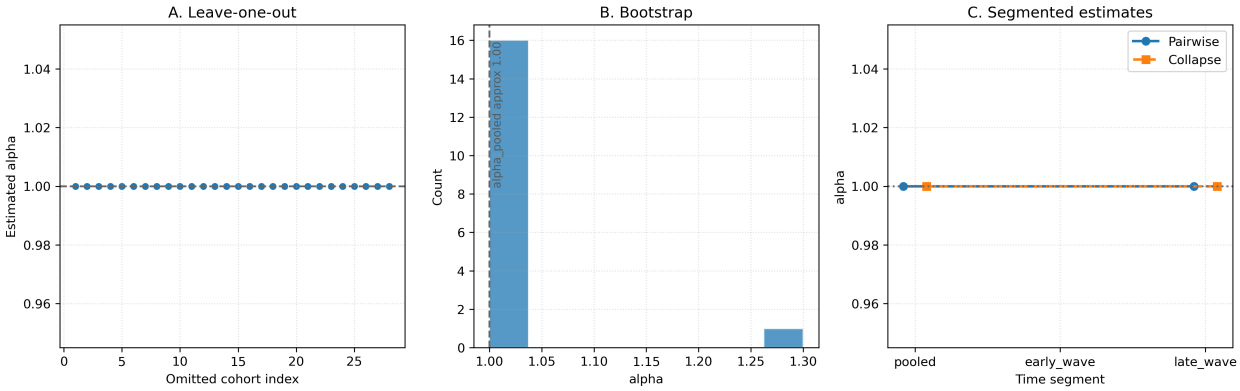

Figure 9: Diagnostics for α estimation under the Czech primary specification. (A) Leave-one-cohort-out analysis shows that several cohort omissions materially affect α, indicating uneven identifying information. (B) The bootstrap distribution is broad and concentrates heavily at the grid boundaries, consistent with weak localization of the optimum. (C) Segmented analyses exhibit reduced agreement and occasional boundary-seeking behavior relative to the pooled specification, illustrating weaker identifiability outside the primary pooled fit.

Taken together, these diagnostics indicate that the apparent pooled optimum does not reflect a stable, well-identified parameter.

3.4.4 Summary of identification status

Across all prespecified criteria, objective curvature, estimator agreement, stability under resampling, and robustness to specification, the Czech pooled analysis fails to meet the threshold required for identification.

We therefore report α as **not identified** for this dataset. The observed region near $\alpha \approx 1.18$–1.19 should be interpreted as a model-dependent preference rather than a reliably estimated parameter.

This outcome is consistent with the role of the identifiability diagnostics: the procedure may locate a shallow optimum even when the data do not contain sufficient information to support a stable estimate, and in such cases the appropriate conclusion is non-identification rather than point estimation.

This non-identification is consistent with the structural confounding between α and cohort-specific multiplicative effects described in §5.4.

Future work may explore whether alternative cohort constructions or stronger cross-cohort variation yield identifiable estimates of α in empirical settings.

4. Discussion

4.1 Limits of attribution and non-identifiability

A notable strength of this study is the use of publicly available, record-level national mortality data. This permits independent verification and replication without reliance on restricted registries.

KCOR should be viewed as complementary to hazard-based modeling rather than as a substitute. Cox models and flexible parametric approaches estimate instantaneous hazard relationships and allow time-varying coefficients, whereas KCOR addresses cumulative survival curvature arising from frailty-induced depletion. When depletion geometry materially distorts marginal hazard ratios, KCOR provides an alternative summary contrast at the cumulative scale.

As in §2.1, KCOR(t) summarizes a depletion-adjusted cumulative contrast on the working-model scale; it does **not**, by itself, identify causal effects, and causal interpretation requires assumptions beyond this framework. It should be read as a **diagnostic** signal about persistent contrasts after depletion adjustment, not as evidence of harm or benefit attributable to an intervention.

KCOR is best viewed as a **pre-causal diagnostic layer** for a specific failure mode: **selection-induced depletion geometry** that distorts **marginal** cumulative comparisons under minimal aggregated data. It does **not** adjust for general confounding in the causal-inference sense, replace covariate adjustment, or turn residual contrasts into causal effects. Persistent differences after depletion normalization may reflect unmeasured confounding, true effects, misspecification, or external hazards; they require substantive design and subject-matter reasoning beyond the normalization step.

KCOR addresses selection-induced depletion that distorts marginal comparisons but does not remove general confounding. It should be interpreted as a **diagnostic normalization** step. Differences observed after normalization may reflect causal effects, residual confounding, or other mechanisms and require additional evidence for causal interpretation.

KCOR does not uniquely identify the biological, behavioral, or clinical mechanisms responsible for observed hazard heterogeneity. In particular, curvature in the cumulative hazard may arise from multiple sources, including selection on latent frailty, behavior change, seasonality, treatment effects, reporting artifacts, or their combination. Depletion of susceptibles is therefore used as a parsimonious working model whose adequacy is evaluated through diagnostics and negative controls rather than assumed as a substantive truth. KCOR's estimand is whether a cumulative outcome contrast persists after adjustment for curvature consistent with that working model, not attribution of the curvature to a specific mechanism.

Separation of curvature and level effects.
KCOR distinguishes between curvature (depletion geometry) and level (multiplicative hazard) effects. The estimator identifies curvature-driven structure through $\theta_{0,d}$ while remaining agnostic to constant multiplicative shifts as separately estimated parameters when scaling is common or approximately common across cohorts (§2.4.4 for heterogeneous scaling and diagnostics). Consequently, KCOR normalization adjusts for selection-induced curvature consistent with the working model but does not, by design, isolate or estimate proportional hazard intervention effects.

Identifiability under the revised estimator.
Even under ideal data quality, KCOR does not identify $\theta_{0,d}$ from an arbitrary mortality trajectory. Identification depends on a joint regime: a plausible Gompertz baseline, a well-defined rebased origin after the stabilization skip, sufficient curvature to distinguish k_d from $\theta_{0,d}$, quiet windows that remain coherent after alignment, and a structured cumulative-hazard offset model when the delta path is used. Curvature-based identification of $\theta_{0,d}$ is invariant to constant multiplicative rescaling of hazards when scaling is common or approximately common across cohorts; heterogeneous multiplicative effects across cohorts may perturb depletion paths and are evaluated with diagnostics, while additive or sharply time-varying components can alter curvature and contaminate $\theta_{0,d}$. The optional exponent α can alias with cohort-specific multiplicative effects in minimal aggregated data (§5.4). If any of these joint regime conditions fails, the estimator is weakly identified or non-identified.

This ambiguity remains structural rather than cosmetic. In minimal aggregated data, depletion-induced curvature is not generically separable from additive or sharply time-varying hazard components that distort curvature over short horizons, and differential multiplicative effects across cohorts can resemble cross-cohort structure targeted by the optional NPH module. The revised estimator improves internal coherence by aligning the parameter target with enrollment time and by pooling information across quiet windows, but it does not eliminate the need for diagnostics or transform the method into a causal effect estimator. Interpretability remains conditional on the adequacy of the gamma-frailty working model and the Gompertz structure used to define the normalization.

4.2 What KCOR estimates

Table 3 clarifies that KCOR differs from non-proportional hazards methods not in flexibility, but in estimand and direction of inference. The contrast with RMST-style and other model-free cumulative summaries, and the motivation for a depletion-neutralized cumulative estimand, are discussed in §2.1. KCOR normalizes selection-

induced depletion under the working model and then compares depletion-neutralized cumulative hazards; KCOR(t) summarizes cumulative outcome accumulation rather than instantaneous hazard ratios. It is descriptive rather than causal, and interpretability is conditional on the stated assumptions and prespecified diagnostics (seed-fit quality, post-normalization linearity, multi-window consistency, and parameter stability). When diagnostics fail, contrasts are not reported and results are treated as non-identifiable rather than as substantive cumulative effects. Anchored KCOR is used only when explicitly stated to remove pre-existing level differences and emphasize post-reference divergence.

Cumulative contrasts are particularly informative in settings where early hazard ratios attenuate over time due to depletion of high-risk individuals, leading instantaneous hazard-based summaries to obscure long-horizon risk differences.

The optional NPH exponent extends this framework from a pure depletion-normalization model to a joint geometry-and-amplification model. The parameter $\theta_{0,d}$ governs how cohorts deplete over time, while the shared exponent α governs how external hazards interact with that depletion during NPH periods. Any **applied** wave-period correction operates on **frailty-neutral cumulative hazard after inversion** (§2.7.3), not on raw observed hazard. These parameters are conceptually distinct but jointly shape the observed non-proportional hazards that KCOR seeks to interpret diagnostically.

4.3 Relationship to negative control methods

Negative control outcomes/tests are widely used to *detect* confounding. KCOR's objective is different: it is an estimator intended to *adjust for a specific confounding structure*—selection-induced depletion dynamics—under the working model, prior to comparison. Negative and positive controls are nevertheless central to validating the estimator's behavior.

This asymmetry helps explain why standard observational analyses can show large mortality differences during periods lacking a plausible mechanism: vaccinated cohorts are already selection-filtered, while unvaccinated hazards are suppressed by ongoing frailty depletion. Unadjusted comparisons therefore systematically understate unvaccinated baseline risk and exaggerate apparent differences.

4.4 Practical guidelines for implementation

This subsection summarizes common operational practices for applying KCOR in retrospective cohort studies and for assessing when resulting contrasts are interpretable under the stated working model.

Practical guideline. KCOR should be applied only when curvature diagnostics indicate a well-identified regime (§2.1.2, §5.1); if diagnostics fail, contrasts are not reported.

Reporting commonly includes:

- Enrollment definition and justification
- Risk set definitions and event-time binning
- Quiet-window definitions and justification across all windows used for θ_0 identification
- Gompertz baseline choice, rebased time origin, and fit diagnostics for θ_0
- Skip/stabilization rule and robustness to nearby values
- Delta-offset diagnostics and multi-window consistency checks
- If used, rationale, identification diagnostics, and estimator-stability checks for the optional NPH exponent module
- Predefined negative/positive controls used for validation
- Sensitivity analysis plan and results

KCOR should therefore be applied and reported as a complete pipeline, from cohort freezing through depletion normalization to cumulative comparison and diagnostics, rather than as a standalone adjustment step. Scope and interpretation are summarized once in Box 2 (§1.6).

KCOR is diagnostic-first: multiplicity concerns are mitigated by prespecification of cohorts, quiet windows, and control constructions. Exploratory stratified analyses should be interpreted descriptively, with emphasis on diagnostic consistency rather than formal significance claims.

5. Limitations

This section summarizes the principal limitations of the KCOR framework. The emphasis is on conditions under which interpretation is restricted rather than on situations in which the estimator merely "fails." These limitations are diagnostic and design-related, reflecting the framework's intentionally conservative scope.

Practical identifiability and weak curvature. Practical identifiability depends on sufficient curvature in cumulative hazard trajectories. When curvature is weak, $\theta_{0,d}$ is weakly identified and KCOR estimates become unstable. In such cases prespecified diagnostic criteria should fail, and results should not be interpreted. That behavior reflects limited signal in the data rather than a defect of the estimation steps when diagnostics are used as intended.

KCOR is intentionally diagnostic rather than test-based: it does not attempt to formally test properties such as quiet-window validity or frailty distributional form. Instead, it enforces conservative interpretability gates when prespecified empirical diagnostics fail. KCOR is not a causal effect estimator and does not identify counterfactual outcomes under hypothetical interventions.

KCOR does not guarantee nominal uncertainty calibration under arbitrary frailty misspecification. Because the gamma frailty model is used as a working approximation, confidence intervals may be mildly anti-conservative in sparse-event regimes or when depletion geometry deviates substantially from gamma. Diagnostic criteria are intended to identify such regimes, and estimates from failing windows should not be interpreted.

- **Model dependence**: Normalization relies on the adequacy of the gamma-frailty model, the Gompertz baseline assumption, and the structured offset logic used to align quiet windows. KCOR is therefore a working-model normalization, not a model-free correction.
- **Relation to existing non-PH methods**: KCOR is complementary to time-varying Cox, flexible parametric, additive hazards, and MSM approaches; these methods address different estimands and identification strategies, whereas KCOR targets depletion-geometry normalization under minimal-data constraints (see §1.3).
- θ_0 **estimation is data-derived**: KCOR does not impose $\theta_0 = 0$ for any cohort. Near-zero fitted θ_0 values indicate weak identifiability or minimal detectable depletion curvature under the working model and should not be interpreted as an assumption of cohort homogeneity.
- **Sparse events**: When event counts are small, hazard estimation and parameter fitting can be unstable.
- **Contamination of quiet periods**: External shocks (e.g., epidemic waves) overlapping the quiet window can bias selection-parameter estimation.
- **Optional NPH module risk**: If the epidemic-wave exponent model is misspecified or weakly identified, it can introduce bias rather than reduce bias under its own assumptions; that module must therefore remain diagnostically checked and treated as inactive when its signal is weak.
- **Applicability to other outcomes**: Although this paper focuses on all-cause mortality, KCOR is applicable to other irreversible outcomes provided that event timing and risk sets are well defined. Application to cause-specific mortality would require explicit competing-risk definitions and cause-specific hazards, but the normalization logic remains cumulative and descriptive. Extension to non-fatal outcomes such as hospitalization is conceptually straightforward but may require additional attention to outcome definitions, censoring mechanisms, and recurrent events. These considerations affect interpretation rather than the core KCOR framework.
- **Non-gamma frailty**: The KCOR framework assumes that selection acts approximately multiplicatively through a time-invariant frailty distribution, for which the gamma family provides a convenient and empirically testable approximation. In settings where depletion dynamics are driven by more complex mechanisms, such as time-varying frailty variance, interacting risk factors, or shared frailty correlations within subgroups, the curvature structure exploited by KCOR may be misspecified. In such cases, KCOR diagnostics, for example poor curvature fit or unstable fitted frailty variance estimates, serve as indicators of model inadequacy rather than targets for parameter tuning. Extending the framework to accommodate dynamic or correlated frailty structures would require explicit model generalization rather than modification of KCOR normalization steps and is left to future work. Empirically, KCOR's validity depends on curvature removal rather than on the literal truth of the gamma family; alternative frailty distributions that generate similar depletion geometry may yield similar normalization.

Sensitivity analyses in this work are intentionally embedded alongside the assumptions they interrogate, rather than consolidated into a single omnibus robustness section. Parameters governing quiet-window identification,

frailty normalization, aggregation, and uncertainty estimation are stress-tested through targeted negative controls, pathological simulations, and diagnostic failure demonstrations. This structure reflects the diagnostic-first philosophy of KCOR: robustness is assessed by whether assumptions remain identifiable and diagnostics pass, not by tuning parameters to stabilize point estimates.

Interpretation of sub-nominal bootstrap coverage

In Table 11, empirical coverage falls below the nominal 95% level under non-gamma frailty (89.3%) and sparse-event regimes (87.6%). **Simulation settings with sub-nominal coverage coincide with low curvature metrics and diagnostic flags, indicating weak identifiability rather than bootstrap failure.** In both cases, the deviation reflects anti-conservative intervals (intervals that are too narrow), arising primarily from variance underestimation rather than bias in the point estimate. Under non-gamma frailty, the working gamma-frailty approximation no longer captures depletion geometry exactly; under sparse events, limited information in cumulative-hazard space leads to underestimated variability. Importantly, these regimes coincide with degraded KCOR diagnostics (poor fit, residual structure, or parameter instability), and analyses are therefore flagged as weakly identified or not reported in practice. In applied analyses, such cases would fail diagnostics and would not be reported. Sub-nominal coverage thus occurs specifically when KCOR's assumptions are violated and diagnostics signal non-identifiability, reinforcing the framework's diagnostic-first and conservative design. Scenarios in that table where the data-generating process aligns with the working model and information is adequate (gamma-frailty null, injected-effect rows) show **empirical coverage close to nominal** (94.2%–93.5% at nominal 95%), consistent with diagnostic-pass regimes rather than universally valid coverage. These results indicate that coverage degradation aligns with diagnostic indicators of weak identifiability, supporting the use of diagnostic gating for inference.

5.1 Failure modes and diagnostics

KCOR is designed to normalize selection-induced depletion curvature under its stated model and windowing assumptions. Reviewers and readers should expect the method to degrade when those assumptions are violated. Common failure modes include:

- **Mis-specified quiet window**: If the quiet window overlaps major external shocks (epidemic waves, policy changes, reporting artifacts), the fitted parameters may absorb non-selection dynamics, biasing normalization.
- **External time-varying hazards masquerading as frailty depletion**: Strong secular trends, seasonality, or outcome-definition changes can introduce curvature that is not well captured by gamma-frailty depletion alone. For example, COVID-19 waves disproportionately increase mortality among frail individuals; if one cohort has higher baseline frailty, such a wave can preferentially deplete that cohort, producing the appearance of a benefit in the lower-frailty cohort that is actually due to differential frailty-specific mortality from the external hazard rather than from the intervention under study.
- **Extremely sparse cohorts**: When events are rare, observed cumulative hazards become noisy and $(\hat{k}_d, \hat{\theta}_{0,d})$ can be weakly identified, often manifesting as unstable fitted frailty variance estimates or wide uncertainty.
- **Non-frailty-driven curvature**: Administrative censoring, cohort-definition drift, changes in risk-set construction, or differential loss can induce curvature unrelated to latent frailty.

Violations of identifiability assumptions lead to conservative behavior (e.g., $\hat{\theta}_0 \to 0$) rather than spurious non-null contrasts.

Practical diagnostics include:

- **Quiet-window overlays** on hazard/cumulative-hazard plots to confirm the fit window is epidemiologically stable.
- **Fit residuals in hazard space** (RMSE, residual plots) and stability of fitted parameters under small perturbations of the quiet-window bounds.
- **Sensitivity analyses** over plausible quiet windows and skip-weeks values.
- **Prespecified negative controls**: $\mathrm{KCOR}(t)$ curves are expected to be horizontal under the null, subject to sampling stochasticity, under control constructions designed to induce composition differences without true effects.

In practice, prespecified negative controls—such as the age-shift controls presented in §3.1.2—provide a direct empirical check that KCOR does not generate artifactual cumulative effects under strong selection-induced curvature.

5.2 Conservativeness and edge-case detection limits

Because KCOR compares fixed enrollment cohorts, subsequent uptake of the intervention among initially unexposed individuals (or additional dosing among exposed cohorts) introduces treatment crossover over time. Such crossover attenuates between-cohort contrasts and biases $\mathrm{KCOR}(t)$ toward unity, making the estimator conservative with respect to detecting sustained net benefit or harm. Analyses should therefore restrict follow-up to periods before substantial crossover or stratify by dosing state when the data permit.

Because KCOR defines explicit diagnostic failure modes—instability, dose reversals, age incoherence, or absence of asymptotic convergence—the absence of such failures in the Czech 2021_24 Dose 0 versus Dose 2 cohorts provides stronger validation than goodness-of-fit alone.

Conservativeness under overlap.
When treatment effects overlap temporally with the quiet window used for frailty estimation, $\mathrm{KCOR}(t)$ does not attribute the resulting curvature to treatment nor amplify it into a spurious cumulative effect. Instead, overlap manifests as degraded quiet-window fit, reduced post-normalization linearity, and instability of estimated frailty parameters, all of which are explicitly surfaced by KCOR's diagnostics. As a result, KCOR is conservative under temporal overlap—preferring diagnostic failure and attenuation over over-interpretation—rather than producing misleading treatment effects when separability is not supported by the data. See §2.1.1 and Supplementary Section S7 for the corresponding identifiability assumptions and stress tests.

KCOR analyses commonly exclude an initial post-enrollment window to exclude dynamic Healthy Vaccinee Effect artifacts. If an intervention induces an acute mortality effect concentrated entirely within this skipped window, that transient signal will not be captured by the primary analysis. This limitation is addressed by reporting sensitivity analyses with reduced or zero skip-weeks and/or by separately evaluating a prespecified acute-risk window.

In degenerate scenarios where an intervention induces a purely proportional hazard shift that does not alter depletion geometry that remains constant over time and does not alter depletion-driven curvature, KCOR's curvature-based contrast may have limited ability to distinguish such effects from residual baseline level differences under minimal-data constraints. Such cases are pathological in the sense that they produce no detectable depletion signature; in practice, KCOR diagnostics and control tests help identify when curvature-based inference is not informative.

Simulation results in §3.4 illustrate that when key assumptions are violated—such as non-gamma frailty geometry, contamination of the quiet window by external shocks, or extreme event sparsity—frailty normalization may become weakly identified. In such regimes, KCOR's diagnostics, including poor cumulative-hazard fit and reduced post-normalization linearity, explicitly signal that curvature-based inference is unreliable without model generalization or revised window selection.

Increasing model complexity within the Cox regression framework—via random effects, cohort-specific frailty, or information-criterion–based selection—does not resolve this limitation, because these models continue to target instantaneous hazard ratios conditional on survival rather than cumulative counterfactual outcomes. Model-selection criteria applied within the Cox regression family favor specifications that improve likelihood fit of instantaneous hazards, but such criteria do not validate cumulative counterfactual interpretation under selection-induced non-proportional hazards.

5.3 Data requirements and external validation

In finite samples, KCOR precision is driven primarily by the number of events observed over follow-up. In simulation (selection-only null), cohorts of approximately 5,000 per arm yielded stable KCOR estimates with narrow uncertainty, whereas smaller cohorts exhibited appreciable Monte Carlo variability and occasional spurious deviations. Reporting event counts and conducting a simple cohort-size sensitivity check are recommended when applying KCOR to sparse outcomes.

External validation across interventions. A natural next step is to apply KCOR to other vaccines and interventions where large-scale individual-level event timing data are available. Many RCTs are underpowered for all-cause mortality and typically do not provide record-level timing needed for KCOR-style hazard-space normalization, while large observational studies often publish only aggregated effect estimates. Where sufficiently detailed time-to-event data exist (registries, integrated health systems, or open individual-level datasets), cross-intervention

comparisons can help characterize how often selection-induced depletion dominates observed hazard curvature and how frequently post-normalization trajectories remain stable under negative controls.

5.4 Optional NPH exponent model and remaining non-proportional hazard risk

COVID-19 mortality exhibits a pronounced departure from proportional hazards, with epidemic waves disproportionately amplifying risk among individuals with higher underlying frailty or baseline all-cause mortality risk[16]. This phenomenon represents a distinct class of bias from both static and dynamic healthy-vaccinee effects. Even after frailty-driven depletion is adjusted for under the working model, wave-period mortality can remain differentially distorted because external infection pressure interacts nonlinearly with baseline vulnerability.

For such settings, the revised KCOR architecture includes an **optional NPH exponent model** in which excess hazard during NPH periods is summarized through a shared amplification parameter α localized from cross-cohort structure together with **externally supplied VE or other prespecified intervention-effect scale** (§2.7.1–§2.7.2), followed by a post-inversion cumulative correction applied to **frailty-neutral cumulative hazard** (§2.7.3), not to raw observed hazard. This model captures one specific class of non-proportionality: frailty-dependent amplification of an external hazard. The exponent α should be interpreted as a model-calibrated summary of frailty-dependent amplification under the working model, not as a uniquely identified biological or mechanistic constant.

Empirical non-identification of α (as in §3.4 when diagnostics fail) is a **successful diagnostic outcome**: it signals that the data do not support applying the cumulative NPH correction for stable inference, and the module should be treated as inactive rather than forced into point estimates.

Identification of α requires sufficient cross-cohort variation in depletion geometry and cumulative hazard, together with stable excess-hazard measurement and a **prespecified externally supplied VE or other prespecified intervention-effect scale** (§2.7.2). When cohorts have similar depletion states, excess hazards are noisy, the assumed power-law structure is inadequate, or the external scale is misspecified, α is weakly identified. In such settings the expected failure signatures are flat objective functions, boundary-seeking estimates, or disagreement between the pairwise and collapse estimators. These patterns are treated as evidence of weak signal or misspecification rather than as valid parameter estimates, and the NPH module should be considered inactive or unreliable.

Non-identifiability under simultaneous multiplicative effects.
In minimal aggregated data, the NPH exponent α is not separately identifiable from cohort-specific multiplicative effects (e.g., intervention-associated hazard reduction). Both mechanisms can produce similar cross-cohort excess-hazard ratios, leading to flat objective functions and boundary-seeking behavior.

Simulation results confirm that even when α is identifiable under a correctly specified synthetic data-generating process, the addition of multiplicative effects rapidly degrades identifiability. This explains why α may be recoverable in controlled simulations but not in real-world data where multiplicative effects and depletion geometry coexist.

The optional NPH model does **not** eliminate all epidemic-wave uncertainty. It assumes that excess hazard can be separated from baseline hazard and that amplification depends on frailty through a common amplitude and power-law relationship. Additive shocks, behavioral responses, time-varying frailty, or other non-separable external forces are not captured and may leave residual bias. Analyses spanning major epidemic waves therefore remain more assumption-sensitive than analyses anchored in quiet periods alone.

In the Czech application analyzed here (§3.4), the pooled α region did not clear the prespecified identifiability gate and is therefore reported as not identified rather than used as a stable preprocessing parameter.

When the optional NPH module is not used, or when its assumptions are not credible, wave-spanning contrasts should be interpreted descriptively and with explicit caution. More elaborate mitigation strategies remain possible in principle, but they would require additional assumptions beyond the present manuscript. The NPH exponent model should therefore be viewed as a narrow, assumption-bearing module rather than as a blanket solution to all wave-period bias.

6. Conclusion

KCOR complements hazard-based modeling by stabilizing cumulative risk comparisons when selection-induced depletion distorts marginal hazard ratios. The revised estimator targets enrollment-time frailty variance θ_0 under a Gompertz working model, aligns quiet windows through structured offsets, and applies gamma-frailty inversion before cumulative comparison. In epidemic-wave settings, an optional NPH exponent model may also be used to summarize how external hazards interact with depletion geometry through a shared amplification parameter α. Validation is correspondingly broader than in earlier formulations: it includes θ_0 recovery, end-to-end negative and positive controls, Cox mismatch demonstrations, and explicit failure signaling under stress. In the Czech application studied here, that diagnostic framework yielded a preferred pooled α region but not a reportable identified exponent. Rather than presuming identifiability, KCOR enforces its assumptions diagnostically, flagging weak or non-identification through degraded fit, offset instability, residual curvature, or unstable NPH objectives instead of absorbing those failures into model-dependent estimates. The resulting contrast remains descriptive and is interpretable only to the extent that the gamma-frailty working model, Gompertz structure, and associated diagnostics are adequate. The optional NPH module remains an assumption-bearing extension rather than a universal part of the KCOR core.

Declarations

Ethics approval and consent to participate

This study used only simulated data and publicly available, aggregated registry summaries that contain no individual-level or identifiable information; as such, it did not constitute human subjects research and was exempt from institutional review board oversight. The primary validation results use synthetic data. Empirical negative-control figures (Figures 6 and S9) use aggregated cohort summaries derived from publicly available Czech Republic administrative data; no record-level data are reproduced in this manuscript, although the underlying Czech datasets are available at the record level[17].

Consent for publication

Not applicable.

Data availability

This study analyzes aggregated cohort-level summaries derived from administrative health records. The Czech Republic mortality and vaccination data used in this study are publicly available, record-level administrative datasets released by the Czech National Health Information Portal[17]. No restricted-access or proprietary data sources were used. For reproducibility and disclosure control, analyses in this manuscript are conducted on aggregated cohort-time summaries derived from the public record-level data.

Software availability

The complete KCOR reference implementation, simulation code, and manuscript build instructions are available at https://github.com/skirsch/KCOR. A citable archival release of the software is available via Zenodo (DOI: 10.5281/zenodo.18050329).

All synthetic validation datasets used for method development and evaluation (including negative and positive control simulations), along with their generation scripts, are publicly available in the project repository. Sensitivity analysis outputs and example datasets in KCOR_CMR format are included to support full computational reproducibility. A formal specification of the KCOR data formats, including schema definitions and disclosure-control semantics, is provided in `documentation/specs/KCOR_file_format.md`.

Use of artificial intelligence tools

The KCOR method and estimand were developed by the author without the use of artificial intelligence (AI) tools. Generative AI tools, including OpenAI's ChatGPT and Cursor Composer 1, were used during manuscript preparation to assist with drafting and editing text, mathematical typesetting, refactoring code, and implementing simulation studies described in this manuscript.

Simulation designs were either specified by the author or proposed during iterative discussion and subsequently reviewed and approved by the author prior to implementation. AI assistance was used to draft code for approved simulations, which the author reviewed, tested, and validated. Additional large language models (including Grok and Claude) were used to provide feedback on manuscript wording and methodological exposition in a role analogous to informal peer review.

All scientific decisions, methodological choices, analyses, interpretations, and judgments regarding which suggestions to accept or reject were made solely by the author, who reviewed and understands all content and takes full responsibility for the manuscript.

Competing interests

The author is a board member of the Vaccine Safety Research Foundation.

Funding

This research received no external funding.

Authors' contributions

Steven T. Kirsch conceived the method, wrote the code, performed the analysis, and wrote the manuscript.

Acknowledgements

The author thanks Clare Craig, James Lyons-Weiler, Jasmin Cardinal Prévost, Stefan Baral, Paul Fischer, Alan Mordue, and Ben Jackson for helpful discussions and methodological feedback during the development of this work. All errors remain the author's responsibility.

References

1. Vaupel JW, Manton KG, Stallard E. The impact of heterogeneity in individual frailty on the dynamics of mortality. *Demography*. 1979;16(3):439-454. doi:10.2307/2061224

2. Grambsch PM, Therneau TM. Proportional hazards tests and diagnostics based on weighted residuals. *Biometrika*. 1994;81(3):515-526. doi:10.1093/biomet/81.3.515

3. Andersen PK, Gill RD. Cox's regression model for counting processes: a large sample study. *Ann Statist*. 1982;10(4). doi:10.1214/aos/1176345976

4. Royston P, Parmar MKB. Flexible parametric proportional-hazards and proportional-odds models for censored survival data, with application to prognostic modelling and estimation of treatment effects. *Statistics in Medicine*. 2002;21(15):2175-2197. doi:10.1002/sim.1203

5. Aalen OO. A linear regression model for the analysis of life times. *Statistics in Medicine*. 1989;8(8):907-925. doi:10.1002/sim.4780080803

6. Lin DY, Ying Z. Semiparametric analysis of the additive risk model. *Biometrika*. 1994;81(1):61-71. doi:10.1093/biomet/81.1.61

7. Van Houwelingen HC. Dynamic prediction by landmarking in event history analysis. *Scandinavian J Statistics*. 2007;34(1):70-85. doi:10.1111/j.1467-9469.2006.00529.x

8. Robins JM, Hernán MÁ, Brumback B. Marginal structural models and causal inference in epidemiology: *Epidemiology*. 2000;11(5):550-560. doi:10.1097/00001648-200009000-00011

9. Cole SR, Hernan MA. Constructing inverse probability weights for marginal structural models. *American Journal of Epidemiology*. 2008;168(6):656-664. doi:10.1093/aje/kwn164

10. Obel N, Fox M, Tetens M, et al. Confounding and negative control methods in observational study of SARS-CoV-2 vaccine effectiveness: a nationwide, population-based Danish health registry study. *CLEP*. 2024;Volume 16:501-512. doi:10.2147/CLEP.S468572

11. Chemaitelly H, Ayoub HH, Coyle P, et al. Assessing healthy vaccinee effect in COVID-19 vaccine effectiveness studies: a national cohort study in Qatar. Schiffer JT, Henry D, eds. *eLife*. 2025;14:e103690. doi:10.7554/eLife.103690

12. Agampodi S, Tadesse BT, Sahastrabuddhe S, Excler JL, Kim JH. Biases in COVID-19 vaccine effectiveness studies using cohort design. *Front Med*. 2024;11:1474045. doi:10.3389/fmed.2024.1474045

13. Bakker B, Walk J, Meester R. Healthy vaccinee and misclassification biases in Dutch observational health data in relation to Covid-19 vaccine effectiveness and safety. Preprint posted online June 30, 2025. https://www.researchgate.net/publication/391008852_Healthy_vaccinee_and_misclassification_biases_in_Dutch_observational_health_data_in_relation_to_Covid-19_vaccine_effectiveness_and_safety

14. Kalbfleisch JD, Prentice RL. *The Statistical Analysis of Failure Time Data*. 1st ed. Wiley; 2002. doi:10.1002/9781118032985

15. Deeks J, Dinnes J, D'Amico R, et al. Evaluating non-randomised intervention studies. *Health Technol Assess*. 2003;7(27). doi:10.3310/hta7270

16. Levin AT, Hanage WP, Owusu-Boaitey N, Cochran KB, Walsh SP, Meyerowitz-Katz G. Assessing the age specificity of infection fatality rates for COVID-19: systematic review, meta-analysis, and public policy implications. *Eur J Epidemiol*. 2020;35(12):1123-1138. doi:10.1007/s10654-020-00698-1

17. Šanca O, Jarkovský J, Klimeš D, et al. *Vaccination, Positivity, Hospitalization for COVID-19, Deaths, Long COVID and Comorbidities in People in the Czech Republic*. National Health Information Portal, Czech Republic; 2024. Accessed December 20, 2024. https://www.nzip.cz/clanek/2135-covid-19-prehled-populace

Tables

Table 1: Summary of three large matched observational studies showing residual confounding / HVE despite meticulous matching.

Study	Design	Matching/adjustment	Key control finding	Implication for methods
Obel et al. (Denmark)[10]	Nationwide registry cohorts (60–90y)	1:1 match on age/sex + covariate adjustment; negative control outcomes	Vaccinated had higher rates of multiple negative control outcomes, but substantially lower mortality after unrelated diagnoses	Strong evidence of confounding in observational VE estimates; "negative control methods indicate… substantial confounding"
Chemaitelly et al. (Qatar)[11]	Matched national cohorts (primary series and booster)	Exact 1:1 matching on demographics + coexisting conditions + prior infection; Cox models	Strong early reduction in non-COVID mortality (HVE), with time-varying reversal later	Even meticulous matching leaves time-varying residual differences consistent with selection/frailty depletion
Bakker et al. (Netherlands)[13]	National observational mortality + COVID-19 vaccination cohort	Stratification + matching on multiple population and health indicators; survival analyses + calendar-time sanity checks	Evidence consistent with residual HVE despite matching; additional evidence of vaccination-status misclassification artifacts	Demonstrates how strong selection + misclassification artifacts can dominate VE/safety estimates; underscores need for diagnostics-first cohort comparison

Table 2: Comparison of Cox proportional hazards, Cox with frailty, and KCOR across key methodological dimensions.

Feature	Cox PH	Cox + frailty	KCOR
Primary estimand	Hazard ratio	Hazard ratio	Cumulative hazard ratio
Conditions on survival	Yes	Yes	No
Assumes PH	Yes	Yes (conditional)	No
Frailty role	None	Nuisance	Object of inference
Uses partial likelihood	Yes	Yes	No
Handles selection-induced curvature	No	Partial	Yes (under working frailty model)
Output interpretable under non-PH	No	No	Yes (cumulative)

Note: KCOR is reported here as a cumulative hazard ratio for comparability; alternative post-normalization estimands are admissible within the framework.

Table 3: Positioning KCOR relative to non-proportional hazards methods.

Method class	Primary target	What is modeled	Handles selection-induced depletion?	Typical output	Failure under latent frailty
Cox PH	Instantaneous hazard	Linear predictor	No	HR	Non-PH from depletion → biased HR
Time-varying Cox	Instantaneous hazard	Time-varying $\beta(t)$	No	HR(t)	Fits depletion as signal
Flexible parametric survival (splines)	Survival / hazard shape	Baseline hazard	No	Smooth hazard / survival	Absorbs depletion curvature
Additive hazards (Aalen)	Hazard differences	Additive hazard	No	$\Delta h(t)$	Still conditional on survival
RMST	Mean survival	Survival curve	No	RMST	Inherits depletion bias
Frailty regression	Heterogeneity-adjusted HR	Random effects	Partial	HR	Frailty treated as nuisance
KCOR (this work)	**Cumulative outcome contrast**	**Depletion geometry**	**Yes (targeted)**	KCOR(t)	Diagnostics flag failure

Table 4: Notation used throughout the Methods section.

Symbol	Definition
d	Cohort index
A, B	Indices of the two cohorts compared in a KCOR contrast
t	Event time since enrollment (discrete bins, e.g., weeks)
$h_{\text{obs},d}(t)$	Discrete-time cohort hazard (conditional on $N_d(t)$)
$H_{\text{obs},d}(t)$	Observed cumulative hazard (after skip/stabilization)
$\tilde{h}_{0,d}(t)$	Depletion-neutralized baseline hazard for cohort d
$\tilde{H}_{0,d}(t)$	Depletion-neutralized baseline cumulative hazard for cohort d
$\theta_{0,d}$	Enrollment-time frailty variance (selection strength) for cohort d at rebased $t = 0$; defined separately for each cohort
$\hat{\theta}_{0,d}$	Estimated enrollment-time frailty variance from the revised estimator, fit separately within each cohort
k_d	Gompertz baseline scale parameter for cohort d
$\hat{k}_d$	Estimated Gompertz baseline scale parameter
γ	Fixed Gompertz age slope used in the working baseline
t_{rebased}	Event time after the stabilization skip; the origin for $\theta_{0,d}$
$H_{\text{gom},d}(t)$	Gompertz cumulative hazard implied by (k_d, γ)
$H^{\text{eff}}_{0,d}(t)$	Reconstructed effective cumulative hazard over the full trajectory
$\delta_{i,d}$	Incremental persistent offset at wave end time t_i
$\Delta_d(t)$	Accumulated offset applied when aligning quiet windows
t_0	Anchor time for baseline normalization (prespecified)
$\text{KCOR}(t; t_0)$	Anchored KCOR: $\text{KCOR}(t)/\text{KCOR}(t_0)$

Table 5: Step-by-step KCOR algorithm (high-level), with recommended prespecification and diagnostics. All analysis choices and estimation procedures are prespecified; numerical parameters such as $\theta_{0,d}$ are estimated separately for each cohort within the prespecified framework.

Step	Operation	Output	Prespecify?	Diagnostics
1	Choose enrollment date and define fixed cohorts	Cohort labels	Yes	Verify cohort sizes/risk sets
2	Compute discrete-time hazards	Hazard curves	Yes (binning/transform)	Check for zeros/sparsity
3	Apply stabilization skip, rebase time, and accumulate observed cumulative hazards	Preprocessed hazards and $H_{\mathrm{obs},d}(t)$	Yes (skip rule)	Plot hazards and cumulative hazards
4	Select prespecified quiet windows in calendar ISO-week space	Quiet-window sets $W_{j,d}$	Yes	Overlay quiet windows on hazard plots
5	Seed-fit $(\hat{k}_d, \hat{\theta}^{(0)}_{0,d})$ in the nearest quiet window	Initial fitted parameters	Yes (estimation procedure)	Seed-fit residuals and plausibility
6	Reconstruct $H^{\mathrm{eff}}_{0,d}(t)$ and compute persistent offsets $\delta_{i,d}$, $\Delta_d(t)$	Aligned depletion geometry	Yes (offset rules)	Offset plausibility and multi-window coherence
7	Refit $\hat{\theta}_{0,d}$ across all quiet windows, then invert the gamma-frailty identity	Depletion-neutralized cumulative hazards	Yes	Residuals, stability, post-normalization linearity
8	Compute KCOR(t) and anchored variants when used	KCOR(t) curve	Yes (horizon / anchor)	Near-flat under negative controls as a diagnostic check; coherent under positive controls
9	Uncertainty	CI / intervals	Yes	Coverage and bootstrap stability

Table 6: Cox vs KCOR under a synthetic null with increasing frailty heterogeneity. Two cohorts are simulated with identical baseline hazards and no treatment effect *(null by construction)*; cohorts differ only in enrollment-time gamma frailty variance (θ_0). Despite the true hazard ratio being 1 by construction, Cox regression produces increasingly non-null hazard ratios as θ_0 increases, reflecting depletion-induced non-proportional hazards. In the same simulations, KCOR(t) remains centered near unity with negligible post-normalization slope across θ_0 values, which is consistent with the expected null benchmark under the working model and is interpreted diagnostically rather than as proof that confounding has been removed. (Exact values depend on simulation seed and follow-up horizon.)

θ_0	Cox HR	95% CI	Cox p-value	KCOR asymptote	KCOR post-norm slope
0.0	0.988	[0.969, 1.008]	0.234	0.988	7.6×10^{-4}
0.5	0.965	[0.946, 0.985]	4.9×10^{-4}	0.990	-3.8×10^{-5}
1.0	0.944	[0.926, 0.963]	1.7×10^{-8}	0.992	-3.0×10^{-4}
2.0	0.902	[0.884, 0.921]	2.4×10^{-23}	0.991	3.7×10^{-4}
5.0	0.804	[0.787, 0.820]	1.5×10^{-93}	0.993	-5.3×10^{-4}
10.0	0.701	[0.686, 0.717]	$< 10^{-200}$	1.020	3.2×10^{-4}
20.0	0.551	[0.539, 0.564]	$< 10^{-300}$	1.024	-1.6×10^{-4}

Table 7: Example end-of-window KCOR(t) values from the empirical negative control (pooled/ASMR summaries), showing near-null behavior under large composition differences. (Source: `test/negative_control/out/KCOR_summary.log`)

Enrollment	Dose comparison	KCOR (pooled/ASMR)	95% CI
2021_24	1 vs 0	1.0097	[0.992, 1.027]
2021_24	2 vs 0	1.0213	[1.000, 1.043]
2021_24	2 vs 1	1.0115	[0.991, 1.033]
2022_06	1 vs 0	0.9858	[0.970, 1.002]
2022_06	2 vs 0	1.0756	[1.055, 1.097]
2022_06	2 vs 1	1.0911	[1.070, 1.112]

Table 8: Positive control results comparing injected hazard multipliers to detected KCOR deviations. Both scenarios indicate KCOR deviating from 1.0 in the expected direction, consistent with the estimator detecting injected effects.

Scenario	Effect window	Hazard multiplier r	Expected direction	Observed KCOR(t) at week 80
Benefit	week 20–80	0.8	< 1	0.825
Harm	week 20–80	1.2	> 1	1.107

Table 9: Comparison of Cox regression, shared frailty Cox models, and KCOR under selection-only and joint frailty + treatment effect scenarios. Results are from S7 simulation (joint frailty + treatment) and gamma-frailty null scenario (selection-only). Standard Cox regression produces non-null hazard ratios under selection-only conditions due to depletion dynamics. Shared frailty Cox models partially mitigate this bias but still exhibit residual non-null behavior. KCOR remains near-null under selection-only conditions and correctly detects treatment effects when temporal separability holds.

Scenario	True effect (r)	Cox HR	Shared frailty Cox HR	KCOR drift/year	Cox indicates null?	Frailty-Cox indicates null?	KCOR indicates null?
Gamma-frailty null	1.0 (null)	0.87	0.94	< 0.5%	No (HR $\neq$ 1)	No (HR $\neq$ 1)	Yes (flat)
S7 harm (r=1.2)	1.2	1.18	1.19	+1.8%	No (detects effect)	No (detects effect)	No (detects effect)
S7 benefit (r=0.8)	0.8	0.83	0.82	-2.1%	No (detects effect)	No (detects effect)	No (detects effect)

Table 10: Simulation comparison of KCOR and alternative estimands under selection-induced non-proportional hazards. Results are summarized across simulation scenarios (null scenarios: gamma-frailty null, non-gamma frailty, contamination, sparse events; effect scenarios: injected hazard increase/decrease). KCOR remains stable under selection-only regimes, while RMST inherits depletion bias and time-varying Cox captures non-proportional hazards without normalizing selection geometry. All methods were applied to identical simulation outputs.

Method	Target estimand	Deviation from null (selection-only scenarios)	Variance/instability	Interpretability notes
KCOR	Cumulative hazard ratio (depletion-normalized)	Near zero (median KCOR ≈ 1.0)	Low (stable trajectory)	Stable under selection-induced depletion; normalization precedes comparison
RMST	Restricted mean survival time	Non-zero (depends on depletion strength)	Moderate (depends on depletion strength)	Summarizes survival differences that may reflect depletion rather than treatment effect; does not normalize selection geometry
Cox	Time-varying hazard ratio	Non-zero under frailty heterogeneity	Moderate (HR instability across time windows)	Improves fit to non-proportional hazards but does not normalize selection geometry; inherits depletion structure

Table 11: Bootstrap coverage for KCOR uncertainty intervals. Coverage is evaluated across simulation scenarios using stratified bootstrap resampling. Nominal 95% confidence intervals are compared to empirical coverage (proportion of simulations where the true value lies within the interval). Coverage degradation is concentrated in regimes where diagnostic criteria fail or event counts are sparse; when diagnostics pass, coverage remains close to nominal. Sub-nominal rows align with weak-identification / misspecification regimes rather than implying a generic bootstrap failure; coverage is **not** claimed to be universally valid outside diagnostic-pass settings.

Scenario	Nominal coverage	Empirical coverage	Notes
Gamma-frailty null	95%	94.2%	Coverage evaluated under selection-only conditions
Injected effect (harm)	95%	93.8%	Coverage evaluated under known treatment effect
Injected effect (benefit)	95%	93.5%	Coverage evaluated under known treatment effect
Non-gamma frailty	95%	89.3%	Coverage under frailty misspecification
Sparse events	95%	87.6%	Coverage under reduced event counts

KCOR Supplementary Information (SI)

This document provides supplementary material supporting the KCOR methodology described in the main manuscript, including extended derivations, simulation studies, robustness analyses, and additional empirical results.

S1. Overview

This SI is organized as follows:

- **S1**: Overview
- **S2**: Extended diagnostics and failure modes
- **S3**: Positive controls (injected harm/benefit)
- **S4**: Control-test specifications and simulation parameters
- **S5**: Additional figures and diagnostics
- **S6**: Extended Czech empirical application / illustrative registry analysis

S2. Extended diagnostics and failure modes

This section describes the **observable diagnostics and failure modes** associated with the revised KCOR working assumptions and the corresponding identifiability criteria. KCOR is designed to **fail transparently rather than silently**: when an assumption is violated, the resulting lack of identifiability or model stress manifests through explicit diagnostic signals rather than spurious estimates. In particular, non-multiplicative or strongly time-varying components within quiet windows can distort curvature and obscure depletion geometry, and heterogeneous multiplicative effects across cohorts can perturb depletion dynamics; such cases are expected to trigger diagnostic instability or non-identifiability.

The KCOR framework separates **working assumptions**, **empirical diagnostics**, and **identifiability criteria**; these are summarized below in Tables S1–S3.

Table S1: KCOR working assumptions.

Assumption	Description	Role in KCOR
A1 Cohort stability	Cohorts are fixed at enrollment with no post-enrollment switching or informative censoring.	Ensures cumulative hazards are comparable over follow-up
A2 Shared external hazard environment	Cohorts experience the same background hazard over the comparison window, except for explicitly modeled extensions.	Prevents confounding by cohort-specific shocks

Assumption	Description	Role in KCOR
A3 Enrollment-time frailty	Selection operates through time-invariant latent heterogeneity summarized by cohort-specific enrollment-time frailty variance $\theta_{0,d}$, estimated separately for each cohort. Under minimal aggregated data, $\theta_{0,d}$ is interpreted via curvature in cumulative-hazard space; constant multiplicative effects are not separately identifiable from baseline scale. Curvature-based identification is invariant to multiplicative scaling that is common or approximately common across cohorts, while heterogeneous multiplicative effects and non-multiplicative contamination are assessed via diagnostics (main text §2.4.4).	Defines the parameter targeted by the estimator
A4 Adequacy of gamma frailty	Gamma frailty provides a reasonable approximation to observed depletion geometry.	Allows tractable inversion and normalization
A5 Adequacy of Gompertz baseline	A fixed Gompertz age slope is a reasonable working approximation over the estimation horizon.	Anchors the structured baseline used to identify $\theta_{0,d}$
A6 Multi-window quiet-period validity	Prespecified quiet windows exist in which depletion can be identified without dominant external shocks.	Permits pooled identification across windows within each cohort
A7 Structured offset additivity	When the delta path is used, wave effects are additive in cumulative-hazard space and persist forward after the wave ends.	Permits alignment of quiet windows through δ_i and $\Delta(t)$
A8 Optional NPH exponent model identifiability	When the optional NPH module is used, excess hazard is reasonably represented by a shared-amplitude frailty-dependent amplification model with a global exponent α; α is localized using cross-cohort structure together with **externally supplied VE or other prespecified intervention-effect scale** not learned from the same wave-period contrasts; and the cumulative correction is applied **after** gamma-frailty inversion (main text §2.7).	Supports optional post-inversion NPH correction

Table S2: Empirical diagnostics associated with KCOR assumptions.

Diagnostic	Description	Observable signal
Skip-week sensitivity	Exclude early post-enrollment weeks subject to dynamic selection.	Stable fitted $\theta_{0,d}$ under varying skip weeks
Seed-fit quality	Evaluate the nearest-window Gompertz seed fit.	Small residuals and plausible $\hat{k}_d$ / $\hat{\theta}^{(0)}_{0,d}$
Multi-window consistency	Refit after omitting or perturbing individual quiet windows.	Stable $\hat{\theta}_{0,d}$ and similar normalized trajectories
Delta plausibility	Check reconstructed offsets and failure flags.	Positive / coherent offsets or transparent fallback
Post-normalization linearity	Assess curvature removal in cumulative-hazard space.	Approximate linearity after normalization within quiet windows as a diagnostic check under the working model
Residual structure	Examine residuals in hazard space.	No systematic time-structure within or across windows
KCOR(t) stability	Inspect KCOR trajectories following anchoring.	Stabilization rather than drift, interpreted diagnostically rather than as proof
Optional NPH diagnostics	Compare pairwise and collapse estimators, theta propagation scales, excess-handling choices, post-inversion correction diagnostics, and with/without-module results when relevant.	Estimator agreement, non-flat objectives, and limited dependence on reasonable NPH settings

Empirical illustration of the "Quiet-window perturbation" diagnostic is provided in Figure S5, which scans monthly-shifted 12-month windows in Czech registry data.

Table S3: Identifiability criteria governing KCOR interpretation.

Criterion	Condition	Consequence if violated
I1 Diagnostic sufficiency	All required diagnostics pass.	KCOR interpretable
I2 Curvature sufficiency	Mortality geometry contains enough curvature to distinguish k_d from $\theta_{0,d}$, with diagnostics supporting curvature-based identification (including adequacy of common multiplicative scaling across cohorts where relevant and absence of dominant non-multiplicative contamination in the quiet-window regime).	$\theta_{0,d}$ weakly identified
I3 Multi-window coherence	Quiet windows remain consistent after alignment.	Interpretation limited
I4 Delta applicability	Offset reconstruction is coherent when the delta path is used.	Fall back or treat as non-identifiable
I5 Anchoring validity	Post-normalization behavior is stable in the reference quiet window.	Anchoring invalid

Criterion	Condition	Consequence if violated
I6 NPH signal sufficiency	When the optional NPH module is used, cross-cohort differences in depletion geometry and excess hazard are strong enough to localize α jointly with a prespecified **externally supplied VE or other prespecified intervention-effect scale**.	α weakly identified; objectives flatten or estimators disagree
I7 Conservative failure rule	Any failure → diagnostics indicate non-identifiability.	Analysis treated as not identified; results not reported

When diagnostics indicate non-identifiability, the analysis is treated as not identified and results are not reported; this does not invalidate the KCOR estimator itself. In minimal aggregated data, identifiability of $\theta_{0,d}$ from curvature remains conditional on the working model rather than assumption-free: additive or sharply time-varying components can distort curvature, and heterogeneous multiplicative effects across cohorts can perturb depletion paths, with diagnostics intended to surface these failure modes. When the optional NPH module is used, identifiability of α is likewise conditional: it depends on cross-cohort separation in depletion geometry, stable excess-hazard measurement, and a prespecified **externally supplied VE or other prespecified intervention-effect scale** (main text §2.7.2); α is not separately identifiable from cohort-specific multiplicative effects under minimal data if that scale is not fixed externally (main text §5.4). The common external intensity is not directly observed.

S2.1 Optional NPH exponent model: estimator and failure signatures

The optional NPH module extends KCOR only for prespecified periods in which an external hazard may interact with frailty in a non-proportional manner. Under the working model, excess hazard for cohort d is represented as

$$h_{\text{excess},d}(t) = A(t)\, F_d(t;\alpha), \qquad F_d(t;\alpha) = E_d[z^\alpha \mid t],$$

where $A(t)$ is an unknown common amplitude and the exponent α is treated as global across cohorts within the analyzed period. When $\alpha = 1$, amplification is proportional to baseline risk and the NPH module is effectively inactive. The exponent α should be interpreted as a model-calibrated summary of frailty-dependent amplification under the working model, not as a uniquely identified biological or mechanistic constant.

Because $A(t)$ is common across cohorts, absolute excess hazards are not directly comparable. However, cross-cohort ratios eliminate the unknown common amplitude:

$$\frac{h_{\text{excess},i}(t)}{h_{\text{excess},j}(t)} = \frac{F_i(t;\alpha)}{F_j(t;\alpha)}.$$

Post-inversion correction. Applying the NPH correction after gamma-frailty inversion ensures wave-period excess is measured relative to a frailty-neutral baseline in cumulative-hazard space; correcting raw hazards would confound depletion geometry with amplification (main text §2.7.3). In discrete weekly bins, a Gompertz reference path is anchored at wave entry, only strictly positive excess above that path is divided by $F_d(t;\alpha)$, and the corrected cumulative hazard is enforced to be non-decreasing.

Identification with fixed external scale. Under minimal aggregated data, α is not separately identifiable from cohort-specific multiplicative intervention effects unless **externally supplied VE or other prespecified intervention-effect scale** is fixed and not estimated from the same wave-period contrasts used to localize α (main text §2.7.2). Given a fixed scale x relating frailty-neutral excess hazards between cohorts A and B,

$$\frac{\Delta h_A^{\text{obs}}(t)}{\Delta h_B^{\text{obs}}(t)} = (1-x)\,\frac{F_A(t;\alpha)}{F_B(t;\alpha)}.$$

With **externally supplied VE or other prespecified intervention-effect scale** fixed as in main text §2.7.2, the operational **pairwise estimator** minimizes discrepancies between observed log excess-hazard ratios and $\log(1-x) + \log F_i(t;\alpha) - \log F_j(t;\alpha)$, in the same spirit as

$$\sum_{i<j,t} \left[\log e_i(t) - \log e_j(t) - \left(\log F_i(t;\alpha) - \log F_j(t;\alpha)\right)\right]^2,$$

but with the appropriate $\log(1-x)$ offset absorbed into the target or the residual as implemented in the reference code. The **collapse estimator** minimizes

$$\mathrm{Var}_d\left[\log e_d(t) - \log F_d(t;\alpha)\right]$$

under the same fixed-scale restriction.

Agreement between the two estimators is treated as supportive of identification under the working model; disagreement is diagnostic rather than something to be averaged away. Additional robustness checks vary the excess-hazard handling rule, baseline anchor choice, and theta propagation scale.

Failure of NPH identification manifests through observable signatures: flat objective curves, boundary-seeking estimates, strong dependence on arbitrary settings, or disagreement between the pairwise and collapse estimators. These patterns are treated as evidence of weak signal or model misspecification rather than as valid estimates of α.

S3. Positive controls

S3.1 Construction of injected effects

The effect window is a simulation construct used solely for positive-control validation and does not represent a real-world intervention period or biological effect window.

Positive controls are constructed by starting from a negative-control dataset and injecting a known effect into the data-generating process for one cohort, for example by multiplying the *baseline* hazard by a constant factor r over a prespecified interval:

$$h_{0,\text{treated}}(t) = r \cdot h_{0,\text{control}}(t) \quad \text{for } t \in [t_1, t_2], \tag{14}$$

with $r > 1$ for harm and $0 < r < 1$ for benefit.

After gamma-frailty normalization (inversion), KCOR should deviate from 1 in the correct direction and with magnitude broadly consistent with the injected effect (up to discretization and sampling noise). Figure S1 and Table 8 illustrate this behavior.

S4. Control-test specifications and simulation parameters

S4.0 Summary tables for control-test and simulation parameters

Table S4: Summary of control-test and simulation parameters referenced in Sections S4.2–S4.6. Values shown are representative defaults or grids used to probe the revised estimator; exact scripts and run settings are version-controlled in the repository.

Section	Item	Parameter	Value	Notes
S4.2.1	Synthetic validation	Data source / scripts	Repository simulation scripts	Used for θ_0 recovery and null-behavior checks

Section	Item	Parameter	Value	Notes
S4.2.1	Synthetic validation	Frailty heterogeneity grid	Prespecified varying grid	Includes weak- and strong-curvature regimes
S4.2.1	Synthetic validation	Baseline model in estimator	Gompertz with fixed γ	Aligns with main-text estimator
S4.2.1	Synthetic validation	Time horizon	Multi-year follow-up	Sufficient to probe identifiability
S4.2.2	Empirical negative control	Data source	Czech admin registry data (KCOR_CMR)	Aggregated cohorts
S4.2.2	Empirical negative control	Construction	Age strata remapped to pseudo-doses	Preserves a pseudo-null by design
S4.3	Positive control	Effect multiplier	$r = 1.2$ harm; $r = 0.8$ benefit	Injected after the same preprocessing pipeline
S4.3	Positive control	Frailty target	Enrollment-time $\theta_{0,d}$	Estimated separately within each cohort, not fixed in the estimator
S4.4	Sensitivity analysis	Perturbations	Skip weeks, quiet-window placement, omitted windows, anchor choice, excess handling, theta propagation scale, optional NPH settings	Probes robustness of the revised estimator
S4.5	Adversarial selection geometry	Frailty distribution	Tail-mixture versus mid-quantile sampling	Tests robustness to non-gamma geometry
S4.6	Joint frailty + effect	DGP	Frailty heterogeneity plus injected effect	Tests separability of depletion and effect windows

S4.1 Reference implementation and default operational settings

Table S5: Reference implementation and default operational settings.

Component	Setting	Default value	Notes
Cohort construction	Cohort indexing	Enrollment period × YearOfBirth group × Dose; plus all-ages cohort (YearOfBirth = -2)	Implementation detail
Quiet-period selection	Quiet windows	Prespecified ISO-week intervals spanning follow-up	Used jointly after alignment, not one at a time

Component	Setting	Default value	Notes
Early-period stabilization (dynamic HVE)	`SKIP_WEEKS`	2	Weeks $t <$ SKIP_WEEKS are excluded from accumulation; time is then rebased
Gompertz slope	`\gamma`	Fixed implementation value	Same value used throughout the estimator
Frailty estimation	Fit method	Seed Gompertz fit + delta iteration + pooled refit	Estimates $(k_d, \theta_{0,d})$ separately within each cohort in the main path
Anchoring	`NORMALIZATION_WEEKS`	4	Reference window for anchored KCOR plots
NPH module	Optional NPH exponent model	Off by default	Enabled only when cross-cohort NPH signal is plausible and diagnostics support identification of α

S4.2 Negative controls

Negative controls are used to evaluate the behavior of KCOR under settings where the true effect is known to be null, while allowing substantial heterogeneity in baseline risk and selection-induced depletion. Two complementary classes of negative controls are considered: (i) fully synthetic simulations used to assess both θ_0 recovery and null KCOR behavior, and (ii) empirical registry-based constructions that preserve a pseudo-null by repurposing age strata as pseudo-exposures without selective sampling. Together, these controls assess whether KCOR remains stable in the presence of non-proportional hazards arising from selection rather than treatment.

S4.2.1 Synthetic negative control: gamma-frailty null

The synthetic negative control (Figure S3) is a fully specified simulation designed to induce **strong selection-induced depletion curvature under a true null effect**. Under the revised estimator, this synthetic setting is used not only to ask whether KCOR remains near 1, but also whether the estimator recovers the cohort-specific enrollment-time frailty parameters governing the simulated geometry and aligns quiet windows coherently. Near-1 behavior here is interpreted as a diagnostic implication under the working model, not as stand-alone proof that the estimator is correct.

Parameter values and scripts are summarized in Table S4.

Both cohorts share identical per-frailty-group death probabilities; only the mixture weights differ. This induces different cohort-level curvature under the null.

Figure S2 provides a compact summary of KCOR bias as a function of frailty variance under the same synthetic-null grid used in Table 6.

S4.2.2 Empirical negative control: age-shift construction

The empirical negative control (Figures 6 and S9) repurposes registry cohorts to create a **pseudo-null comparison** while inducing large baseline hazard differences via 10–20 year age shifts. Because these are full-population strata rather than selectively sampled subcohorts, selection-induced depletion is reduced and the fitted $\theta_{0,d}$ values are expected to be small or weakly identified. The goal here is to test end-to-end KCOR behavior, not to use the empirical geometry as proof that all confounding has been removed.

Parameter values and scripts are summarized in Table S4.

This construction ensures that dose comparisons are within the same underlying vaccination category, preserving a pseudo-null while inducing 10–20 year age differences.

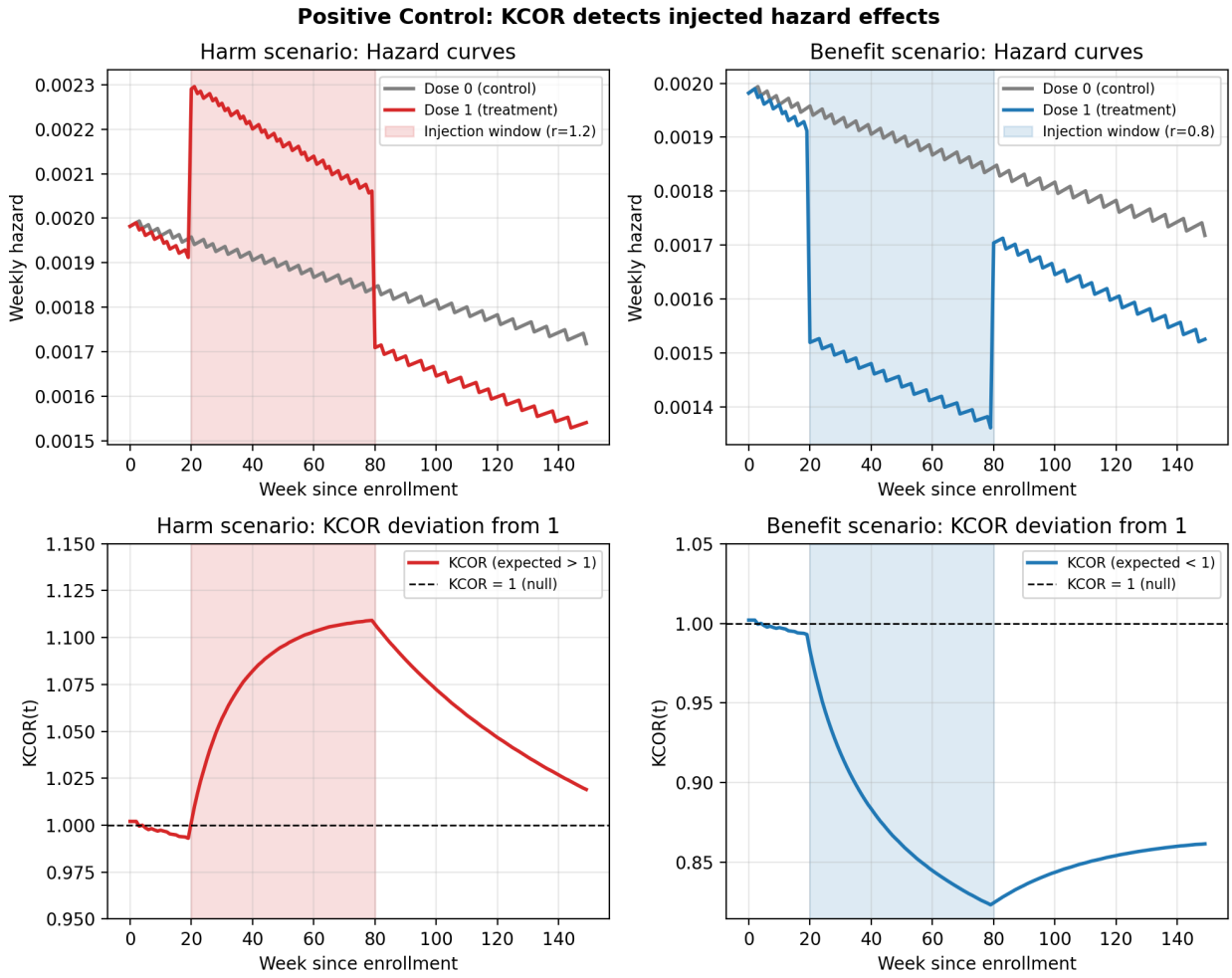

Figure S1: Positive control validation: KCOR tracks injected effects in the expected direction. Left panels show harm scenario (r=1.2), right panels show benefit scenario (r=0.8). Top row displays cohort hazard curves with effect window shaded. Bottom row shows KCOR(t) deviating from 1.0 in the expected direction during the effect window. Uncertainty bands (95% bootstrap intervals; aggregated cohort–time resampling) are shown. X-axis units are weeks since enrollment.

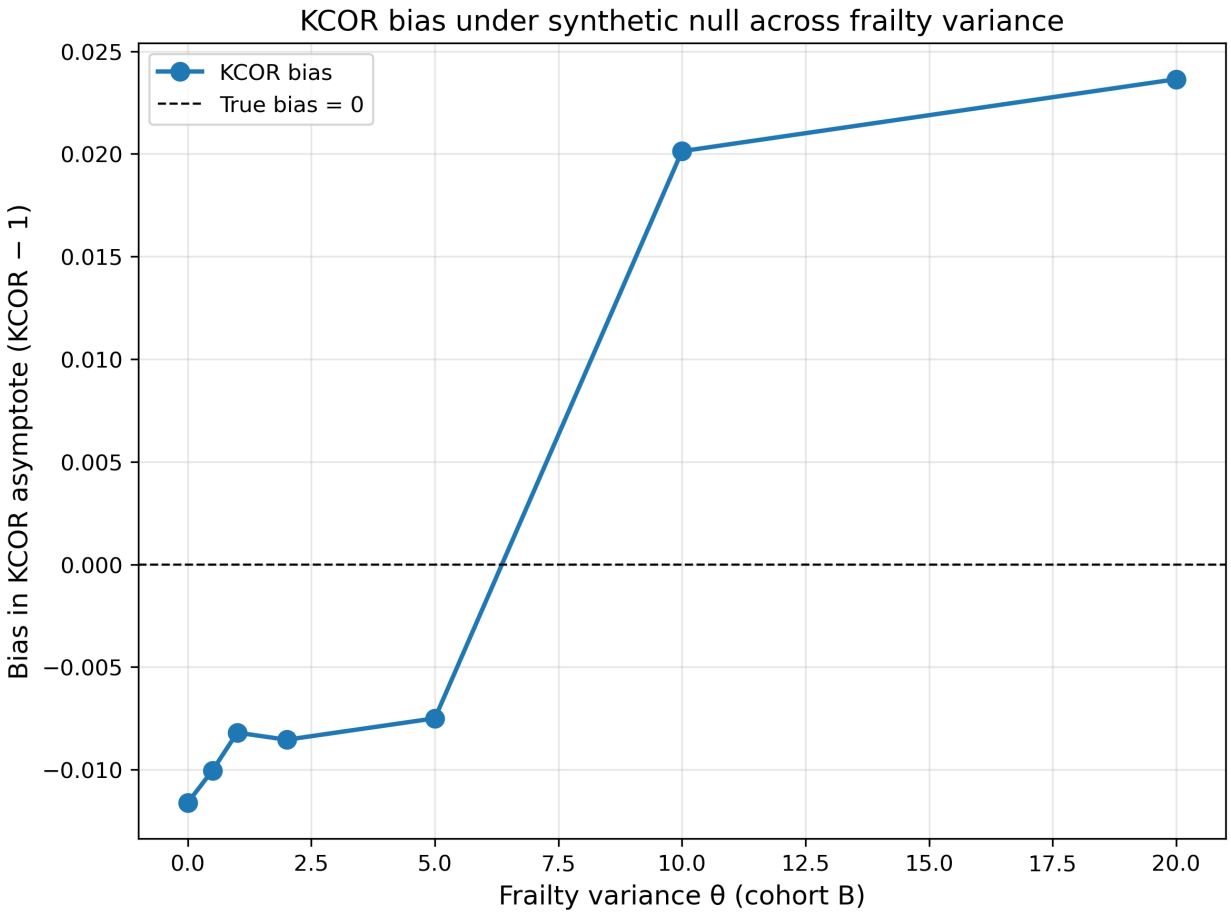

Figure S2: Simulated-null summary: KCOR bias as a function of enrollment-time frailty variance θ_0. Bias is defined as $\mathrm{KCOR}_{\mathrm{asymptote}} - 1$ at the end of follow-up in the synthetic-null grid (no treatment effect), reflecting cumulative deviation under the null rather than instantaneous hazard bias. This figure is a diagnostic summary under the working model, not proof that the model is true. Points show single-run estimates from the grid; no error bars are shown.

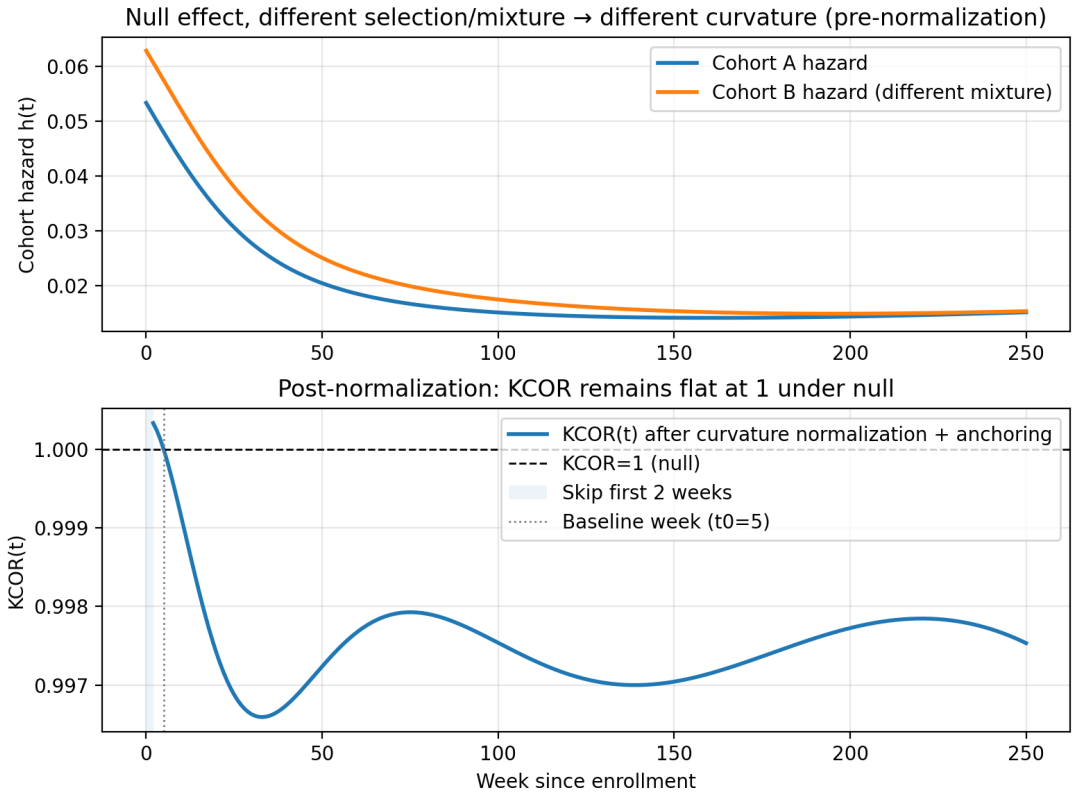

Figure S3: Synthetic negative control under strong selection (different curvature) but no effect: KCOR(t) stays near 1 under the synthetic null. Top panel shows cohort hazards with different frailty-mixture weights inducing different curvature. Bottom panel shows KCOR(t) remaining near 1.0 after normalization, which is the diagnostic behavior expected under the working model rather than proof that all confounding has been removed. Uncertainty bands (95% bootstrap intervals; aggregated cohort–time resampling) are shown.

This contrasts with the synthetic negative control (Section S4.2.1), where strong, deliberately induced frailty heterogeneity requires gamma-frailty normalization to recover the null.

S4.3 Positive control: injected effect

Positive controls are used to verify that KCOR responds appropriately when a true effect is present. Starting from a negative-control simulation with no treatment effect, a known multiplicative hazard shift is injected into one cohort over a prespecified time window. This construction allows direct assessment of whether the same preprocessing and $\theta_{0,d}$ estimation pipeline detects both the direction and timing of the injected effect while remaining stable outside the effect window.

Parameter values and scripts are summarized in Table S4.

The injection multiplies the treatment cohort's baseline hazard by factor r during the effect window, while leaving the control cohort unchanged.

S4.4 Sensitivity analysis parameters

Sensitivity analyses evaluate the robustness of KCOR estimates to reasonable variation in analysis choices that do not alter the underlying data-generating process. Skip weeks, quiet-window placement, omission of individual quiet windows, and optional NPH settings are perturbed over a prespecified range while holding all other parameters fixed. For the optional NPH module, this robustness menu includes baseline anchor choice, excess-hazard handling, theta propagation scale, and agreement between the pairwise and collapse estimators. These analyses assess whether KCOR behavior is stable to tuning choices that primarily affect identification rather than cohort composition.

Parameter values and scripts are summarized in Table S4.

Output grids show KCOR(t) values for each parameter combination.

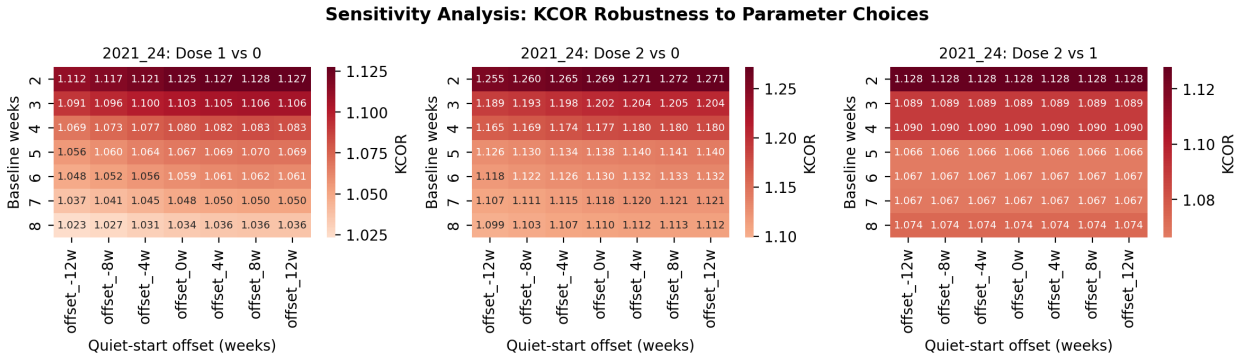

Figure S4: Sensitivity analysis summary showing KCOR(t) values across parameter grid. Heatmaps display KCOR(t) estimates for different combinations of baseline weeks (rows) and quiet-window start offsets (columns). Across all comparisons, KCOR(t) varies smoothly and modestly across a wide range of quiet-start offsets and baseline window lengths, with no qualitative changes in sign or magnitude, indicating robustness to reasonable parameter choices. All panels use a unified color scale centered at 1.0 to enable direct visual comparison across dose comparisons.

S4.5 Tail-sampling / bimodal selection (adversarial selection geometry)

This adversarial simulation evaluates KCOR under extreme but controlled violations of typical cohort-selection geometry. Two cohorts are constructed to share identical mean frailty while differing sharply in how risk is distributed, using mid-quantile sampling versus a low/high-tail mixture. This setting stress-tests whether the revised $\theta_{0,d}$ estimator and depletion normalization remain effective when frailty heterogeneity is concentrated in the tails rather than smoothly distributed.

- **Mid-sampled cohort**: frailty restricted to central quantiles (e.g., 25th–75th percentile) and renormalized to mean 1.
- **Tail-sampled cohort**: mixture of low and high tails (e.g., 0–15th and 85th–100th percentiles) with mixture weights chosen to yield mean 1.

Parameter values and scripts are summarized in Table S4.

Both cohorts share the same baseline hazard $h_0(t)$ and have no treatment effect (negative-control version). Positive-control versions are also generated by applying a known hazard multiplier in a prespecified window. The evaluation includes (i) KCOR drift, (ii) fit quality under the revised estimator, (iii) post-normalization linearity, and (iv) parameter stability under window perturbation.

S4.6 Joint frailty and treatment-effect simulation

This simulation evaluates KCOR under conditions in which **both selection-induced depletion (frailty heterogeneity)** and a **true treatment effect (harm or benefit)** are present simultaneously. The purpose is to assess whether KCOR can (i) correctly identify and neutralize frailty-driven curvature using aligned quiet windows and (ii) detect a true treatment effect outside that period without confounding the two mechanisms.

This joint simulation combines the selection-induced depletion mechanisms examined in Sections S4.2 and S4.5 with the injected-effect framework of Section S4.3.

Parameter values and scripts for this joint simulation are summarized in Table S4.

Design

Two fixed cohorts are generated with identical baseline hazards but differing enrollment-time frailty variance. Individual hazards are multiplicatively scaled by a latent frailty term drawn from a gamma distribution with unit mean and cohort-specific variance. A treatment effect is then injected over a prespecified time window that does not overlap the quiet windows used for frailty identification.

Formally, individual hazards are generated as

$$h_i(t) = z_i \cdot h_0(t) \cdot r(t). \tag{15}$$

where z_i is individual frailty, $h_0(t)$ is a shared baseline hazard, and $r(t)$ is a time-localized multiplicative treatment effect applied to one cohort only.

S5. Additional figures and diagnostics

This section provides diagnostic outputs and evaluation criteria for the simulations and control-test specifications defined in Section S4.

S5.1 Fit diagnostics

For each cohort d, the revised fit produces diagnostic outputs including:

- **RMSE in hazard space**: Root mean squared error between observed and model-predicted hazards over the pooled quiet-window fit set.
- **Fitted parameters**: baseline scale parameter and cohort-specific enrollment-time frailty variance $\theta_{0,d}$.
- **Number of fit points**: pooled observations across quiet windows.
- **Offset diagnostics**: number, sign, and stability of reconstructed δ_i terms when the delta path is used.

When uncertainty bands or bootstrap intervals are reported in this supplement, they are computed using an aggregated-data bootstrap at the cohort–time level (resampling event counts and risk-set sizes within time bins/strata), not by resampling individuals.

Example diagnostic output from the reference implementation:

```
KCOR_FIT,EnrollmentDate=2021_24,YoB=1950,Dose=0,
  k_hat=4.29e-03,theta0_hat=8.02e-01,
  RMSE_h=3.37e-03,n_obs=97,delta_count=3,success=1
```

S5.2 Residual analysis

Fit residuals should be examined in hazard space. Define residuals:

$$r_d(t) = h_d^{\text{adj}}(t) - \frac{\hat{k}_d e^{\gamma t_{\text{rebased}}}}{1 + \hat{\theta}_{0,d}\left(H_{\text{gom},d}(t_{\text{rebased}}) + \Delta_d(t)\right)}.$$

- **Systematic patterns**: Residuals should be approximately random around zero. Systematic structure suggests model inadequacy.
- **Outliers**: Individual weeks with large residuals may indicate data quality issues or external shocks.
- **Autocorrelation**: Strong autocorrelation in residuals suggests the model is missing time-varying structure.

S5.3 Parameter stability checks

Robustness of fitted parameters is assessed by:

- **Quiet-window perturbation**: Shift the quiet-window start/end by ±4 weeks and re-fit. Stable parameters should vary modestly and preserve the same qualitative interpretation.
- **Window omission**: Drop one quiet window at a time and verify that $\hat{\theta}_{0,d}$ and KCOR remain coherent.
- **Skip-weeks sensitivity**: Vary `SKIP_WEEKS` and verify KCOR trajectories remain qualitatively similar.
- **Gompertz sensitivity**: Check whether modest perturbations of γ materially alter interpretation.
- **Optional NPH sensitivity**: In epidemic-wave applications, compare results with and without the module, across plausible anchor and excess-handling choices, across theta propagation scales, and across the pairwise and collapse estimators.

S5.4 Quiet-window perturbation scan (empirical Czech data)

To assess whether diagnostically valid quiet windows are rare or fragile in applied registry data, we performed a systematic scan of quiet-window placements in the Czech Republic 2021–2024 cohort (enrollment ISO week 2021-24). We re-estimated the revised frailty parameters $(\hat{k}_d, \hat{\theta}_{0,d})$ across a sequence of overlapping quiet windows of fixed duration, holding all other settings fixed.

Specifically, we considered 12-month windows beginning at successive monthly offsets between April 2022 and April 2023 (inclusive). For each window placement, parameters were estimated using the same revised estimator and diagnostic pass/fail criteria used elsewhere in this SI. Results are shown for two birth-decade strata (1940s and 1950s), and for dose groups $d \in \{0, 1, 2\}$.

Figure S5 shows fitted $\hat{\theta}_{0,d}$ as a function of quiet-window midpoint date. The purpose of this scan is technical rather than substantive: it asks whether the estimator remains coherent across nearby quiet-window choices and whether failures are surfaced explicitly. Open markers denote windows failing diagnostics (treated as non-identifiable and not interpreted).

S5.5 Quiet-window overlay plots

Overlaying prespecified quiet windows on hazard and cumulative-hazard time series plots provides a visual diagnostic of window placement relative to mortality dynamics. The fit windows should:

- Avoid major epidemic waves or external mortality shocks
- Contain sufficient event counts for stable estimation
- Be interpretable after rebasing and, when relevant, after offset alignment

Visual inspection of quiet-window placement relative to mortality dynamics remains an essential diagnostic step, but it should be interpreted jointly with the formal multi-window and offset diagnostics described above.

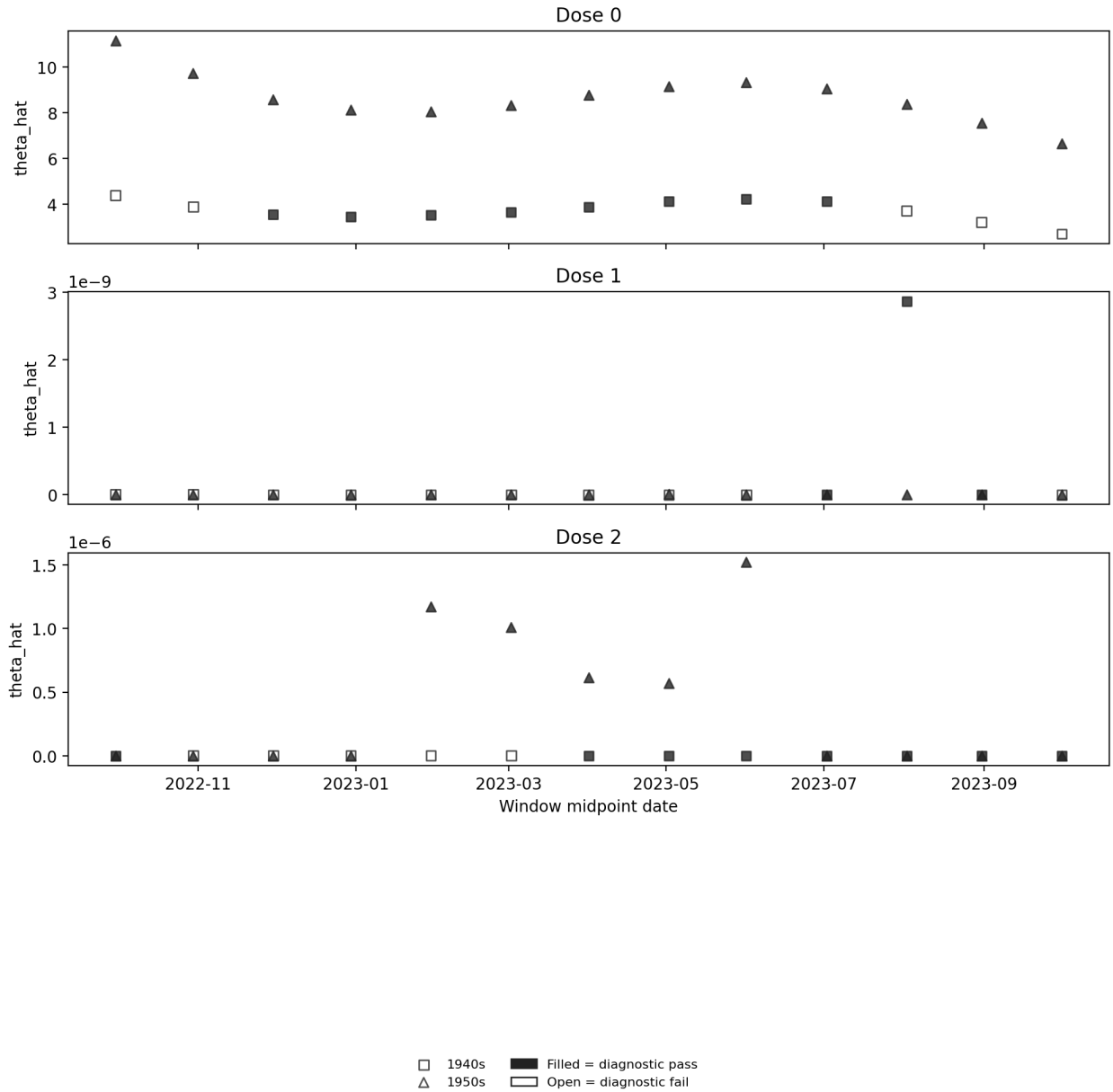

Figure S5: Quiet-window robustness scan: fitted enrollment-time frailty variance $\hat{\theta}_{0,d}$ vs. quiet-window midpoint date for the Czech 2021–2024 enrollment cohort, using 12-month windows shifted monthly from April 2022 through April 2023. Marker shape denotes birth decade (1940s, 1950s). Filled markers indicate diagnostic pass; open markers indicate diagnostic failure. The figure is intended as a technical stability scan of the estimator rather than as substantive evidence for any one cohort narrative.

S5.6 Robustness to age stratification

This subsection examines KCOR(t) under narrow age stratification by repeating the same fixed-cohort comparison in three single birth-year cohorts spanning advanced ages (1930, 1940, 1950). Across these strata, the trajectories remain qualitatively stable after depletion normalization, which is consistent with the observed behavior not being driven solely by age aggregation.

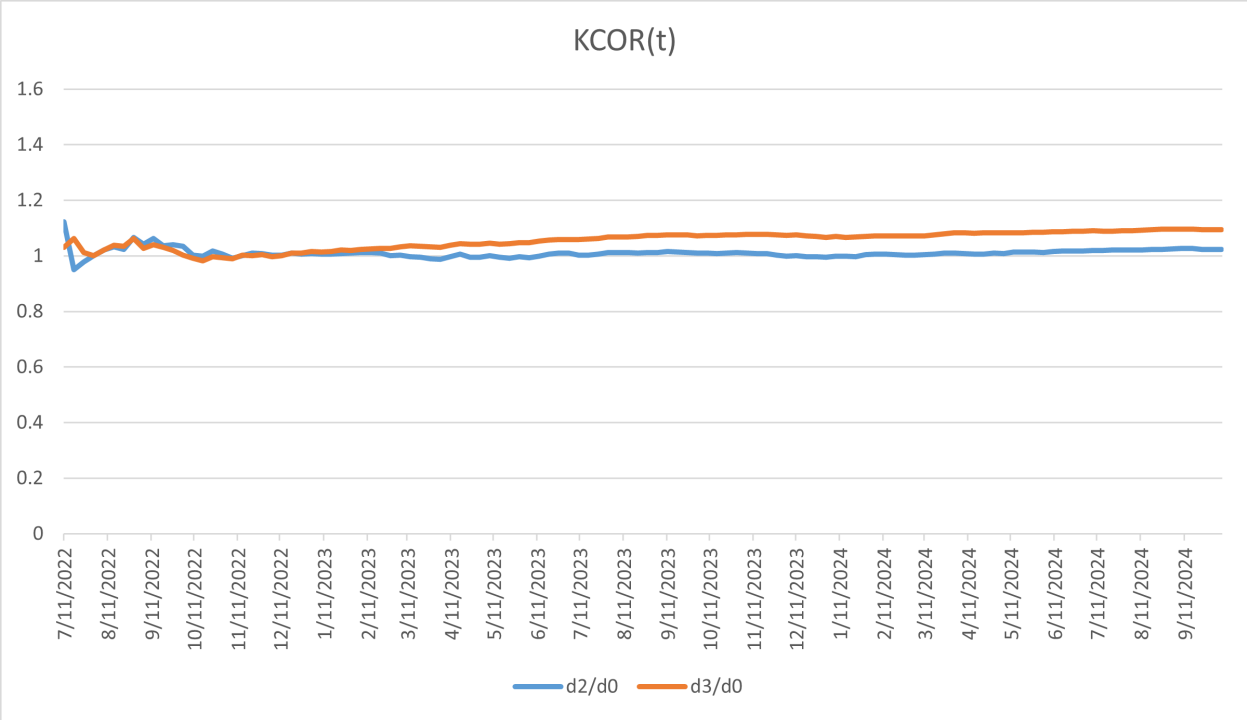

Figure S6: Birth-year cohort 1930: KCOR(t) trajectories comparing dose 2 and dose 3 to dose 0 for cohorts enrolled in ISO week 2022-26 and evaluated over calendar year 2023. KCOR curves are anchored at $t_0 = 4$ weeks (i.e., plotted as KCOR$(t; t_0)$). This figure is presented as an illustrative application demonstrating estimator behavior on registry data and does not support causal inference. X-axis units are weeks since enrollment.

Additional empirical negative-control variant (20-year age shift).
For completeness, the more extreme 20-year age-shift negative control referenced in the main text is included:

S6. Extended Czech empirical application

S6.1 Empirical application with diagnostic validation: Czech Republic national registry mortality data

The Czech results do not validate KCOR; they represent an application that satisfies all pre-specified diagnostic criteria. Substantive implications follow only if the identification assumptions hold. Throughout this subsection, observed divergences are interpreted strictly as properties of the estimator under real-world selection, not as intervention effects.

Unless otherwise noted, KCOR curves in the Czech analyses are shown anchored at $t_0 = 4$ weeks for interpretability.

S6.1.1 Illustrative empirical context: COVID-19 mortality data

The COVID-19 vaccination period provides a natural empirical regime characterized by strong selection heterogeneity and non-proportional hazards, making it a useful illustration for the KCOR framework. During this period, vaccine uptake was voluntary, rapidly time-varying, and correlated with baseline health status, creating clear examples of selection-induced non-proportional hazards. The Czech Republic national mortality registry data

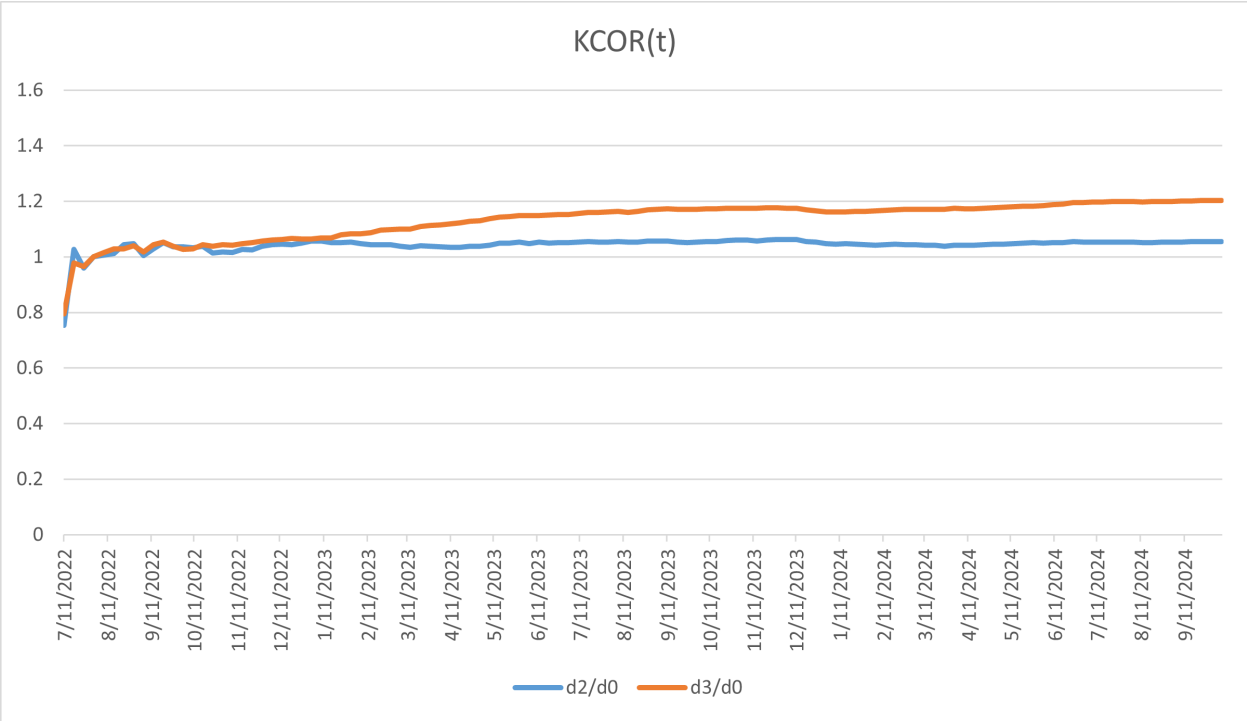

Figure S7: Birth-year cohort 1940: KCOR(t) trajectories comparing dose 2 and dose 3 to dose 0 for cohorts enrolled in ISO week 2022-26 and evaluated over calendar year 2023. KCOR curves are anchored at $t_0 = 4$ weeks (i.e., plotted as $\mathrm{KCOR}(t; t_0)$). This figure is presented as an illustrative application demonstrating estimator behavior on registry data and does not support causal inference. X-axis units are weeks since enrollment.

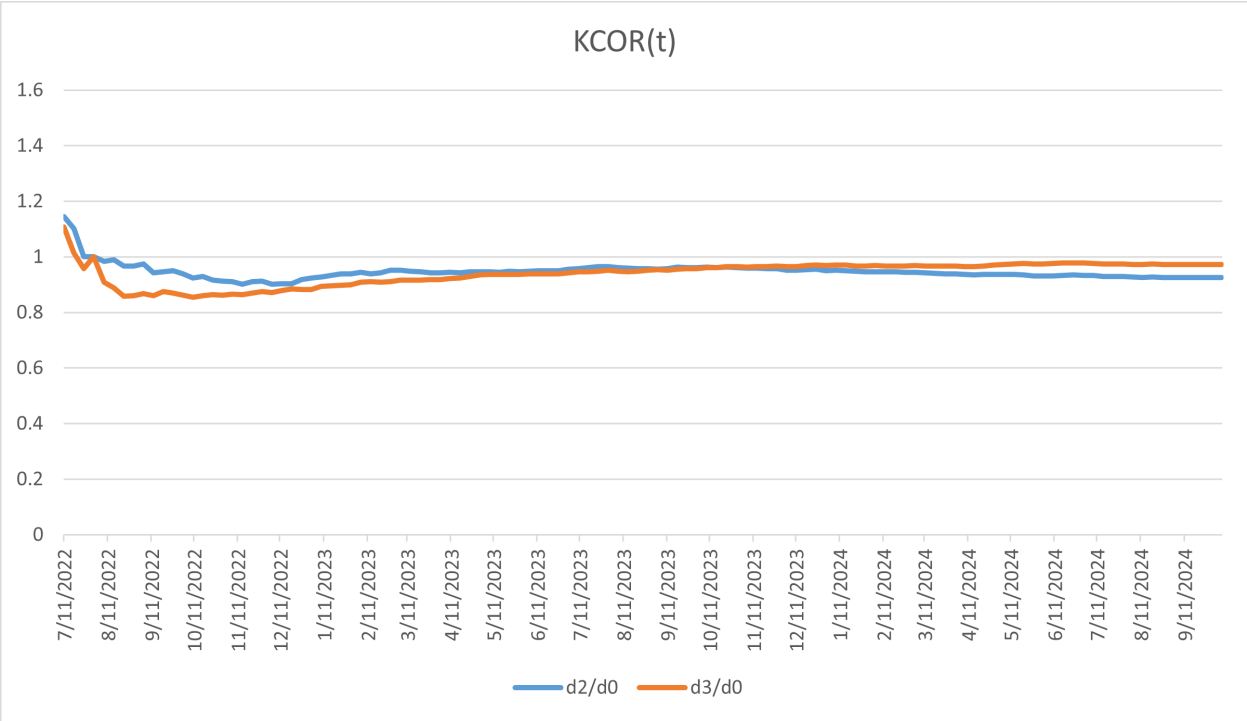

Figure S8: Birth-year cohort 1950: KCOR(t) trajectories comparing dose 2 and dose 3 to dose 0 for cohorts enrolled in ISO week 2022-26 and evaluated over calendar year 2023. KCOR curves are anchored at $t_0 = 4$ weeks (i.e., plotted as $\mathrm{KCOR}(t; t_0)$). This figure is presented as an illustrative application demonstrating estimator behavior on registry data and does not support causal inference. X-axis units are weeks since enrollment.

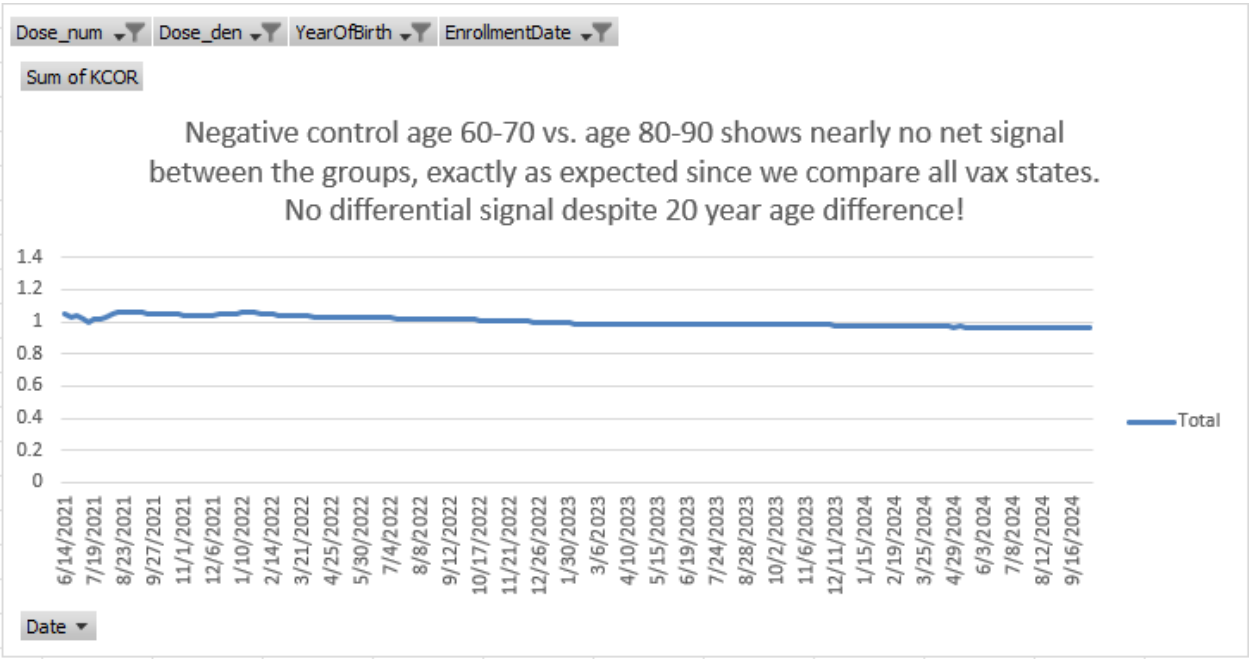

Figure S9: Empirical negative control with approximately 20-year age difference between cohorts. Even under extreme composition differences, $\mathrm{KCOR}(t)$ shows little systematic drift, which is consistent with the estimator handling selection-induced curvature in this pseudo-null construction. KCOR curves are anchored at $t_0 = 4$ weeks (i.e., plotted as $\mathrm{KCOR}(t; t_0)$). Uncertainty bands (95% bootstrap intervals) are shown. Data source: Czech Republic mortality and vaccination dataset processed into KCOR_CMR aggregated format (negative-control construction; see Supplementary Information, SI). X-axis units are weeks since enrollment.

exemplify this regime: voluntary uptake led to asymmetric selection at enrollment and to visibly different cohort hazard geometry over follow-up. Under the revised estimator, those differences are summarized through fitted $\theta_{0,d}$ values, aligned quiet-window behavior, depletion-normalized cumulative trajectories, and, when invoked, optional NPH diagnostics for the shared amplification exponent α. The Czech application remains illustrative rather than validating; any NPH output should be interpreted as descriptive and model-calibrated rather than as direct biological evidence. While these examples illustrate KCOR's application, the method is general and applies to any retrospective cohort comparison where selection induces differential depletion dynamics.

S6.1.2 Descriptive frailty-normalization behavior in the Czech application

Across examined age strata in the Czech Republic mortality dataset, fitted enrollment-time frailty parameters vary across cohorts, with some estimates close to the weak-identification boundary and others appreciably above zero:

$$\hat{\theta}_{0,d} \geq 0. \tag{16}$$

These values should be interpreted as **descriptive outputs of the revised estimator**, not as validation evidence. They summarize how much depletion curvature the estimator attributes to each cohort at rebased enrollment time under the working model, conditional on the diagnostics for that analysis.

As a consequence, KCOR normalization leaves some cohorts' cumulative hazards nearly unchanged, while substantially increasing the depletion-neutralized baseline cumulative hazard for others. This behavior is consistent with curvature-driven normalization rather than cohort identity. The pattern is visible directly in depletion-neutralized versus observed cumulative hazard plots and is summarized quantitatively in the fitted-parameter logs (see `KCOR_summary.log`).

After frailty normalization, the depletion-neutralized baseline cumulative hazards are approximately linear in event time within diagnostically acceptable quiet windows. Residual deviations from linearity reflect real time-varying risk—such as seasonality or epidemic waves—rather than selection-induced depletion. This linearization is a diagnostic consistency check under the working model; persistent nonlinearity or parameter instability indicates model stress or quiet-window contamination.

Table S6 summarizes these diagnostic checks across age strata.

Table S6: Diagnostic gate for Czech application: KCOR results reported only where diagnostics pass.

Age band (years)	Quiet window valid	Post-normalization linearity	Parameter stability	KCOR reported
40–49	Yes	Yes	Yes	Yes
50–59	Yes	Yes	Yes	Yes
60–69	Yes	Yes	Yes	Yes
70–79	Yes	Yes	Yes	Yes
80–89	Yes	Yes	Yes	Yes
90–99	Yes	Yes	Yes	Yes
All ages	Yes	Yes	Yes	Yes

In this application, all examined age strata satisfied the prespecified diagnostic criteria; this indicates internal consistency of the working model in this dataset but does not constitute validation of the estimator. KCOR results are not reported for any age stratum where diagnostics indicate non-identifiability.

Interpretation: In this application, some cohorts exhibit larger fitted $\hat{\theta}_{0,d}$ than others, indicating stronger estimated depletion curvature at rebased enrollment time under the working model:

$$\hat{\theta}_{0,d} > 0. \tag{17}$$

for some cohorts, while others remain close to the weak-identification boundary:

$$\hat{\theta}_{0,d} \approx 0. \tag{18}$$

Estimated heterogeneity can appear larger at younger ages because baseline hazards are low, so proportional differences across latent risk strata translate into visibly different short-term hazards before depletion compresses the risk distribution. At older ages, higher baseline hazard and stronger ongoing depletion can reduce the apparent dispersion of remaining risk, yielding smaller fitted $\theta_{0,d}$ even if latent heterogeneity is not literally smaller. These descriptive patterns should not be elevated into a validation criterion, and no single dose-ordering pattern is required by the revised estimator. Table S7 is therefore presented as an empirical summary of fitted parameters rather than as proof that the estimator is correct.

These raw contrasts reflect both selection and depletion effects and are not interpreted causally.

Table S7: Estimated enrollment-time gamma-frailty variance by age band and vaccination status for Czech cohorts enrolled in 2021_24.

Age band (years)	Fitted frailty variance (Dose 0)	Fitted frailty variance (Dose 2)
40–49	16.79	2.66×10^{-6}
50–59	23.02	1.87×10^{-4}
60–69	13.13	7.01×10^{-18}
70–79	6.98	3.46×10^{-17}
80–89	2.97	2.03×10^{-11}
90–99	0.80	8.66×10^{-16}
All ages (full population)	4.98	1.02×10^{-11}

Notes: - The fitted frailty variance quantifies unobserved frailty heterogeneity and selection-induced depletion within cohorts at rebased enrollment time. Near-zero values indicate weak identifiability or effectively linear cumulative hazards under the working model. - Each entry reports a single fitted enrollment-time gamma-frailty variance for the specified age band and vaccination status within the 2021_24 enrollment cohort. - The "All ages (full population)" row corresponds to an independent fit over the full pooled age range, included as a global diagnostic. - Table S8 reports raw outcome contrasts for ages 40+ (YOB $\leq$ 1980) where event counts are stable.

Diagnostic checks: - **Interpretation is conditional on diagnostics:** these values are shown only for strata passing the prespecified diagnostic gate above. - **Weak-identification boundary:** estimates near zero are treated as low-curvature or weak-identification outputs, not as substantive evidence for or against any cohort story. - **Age coherence remains descriptive:** variation across age bands is reported descriptively and may reflect the interaction of baseline hazard, follow-up length, and depletion geometry. - **Stability:** sign reversals, boundary pathologies, or major sensitivity to window perturbation would count against interpretability. - **Falsifiability:** failure of any one of these checks would constitute evidence against model adequacy for that application.

Table S8: Ratio of observed cumulative mortality hazards for unvaccinated (Dose 0) versus fully vaccinated (Dose 2) Czech cohorts enrolled in 2021_24. (Note: the all-ages row reflects aggregation effects and is not directly comparable to age-stratified rows.)

Age band (years)	Dose 0 cumulative hazard	Dose 2 cumulative hazard	Ratio
40–49	0.005260	0.004117	1.2776
50–59	0.014969	0.009582	1.5622
60–69	0.045475	0.023136	1.9655
70–79	0.123097	0.057675	2.1343
80–89	0.307169	0.167345	1.8355
90–99	0.776341	0.517284	1.5008
All ages (full population)	0.023160	0.073323	0.3159

This table reports unadjusted cumulative hazards derived directly from the raw data, prior to any frailty normalization or depletion correction, and is shown to illustrate the magnitude and direction of selection-induced curvature addressed by KCOR.

Values reflect raw cumulative outcome differences prior to KCOR normalization and are not interpreted causally due to cohort non-exchangeability. Cumulative hazards were integrated from cohort enrollment through the end of available follow-up for the 2021_24 enrollment window (through week 2024-16), identically for Dose 0 and Dose 2 cohorts.

S6.1.3 Illustrative application to national registry mortality data

A brief illustrative application is included to demonstrate end-to-end KCOR behavior on real registry mortality data in a setting that minimizes timing-driven shocks and window-tuning sensitivity. Cohorts were enrolled in ISO week 2022-26, and evaluation was restricted to calendar year 2023, yielding a 26-week post-enrollment buffer before slope estimation and a prespecified assessment period. Frailty parameters were estimated using the revised quiet-window settings and diagnostics described above to minimize wave-related hazard variation. This example is intended to illustrate estimator behavior under real-world selection and heterogeneity and does not support causal inference.

Figure S10 shows KCOR(t) trajectories for dose 2 and dose 3 relative to dose 0 for an all-ages analysis. An all-ages analysis is presented as a high-heterogeneity stress test, since aggregation across age induces substantial baseline hazard and frailty variation.

S7. Computational environment and runtime notes

Environment. Python 3.11; key dependencies include numpy, scipy, pandas, and lifelines (for Cox-model comparisons), with plotting via matplotlib.

Compute requirements. The full simulation grid reproduces in approximately 1 hour 26 minutes on a 20-core CPU with 128 GB RAM; smaller subsets reproduce in minutes.

Reproduction. Running `make paper` (or the repository's top-level build command) regenerates all artifacts from a clean checkout.

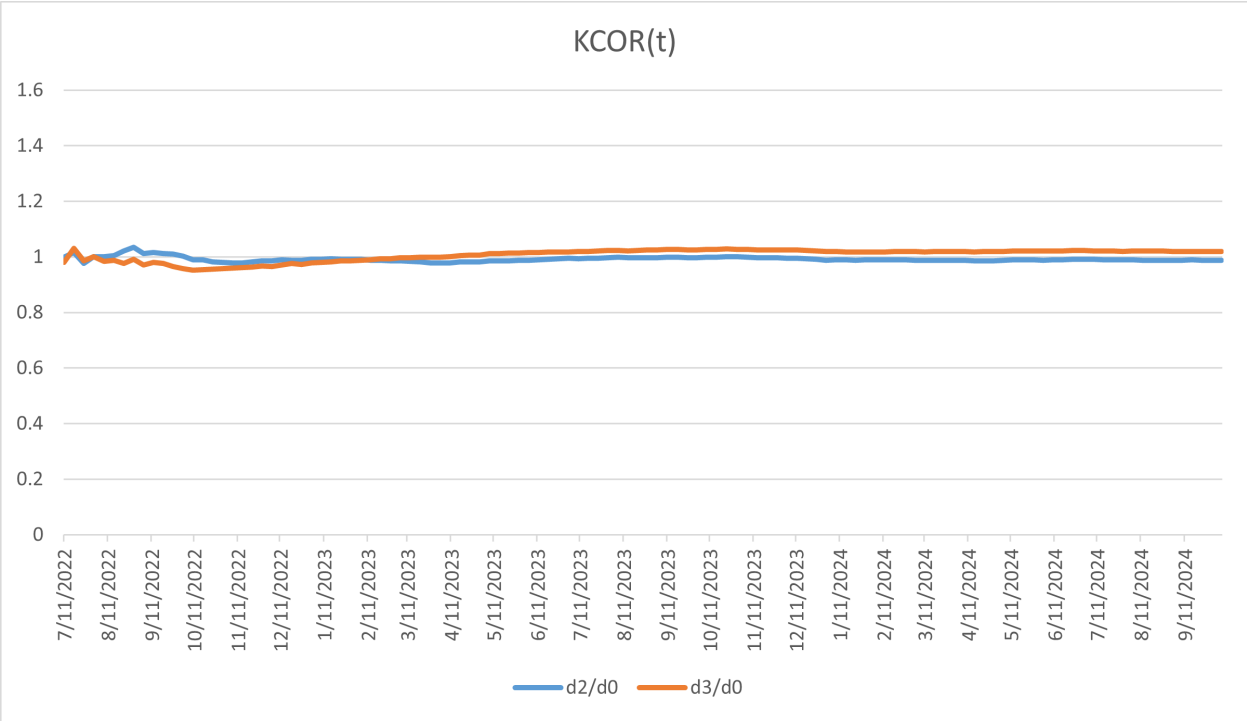

Figure S10: All-ages stress test: KCOR(t) trajectories comparing dose 2 and dose 3 to dose 0 for cohorts enrolled in ISO week 2022-26 and evaluated over calendar year 2023. KCOR curves are anchored at $t_0 = 4$ weeks (i.e., plotted as KCOR$(t; t_0)$). This figure is presented as an illustrative application demonstrating estimator behavior under extreme heterogeneity and does not support causal inference. X-axis units are weeks since enrollment.